Chapter 1

Basic Conception

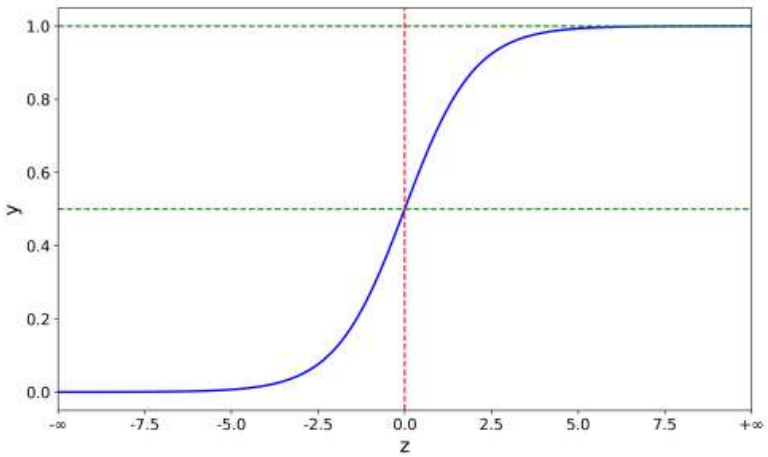
AI发展历程 & 基本要素

人工智能主要任务类别

线性回归 & 逻辑回归

深度学习 & 全连接网络

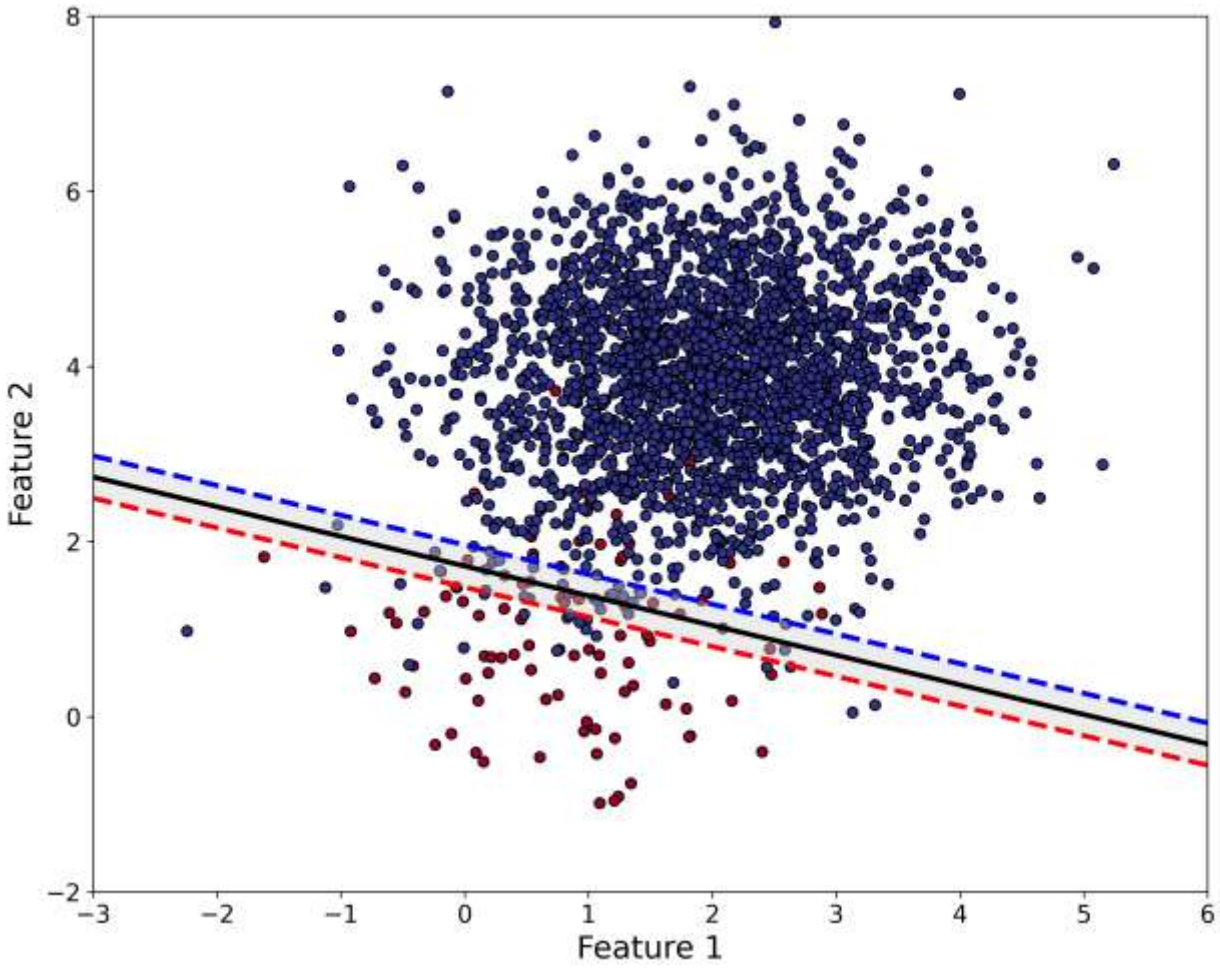
4.1 逻辑回归面临的问题



$$y = \frac{1}{1+e^{-z}}, z = \sum w_i x_i + b$$

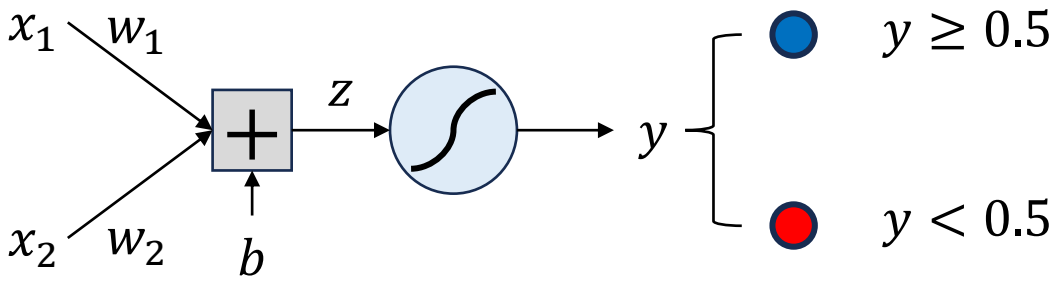
找出一个分界线（超平面）

却无法判别非凸集合



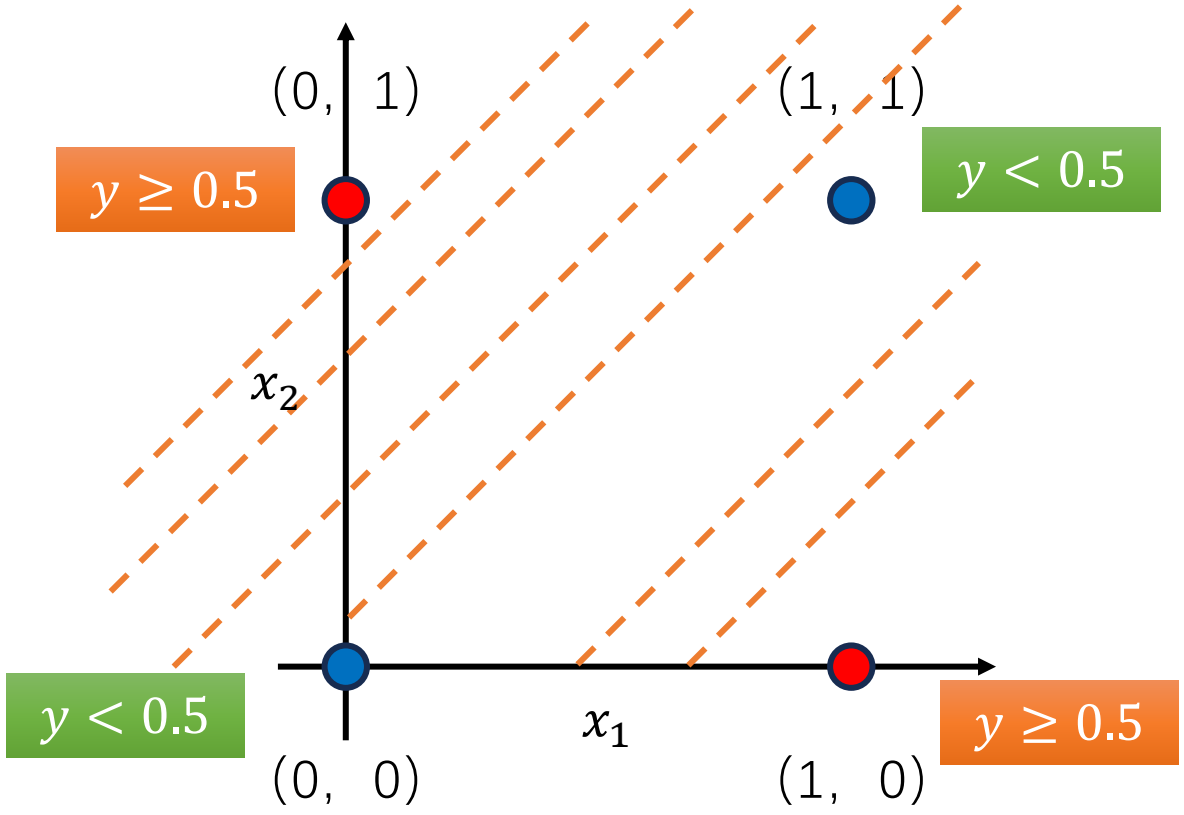
4.1 逻辑回归面临的问题

$$z = w_1x_1 + w_2x_2 + b$$



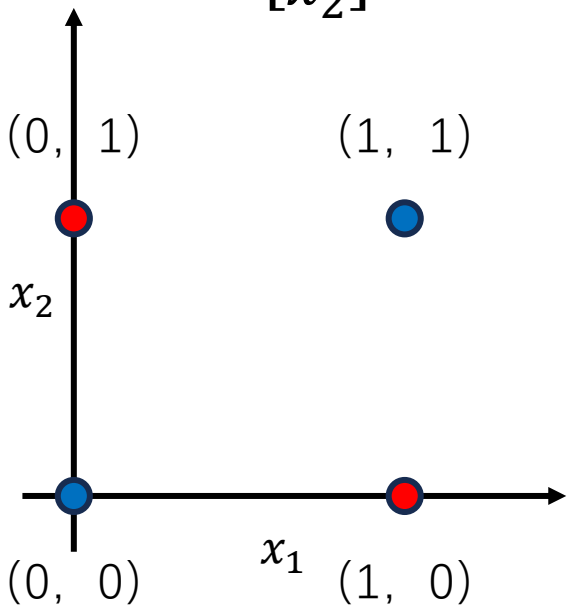
Feature 1 (x_1)	Feature 2 (x_2)	Label (y)
0	0	
1	0	
1	1	
0	1	

就仅有这么一点实力吗？



4.1 逻辑回归面临的问题 ➢ 特征变换

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



x_1	x_2	y
0	0	
1	0	
1	1	
0	1	

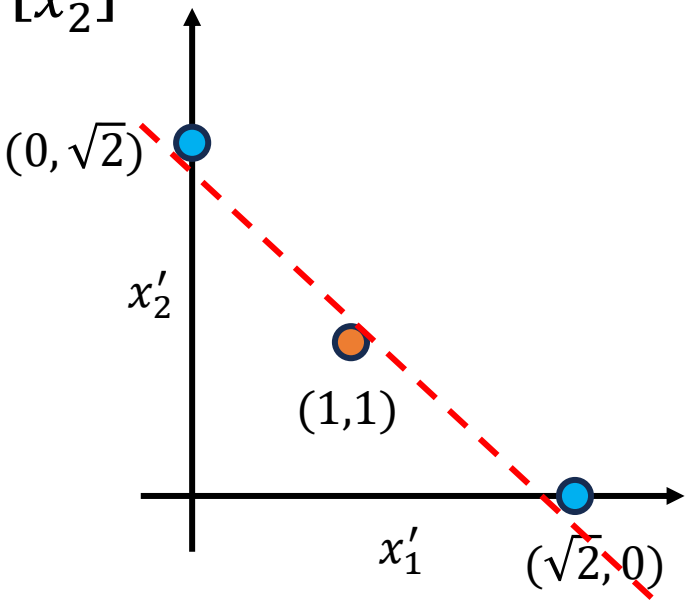
变换特征坐标系

Rule 1: x_1' 为距（0，0）的距离

Rule 2: x_2' 为距（1，1）的距离

如何让机器自动地进行特征变换?

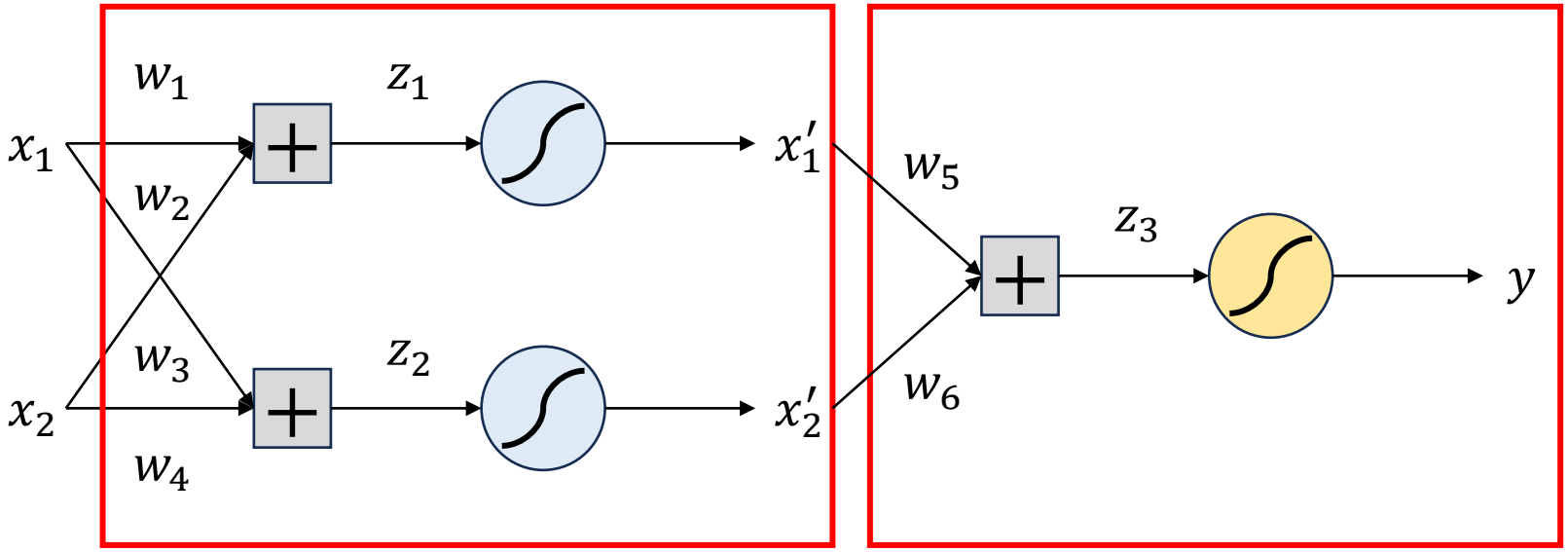
$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix}$$



x_1'	x_2'	y
0	$\sqrt{2}$	
1	1	
$\sqrt{2}$	0	
1	1	

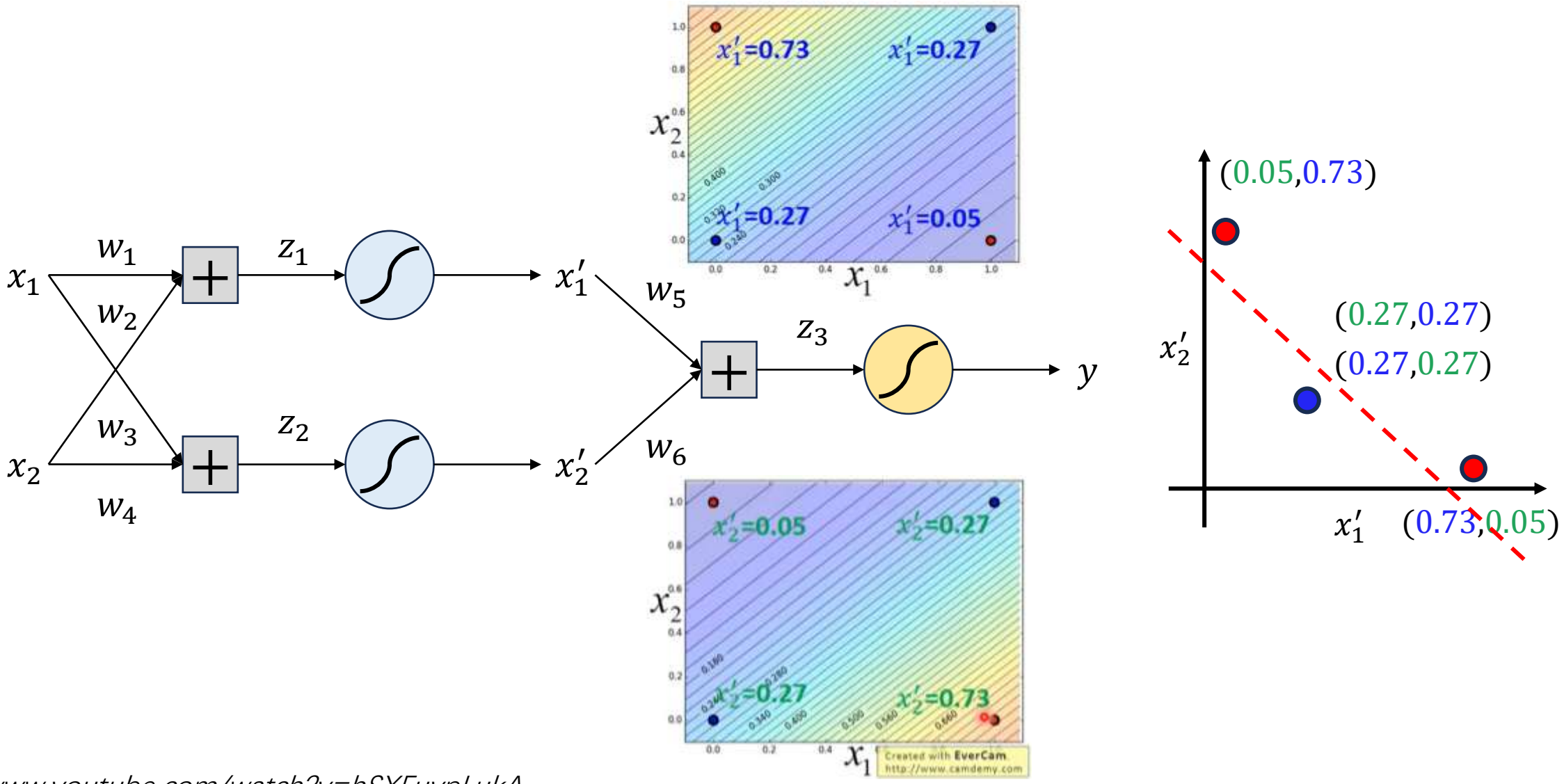
4.1 逻辑回归面临的问题 ➢ 特征变换

让机器自动进行特征变换的方法——级联多个函数



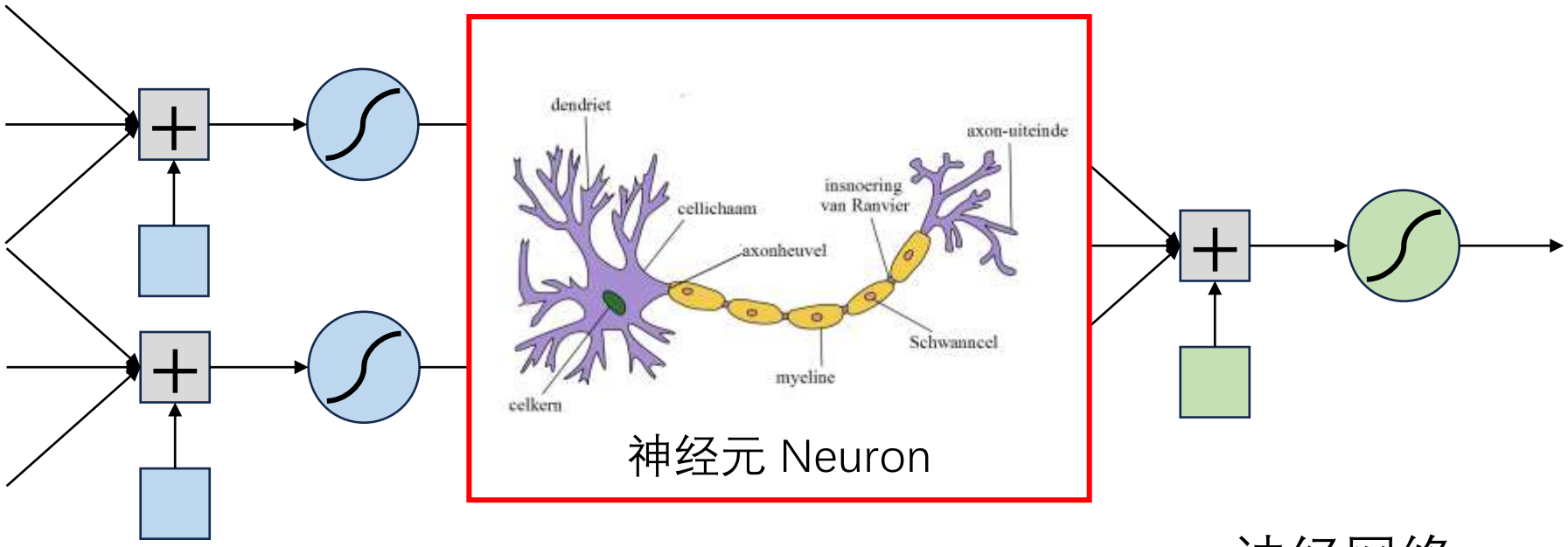
让机器自动地进行特征变换　　分类

4.1 逻辑回归面临的问题 ➢ 特征变换



https://www.youtube.com/watch?v=hSXFuypLukA

深度学习
Deep Learning



神经元 Neuron

神经网络
Neural Network

关于深度学习的故事

- 1958：感知机 Perceptron（linear model）
- 1969：Perceptron has limitation
- 1980s：多层感知机 Multi-layer Perceptron
  - Do not have significant difference from DNN today
- 1986：反向传播算法 Backpropagation
- 1989：1 hidden layer is “good enough”, why deep?
- 2010：IMAGENET：最大的数据集
- 2011：Start to be popular in speech recognition
  - 发现GPU效果非常好
- 2012：Win ILSVRC image competition
  - 深度学习+GPU=碾压之势

Geoffrey Hinton
图灵\诺贝尔物理学奖得主

李飞飞
ILSVRC竞赛创始人

黄仁勋
Nvidia CEO

Alex Krizhevsky

Ilya Sutskever

那剩下的工作就是通过深度学习

《刀剑神域 -序列之争-》

4.2 深度神经网络模型

步骤1

一大堆数据

步骤2

一堆模型

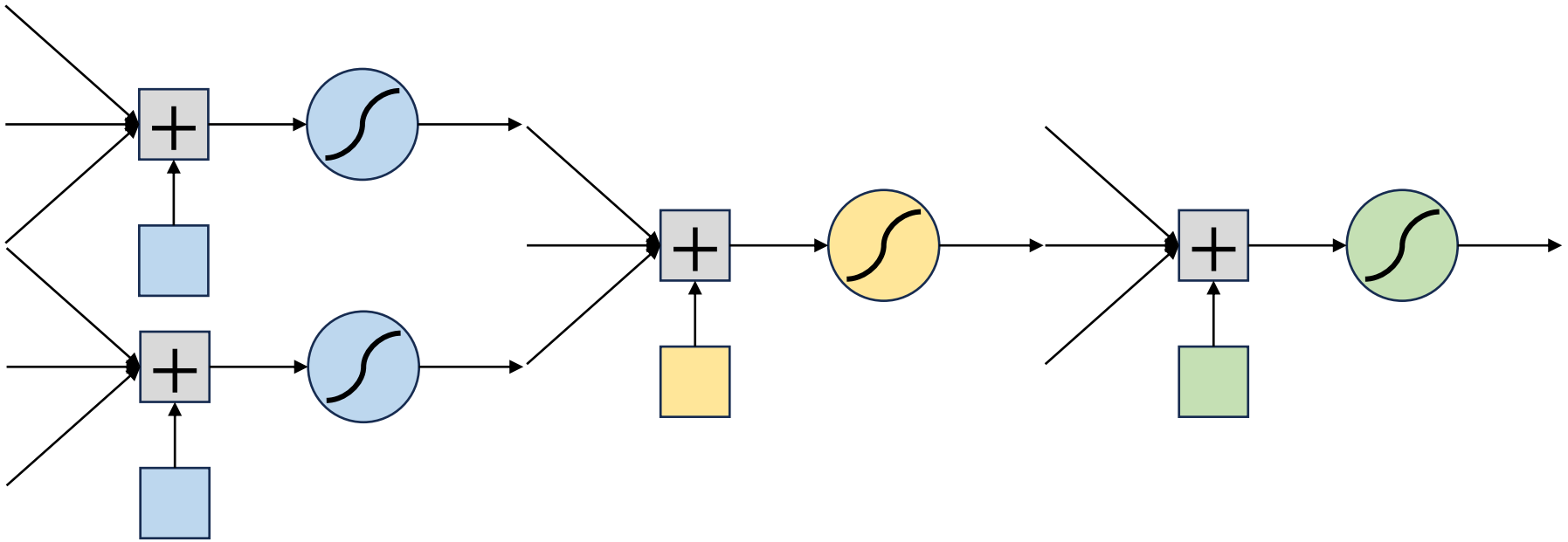
深度神经网络

步骤3

选择最优模型

《钟点工》2000年春节联欢晚会

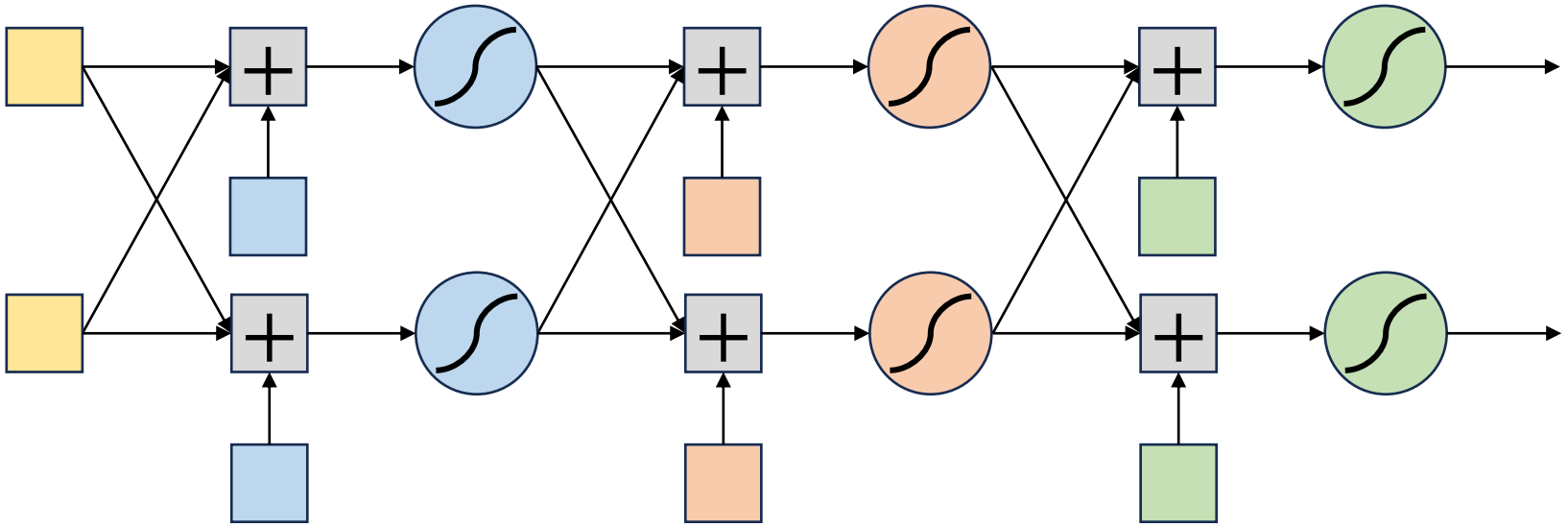
4.2 深度神经网络模型



通过改变神经元的连接方式

得到不同的神经网络

4.2 深度神经网络模型——全连接模型



节点完全连接

全连接神经网络
Fully-Connected Network (FCN)

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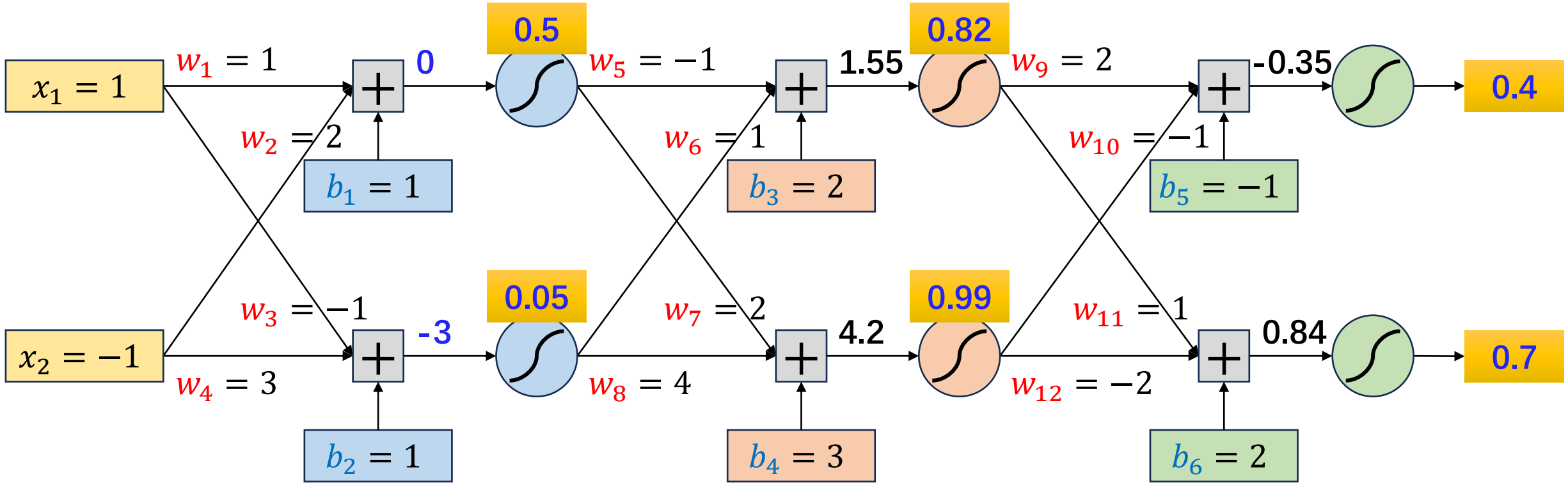
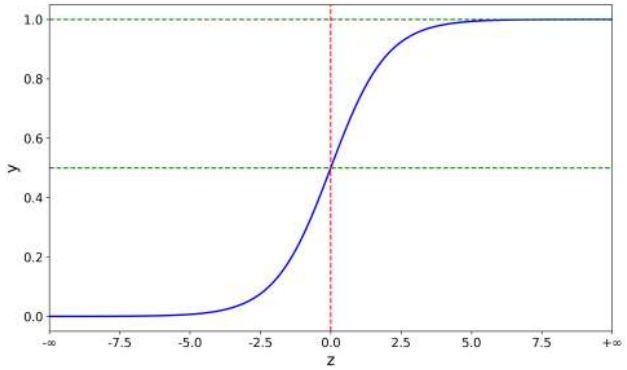


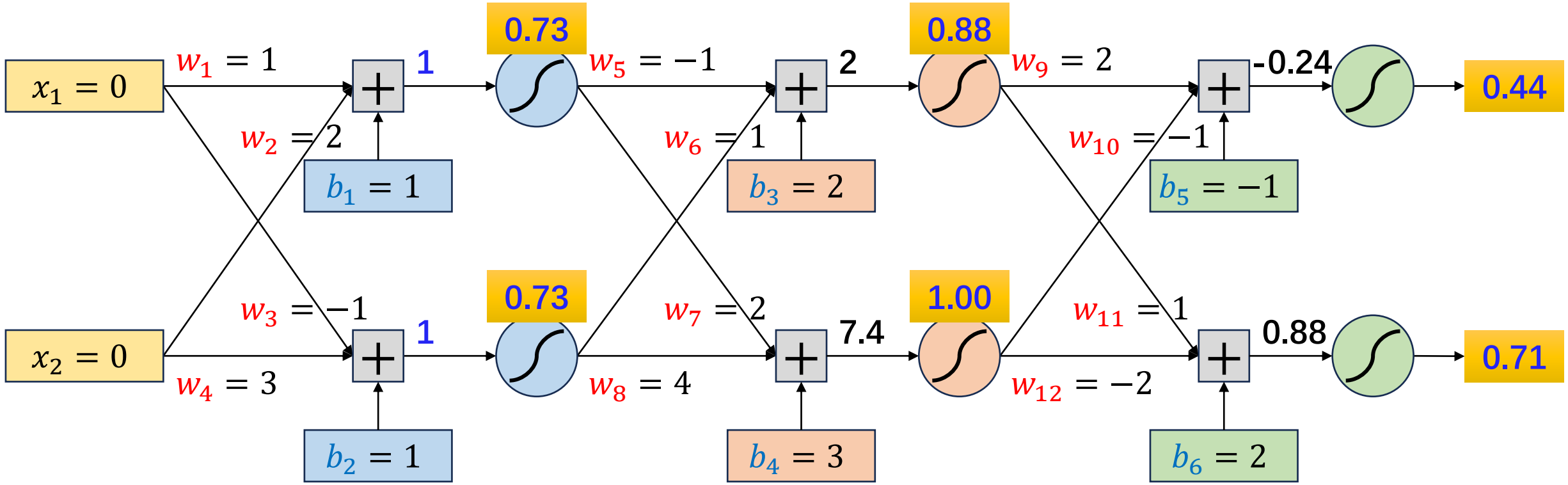
全连接神经网络
Fully-Connected Network (FCN)



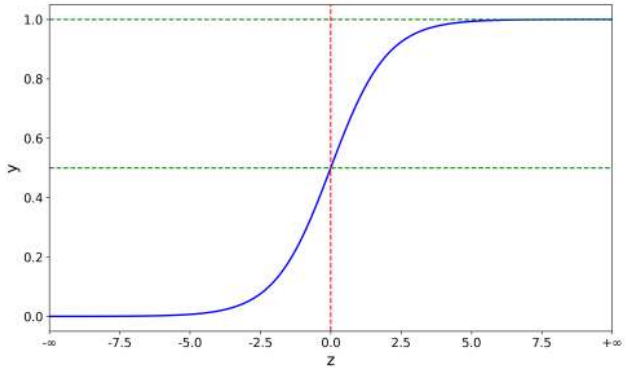
$$f(x_1, x_2) \rightarrow (x_1', x_2')$$

$$(1, -1) \rightarrow (0.4, 0.7)$$

4.2 深度神经网络模型——全连接模型



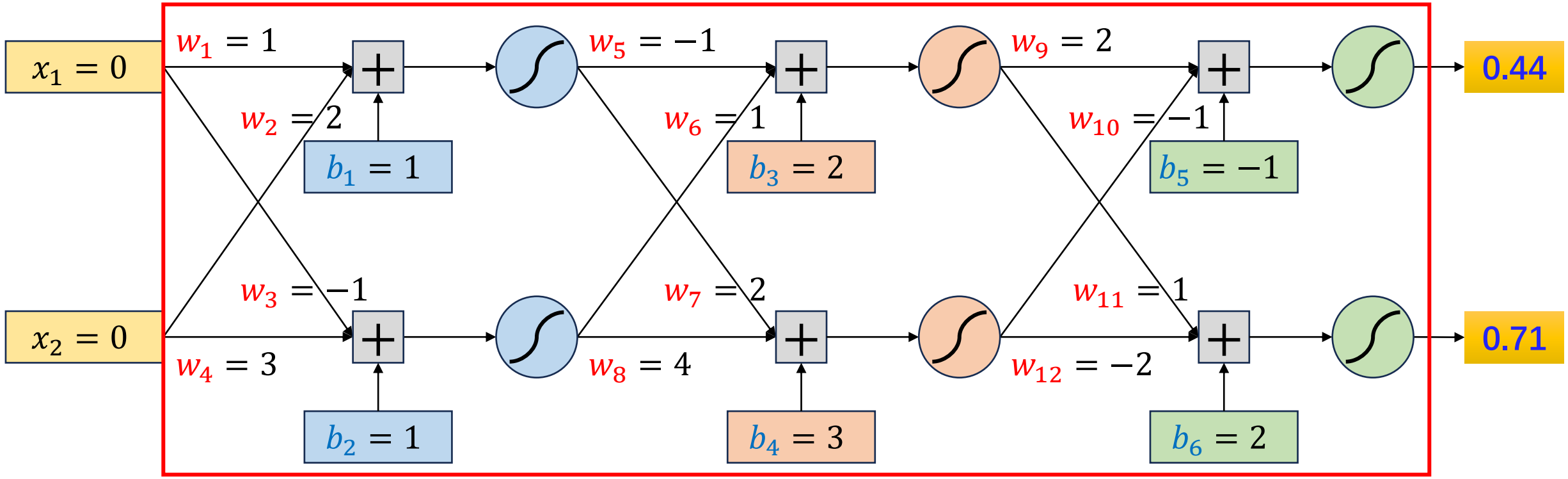
全连接神经网络
Fully-Connected Network (FCN)



$$f(x_1, x_2) \rightarrow (x_1', x_2')$$

$$(0, 0) \rightarrow (0.44, 0.71)$$

4.2 深度神经网络模型——全连接模型

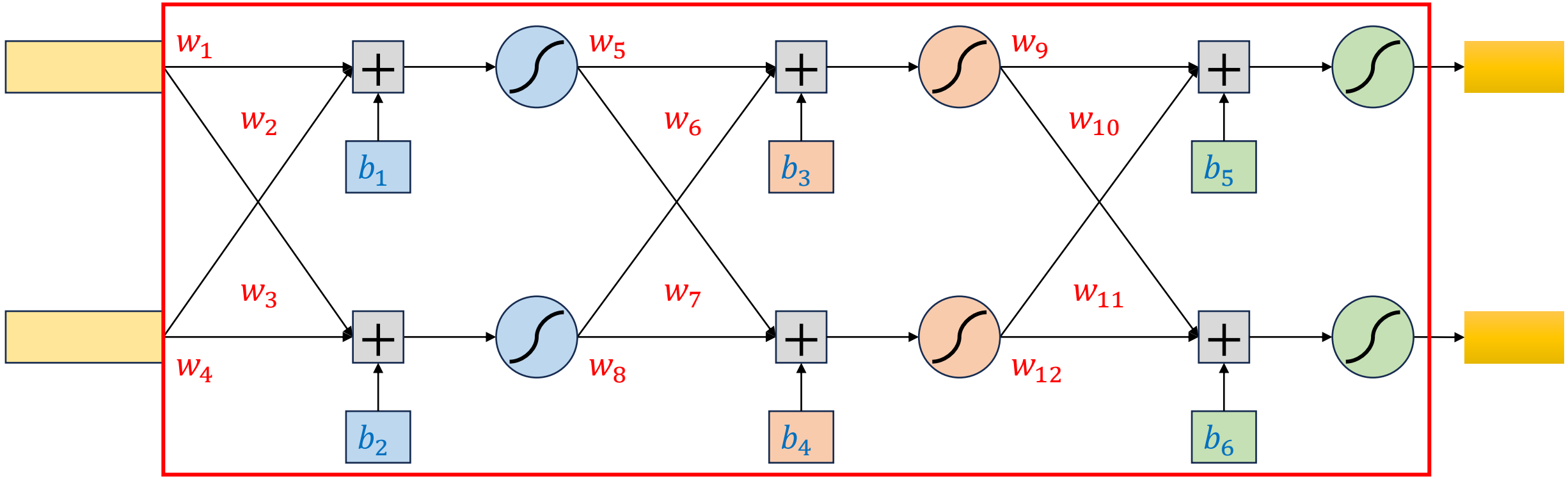


全连接神经网络
Fully-Connected Network (FCN)

这是一个函数$f(\boldsymbol{x})$，参数的集合用 $\boldsymbol{\theta}$ 表示，此时

$\boldsymbol{\theta} = \{w_1, w_2, \dots, w_8, b_1, b_2, \dots, b_6\}$ 有确定的值

4.2 深度神经网络模型——全连接模型

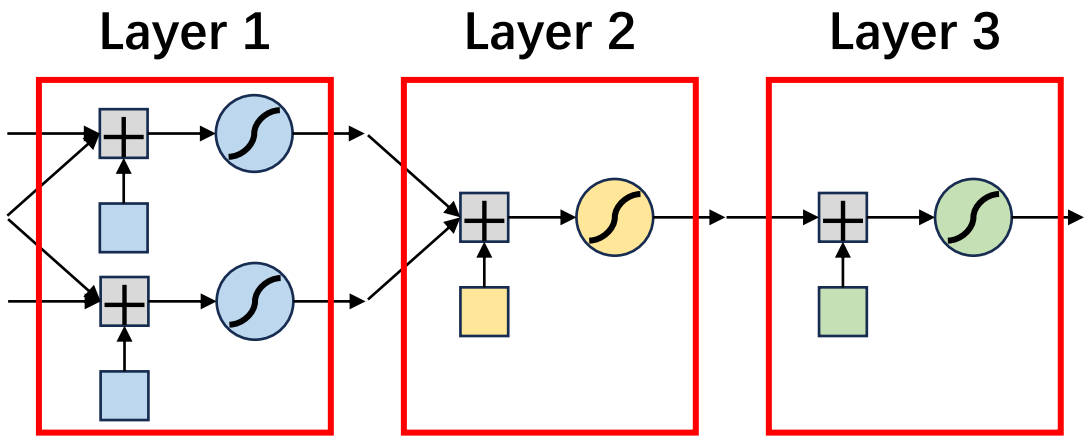


全连接神经网络
Fully-Connected Network (FCN)

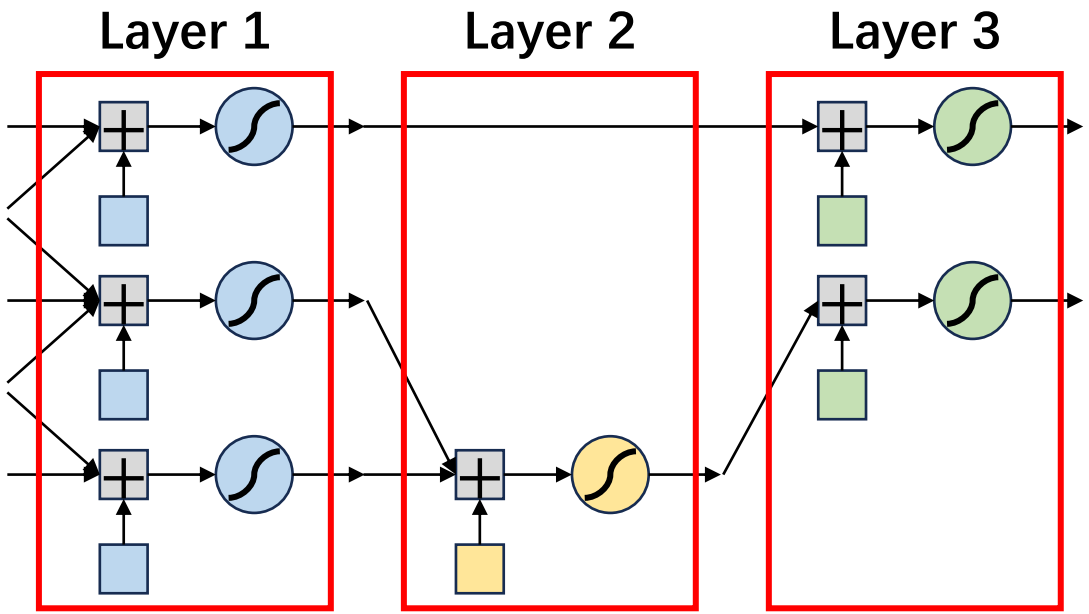
$\boldsymbol{\theta}$ 未确定时，定义的框架表示一个**函数集合**

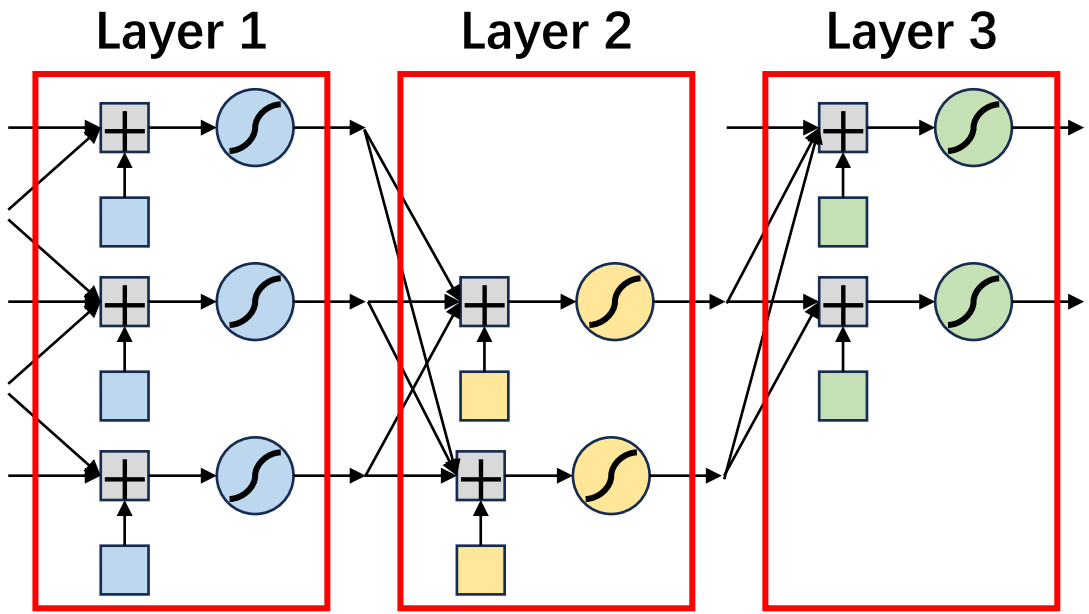
简单的机制使得**神经网络的规模极易扩展**

4.3 深度神经网络的设计



信息在网络中逐层传播





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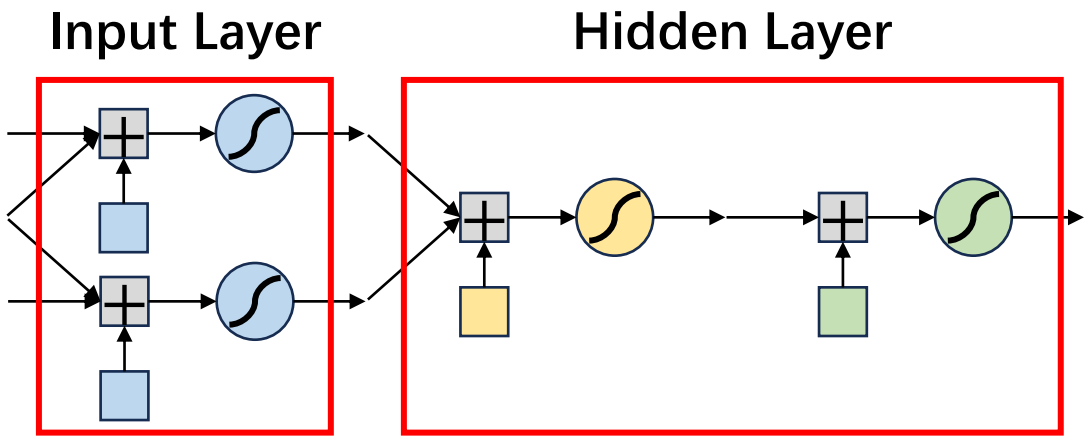

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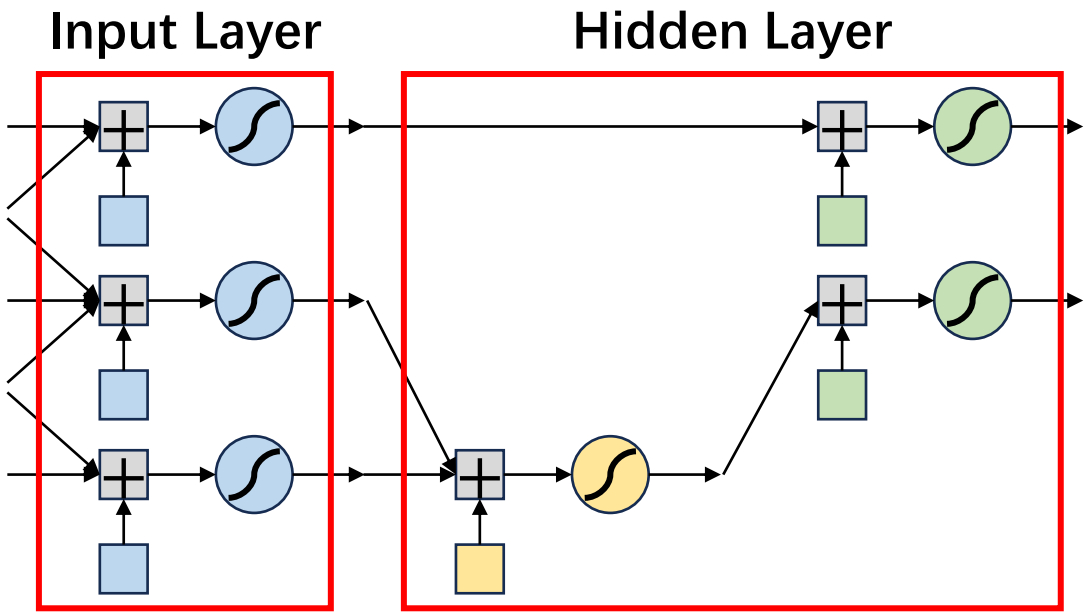
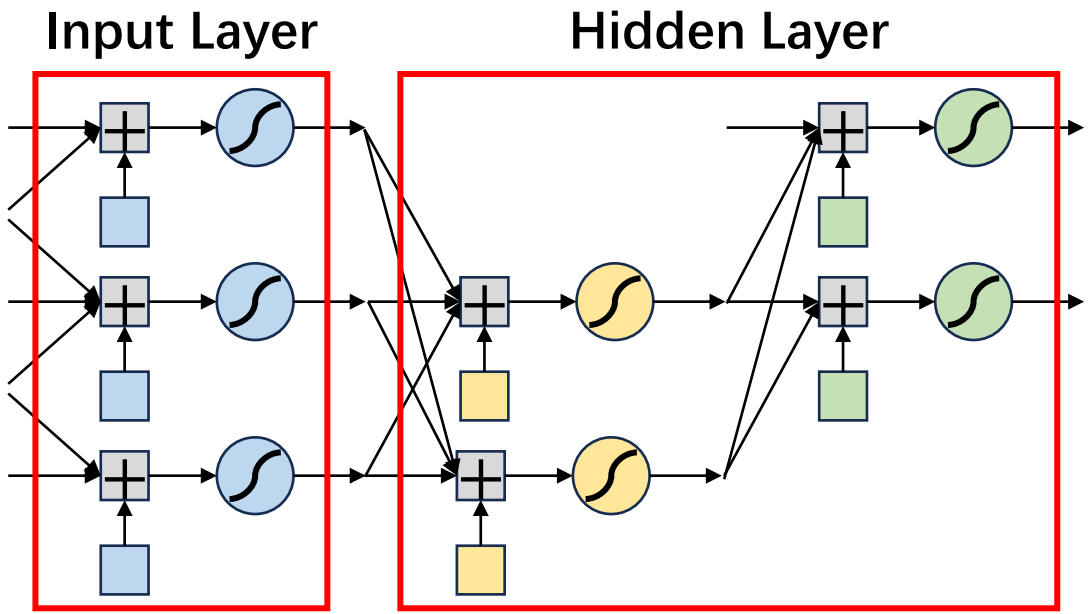
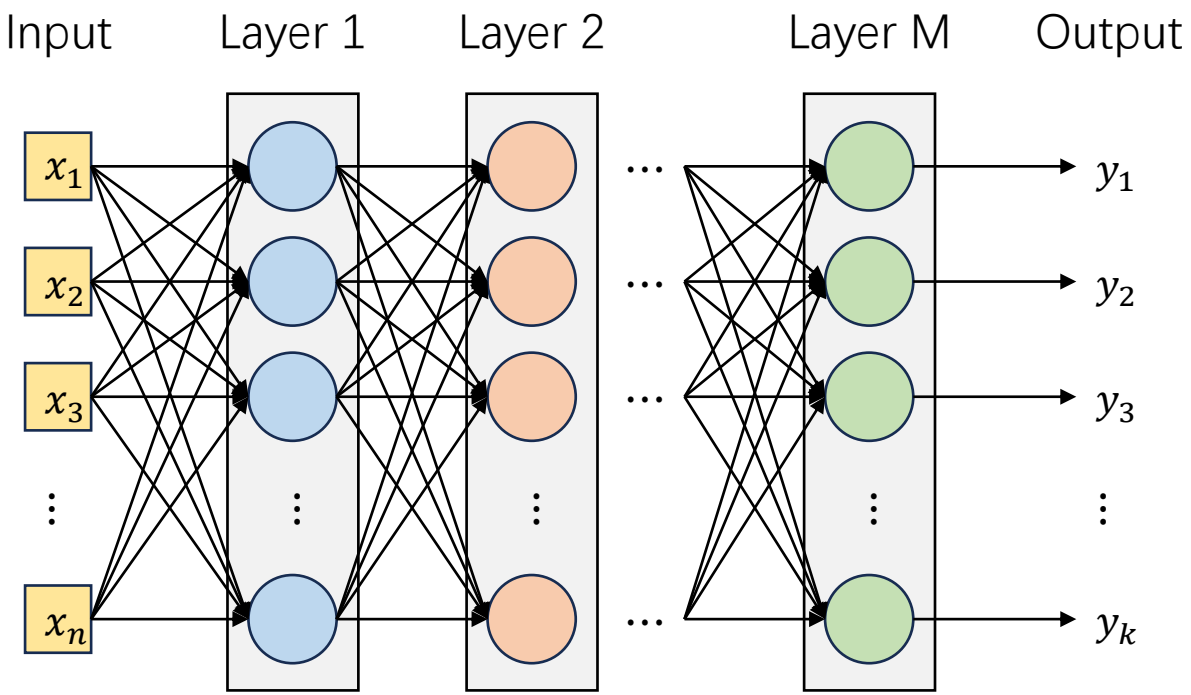




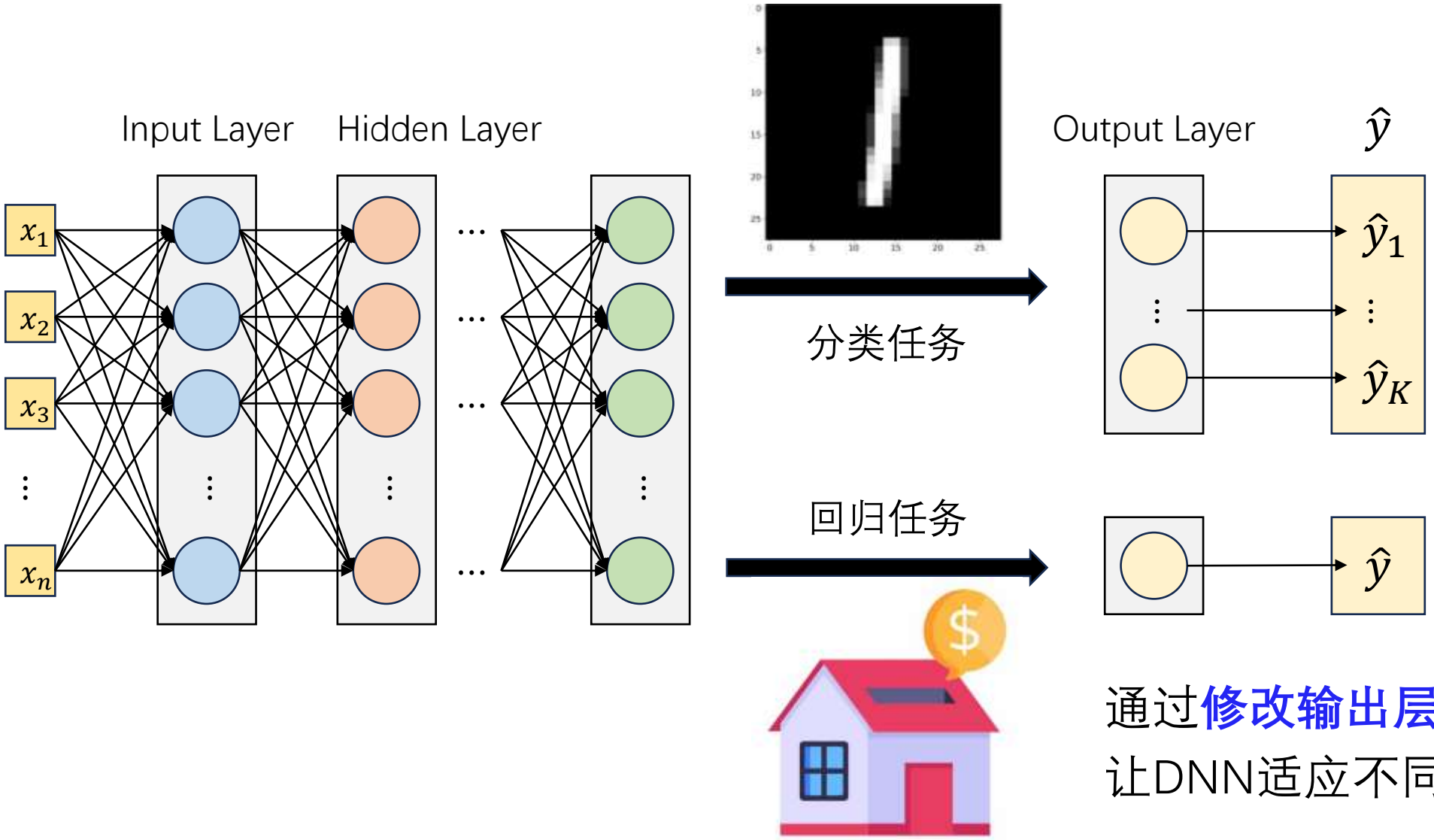
4.3 深度神经网络的设计

问题 1：DNN是“回归”还是“分类”？

DNN是一种模型结构
并不局限于具体问题



4.3 深度神经网络的设计



通过**修改输出层**就可以
让DNN适应不同的任务

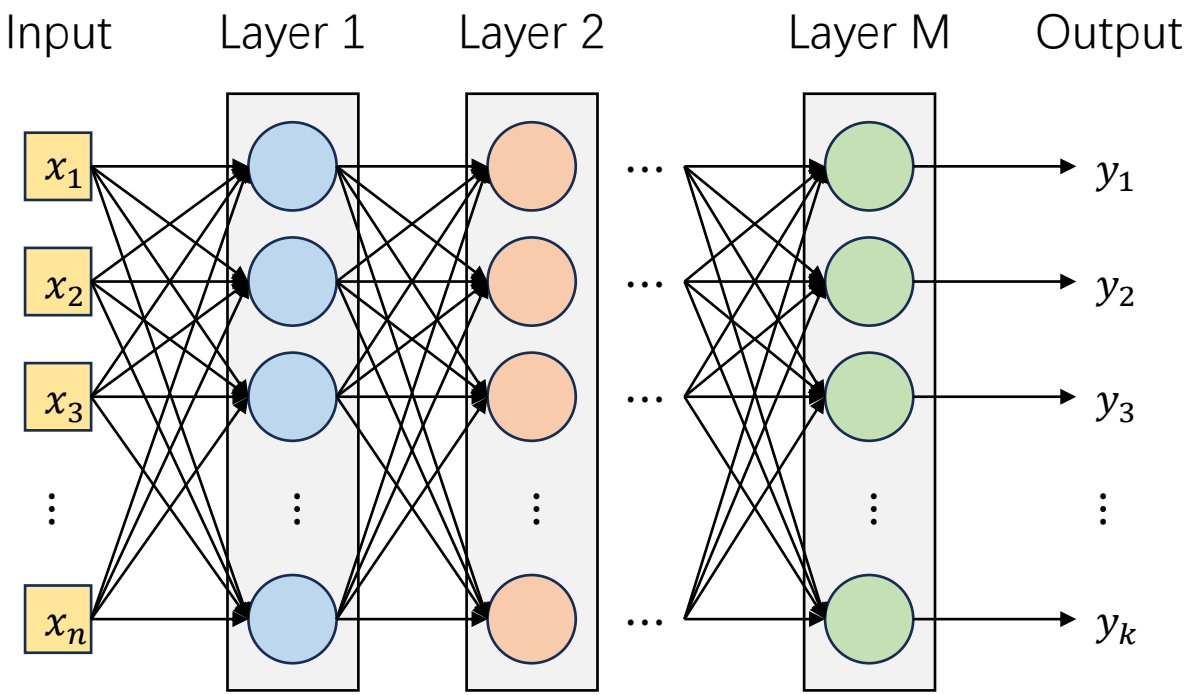
4.3 深度神经网络的设计

问题 1：DNN是“回归”还是“分类”？

DNN是一种模型结构
并不局限于具体问题

问题 2：模型结构对任务表现有影响吗？

肯定的



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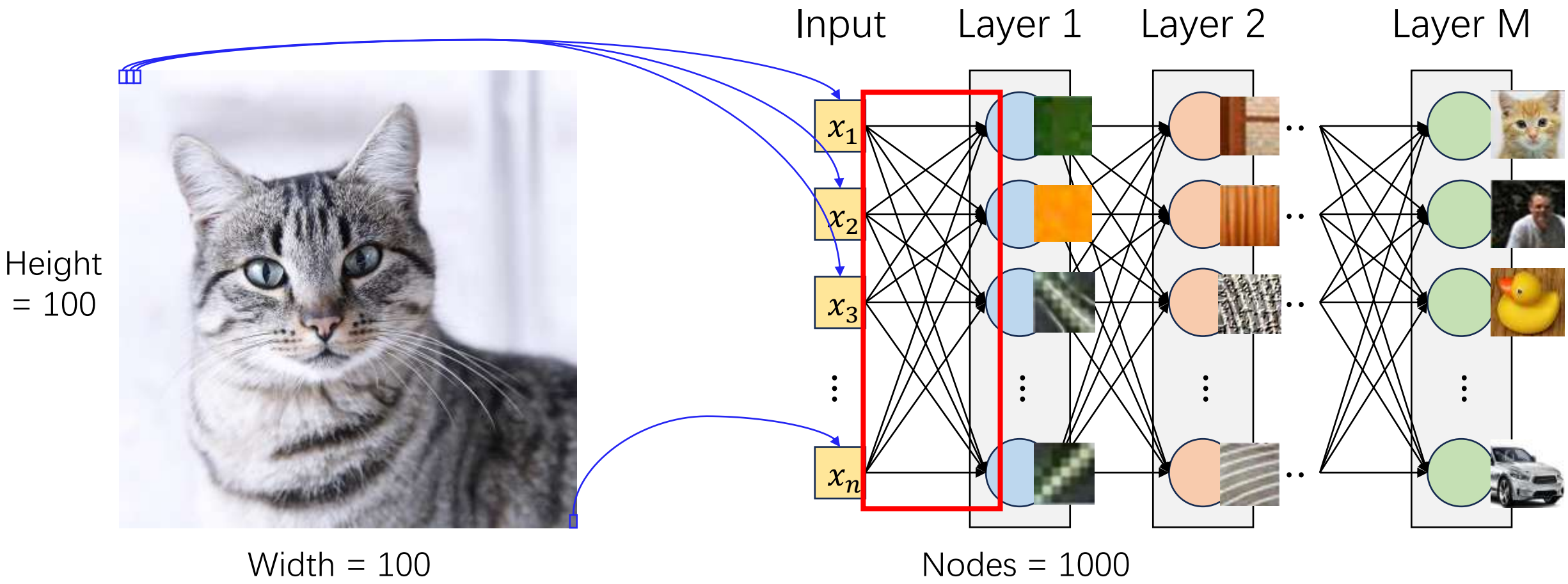


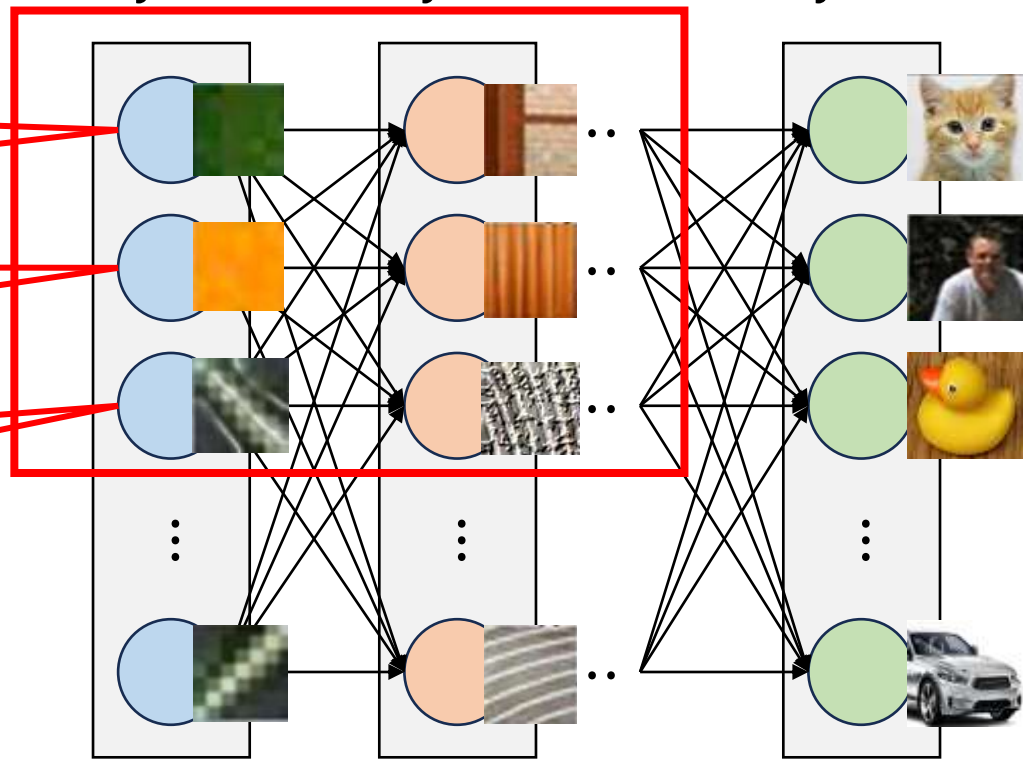
第一层参数：100 X 100 X 3 X 1000 = 3 000 0000

第二层参数：1000 X 1000 = 100 0000

直接使用FCN，计算量非常大

4.3 深度神经网络的设计

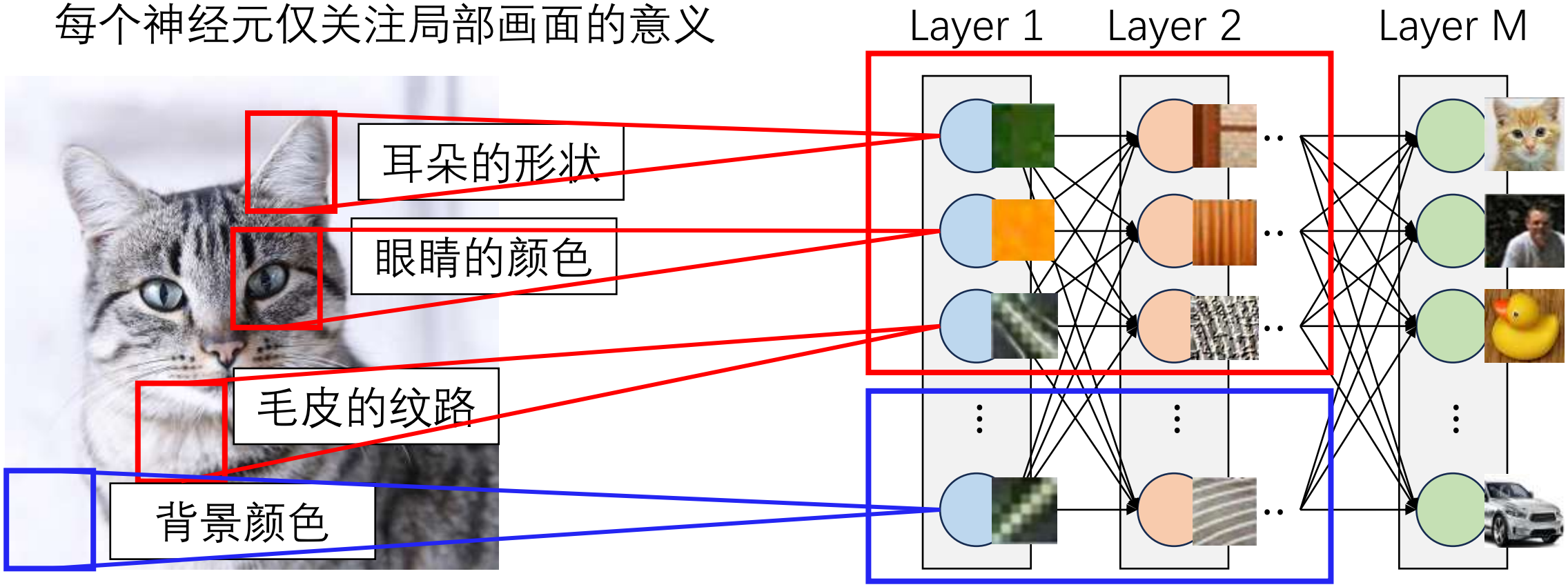
每个神经元仅关注局部画面的意义



对于图片分类任务，可能仅需要关注部分特征

4.3 深度神经网络的设计

每个神经元仅关注局部画面的意义



然而，有些关注点很奇快，甚至会造成反效果

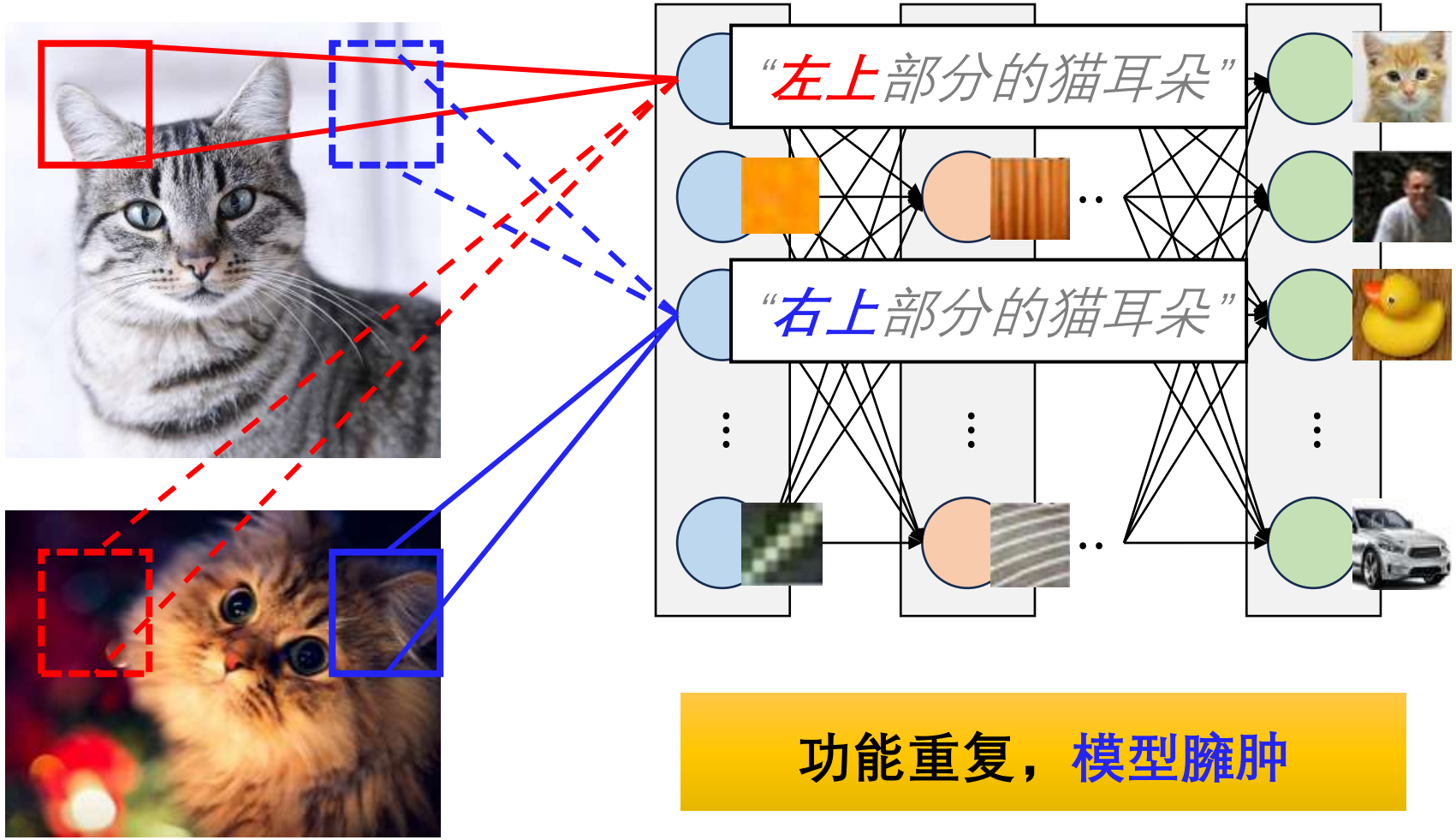
对于图片分类任务，可能仅需要关注部分特征

4.3 深度神经网络的设计

例如，同样模式在不同的图片中的位置不一

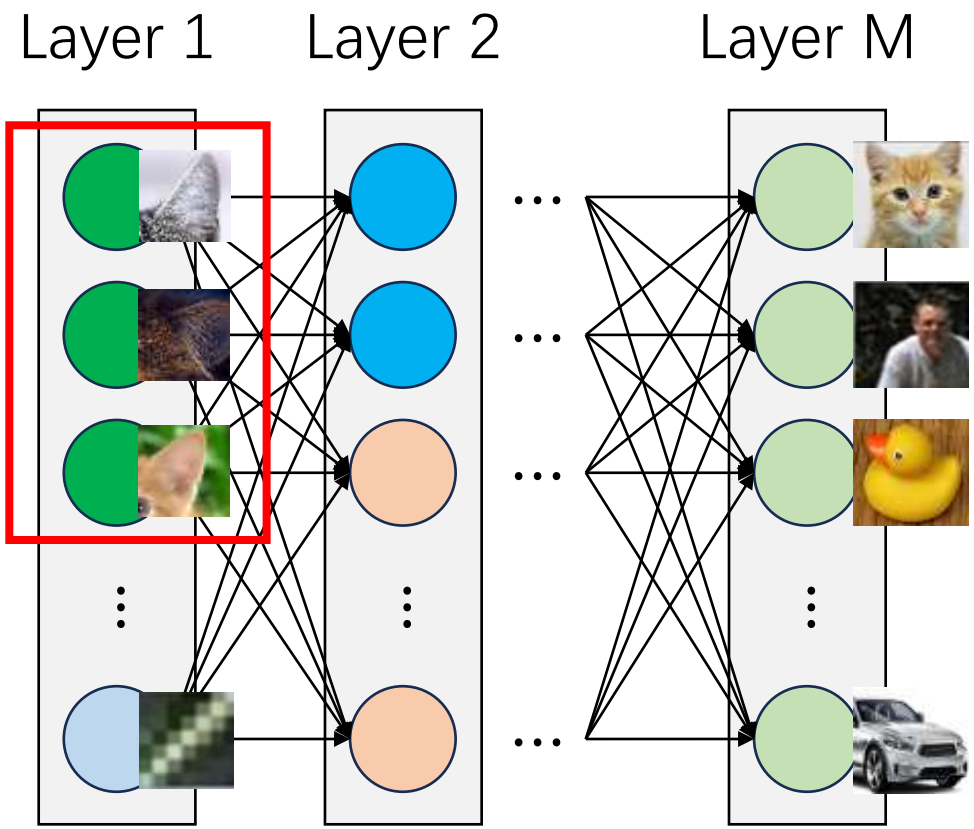
“左上部分有耳朵，是猫的可能性增加；但是，右上角没有耳朵，所以有可能不是猫”

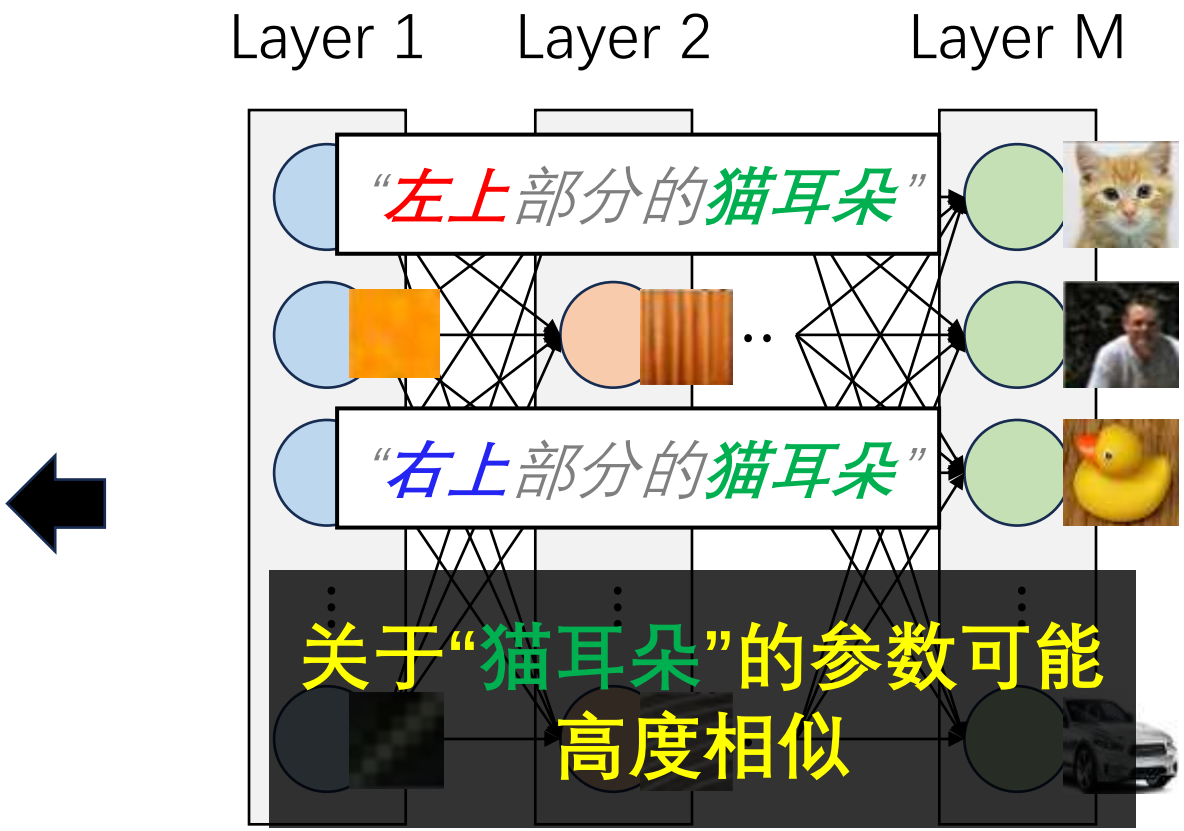
“左上部分没有耳朵，应该不是猫；但是，右上角有耳朵，可是猫”



功能重复，模型臃肿

4.3 深度神经网络的设计

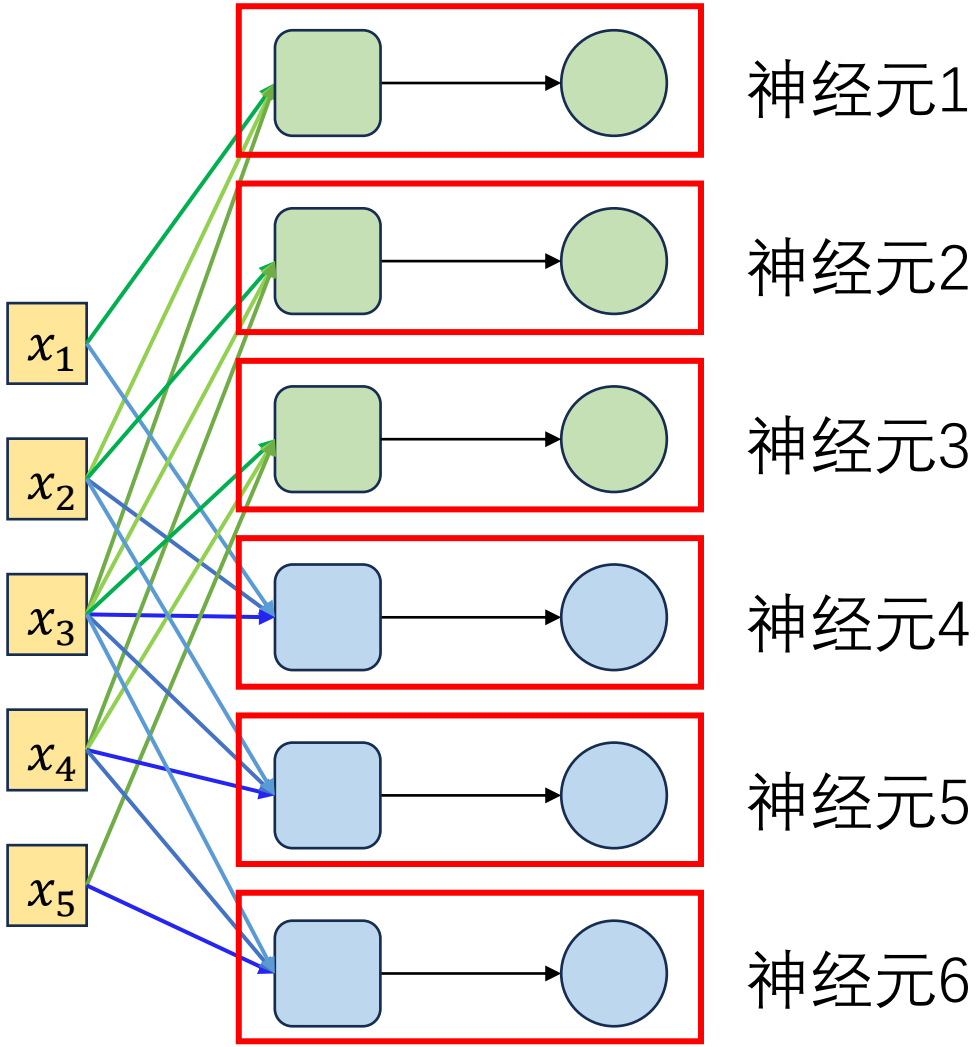


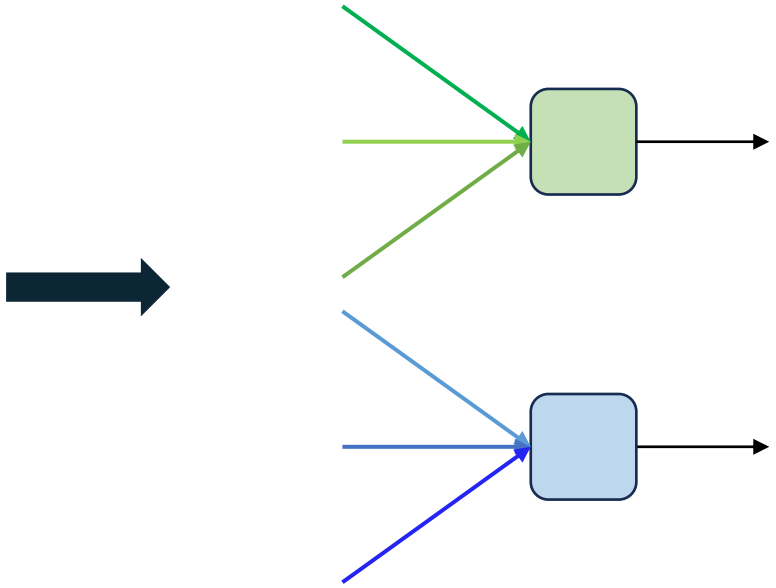


让具有相似功能的节点共享参数

功能重复，模型臃肿

4.3 深度神经网络的设计

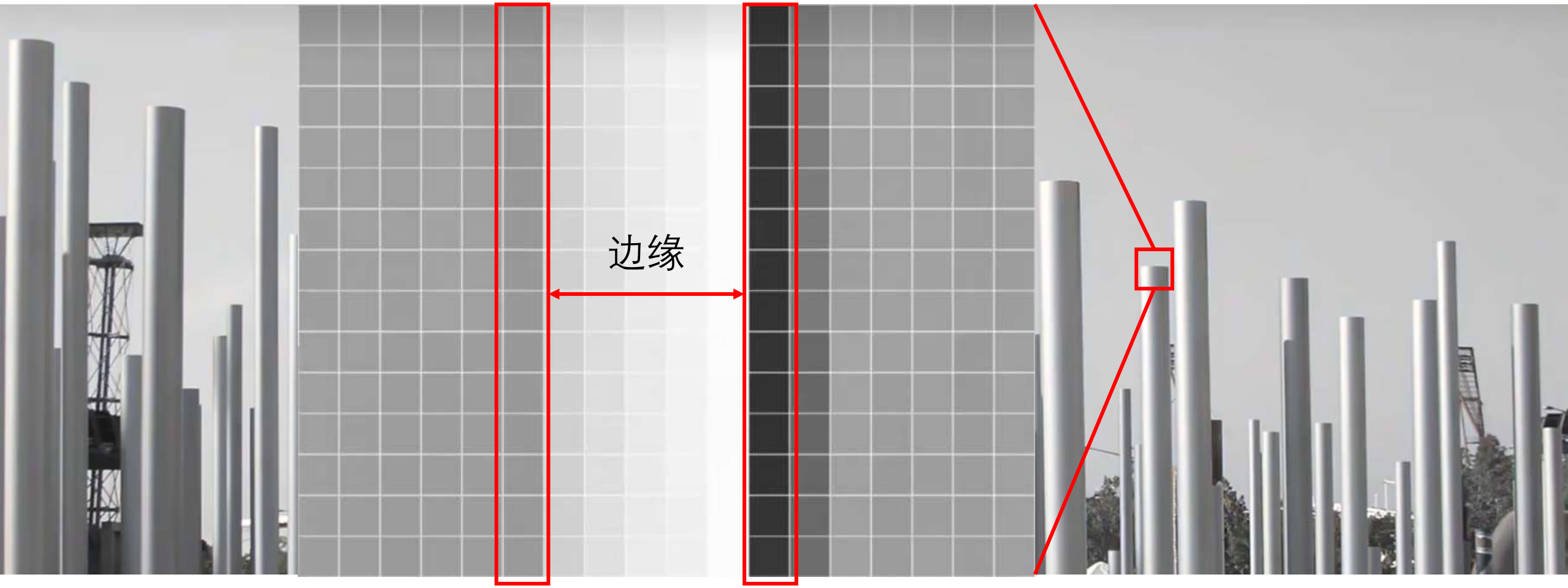




共享参数让模型更简洁高效

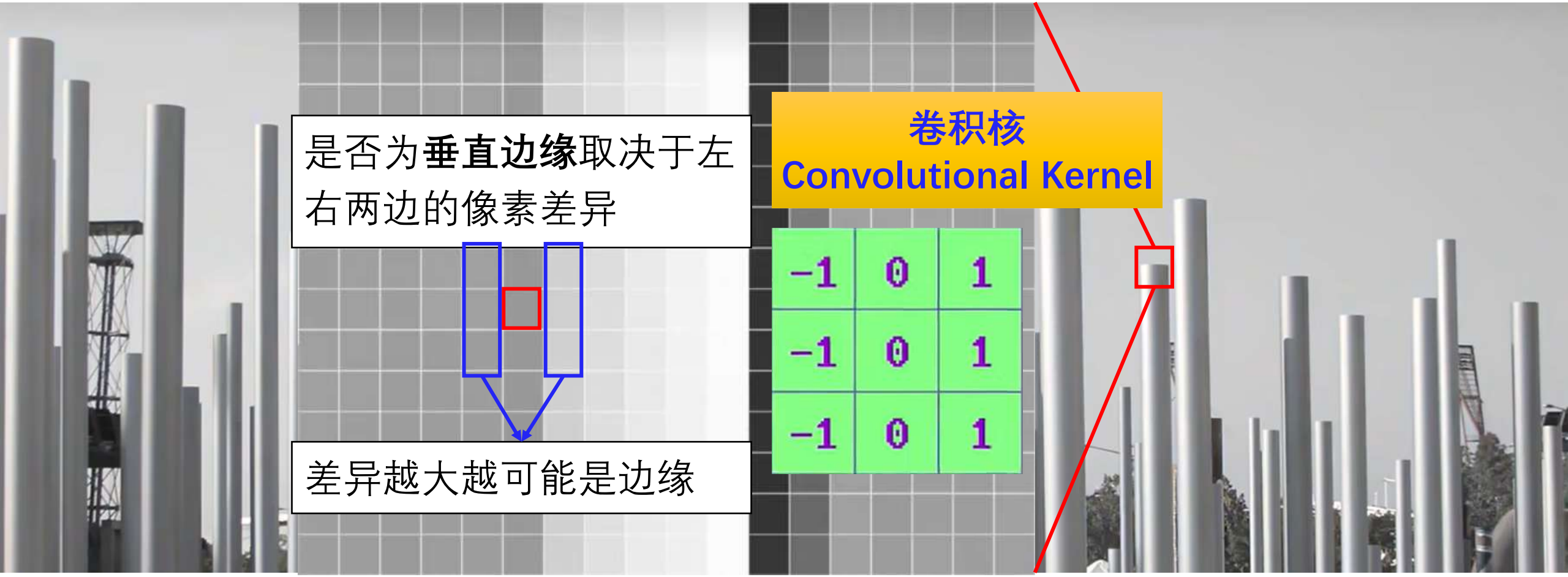
接下来的问题，如何让神经元“看”整个图像？

4.3 深度神经网络的设计



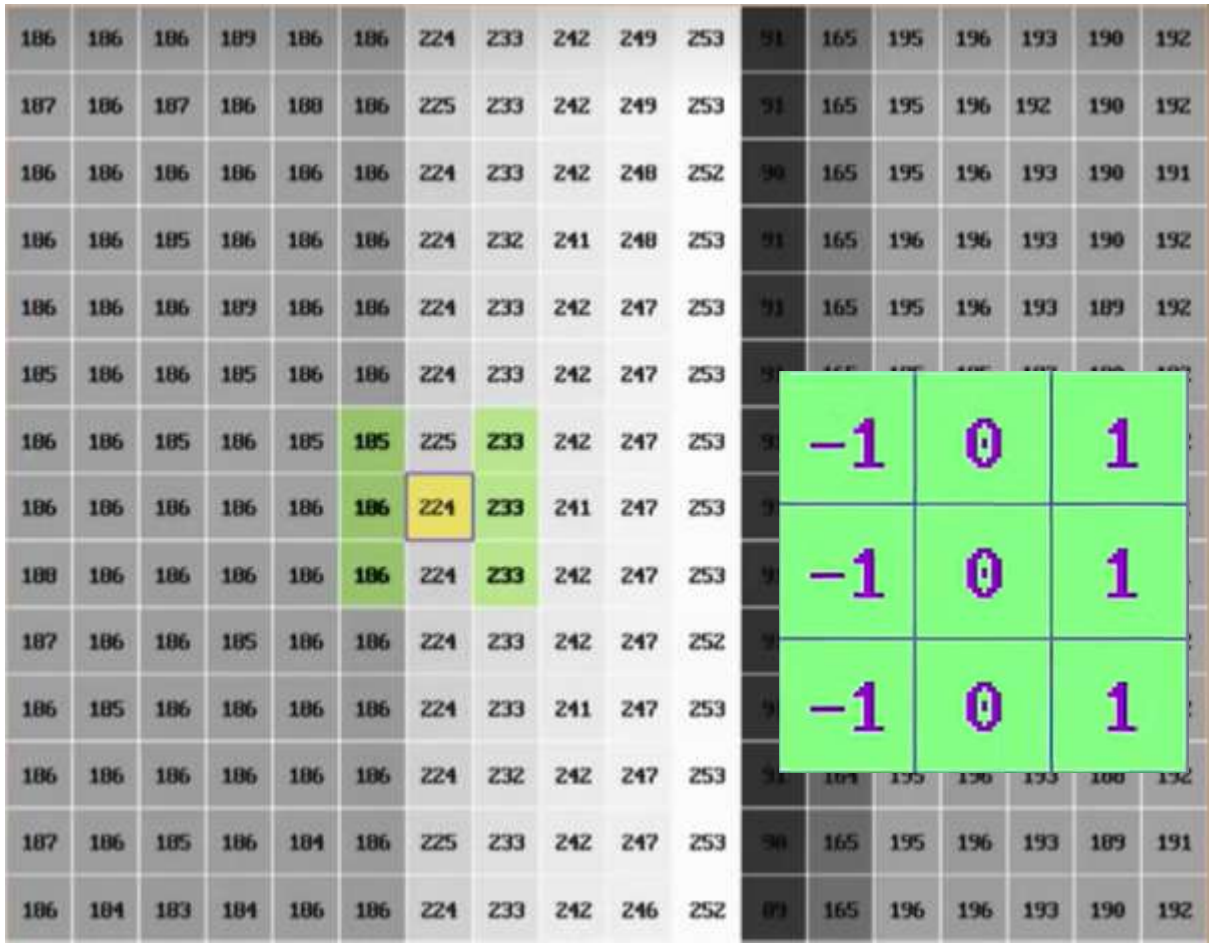
看看边缘检测怎么做

4.3 深度神经网络的设计



看看边缘检测怎么做 **定义检测块**

4.3 深度神经网络的设计

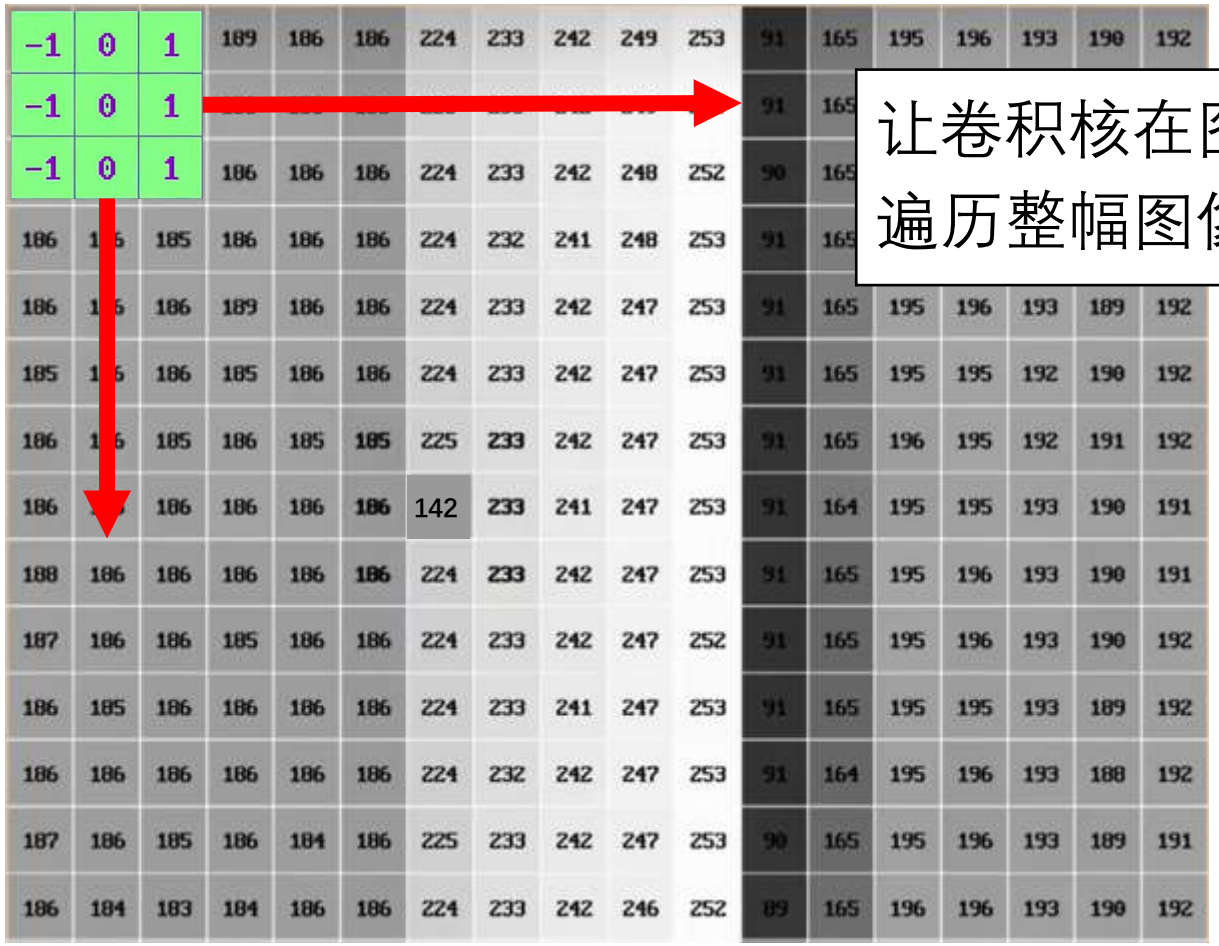


$$
\begin{aligned}
&185 \times (-1) \\
&+ 186 \times (-1) \\
&+ 186 \times (-1) \\
&+ 225 \times 0 \\
&+ 224 \times 0 \\
&+ 224 \times 0 \\
&+ 223 \times 1 \\
&+ 223 \times 1 \\
&+ 223 \times 1 \\
&= 142
\end{aligned}
$$

看看边缘检测怎么做

定义检测块

4.3 深度神经网络的设计



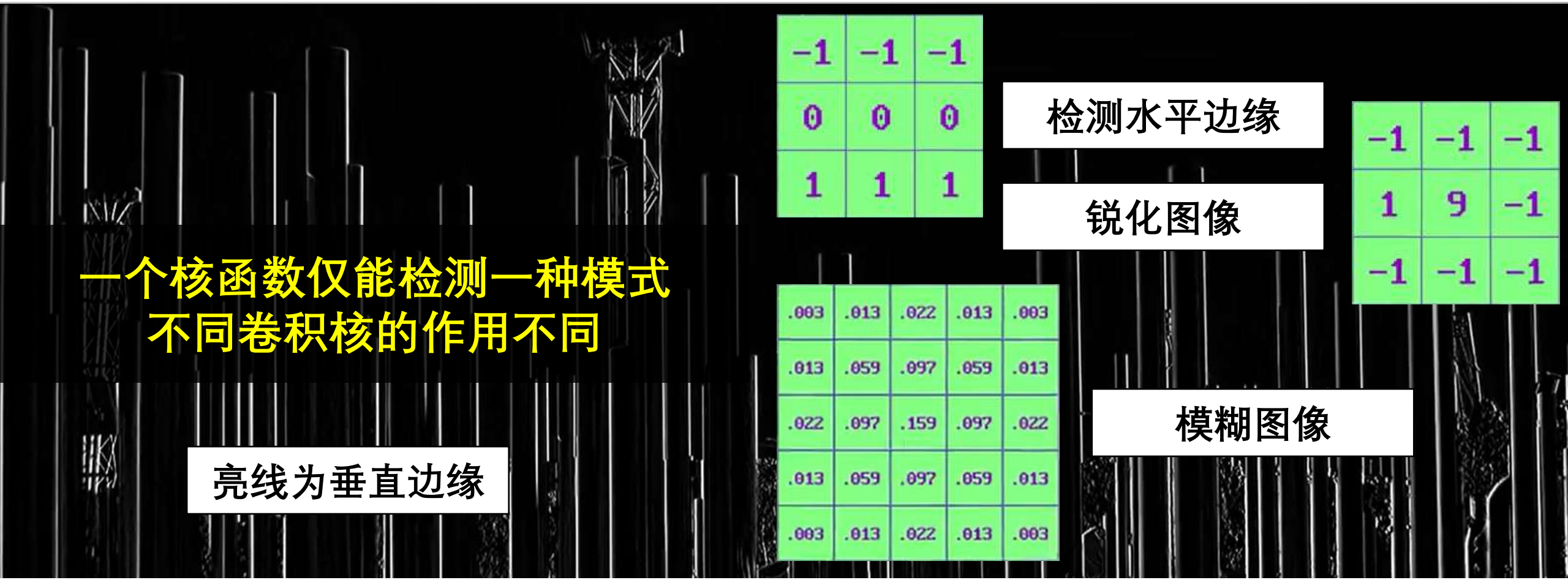
让卷积核在图像上滑动，遍历整幅图像

看看边缘检测怎么做

定义检测块

卷积操作：遍历整幅图，使用卷积核处理

4.3 深度神经网络的设计



一个核函数仅能检测一种模式
不同卷积核的作用不同

-1	-1	-1
0	0	0
1	1	1

检测水平边缘

-1	-1	-1
1	9	-1
-1	-1	-1

锐化图像

.003	.013	.022	.013	.003
.013	.059	.097	.059	.013
.022	.097	.159	.097	.022
.013	.059	.097	.059	.013
.003	.013	.022	.013	.003

模糊图像

亮线为垂直边缘

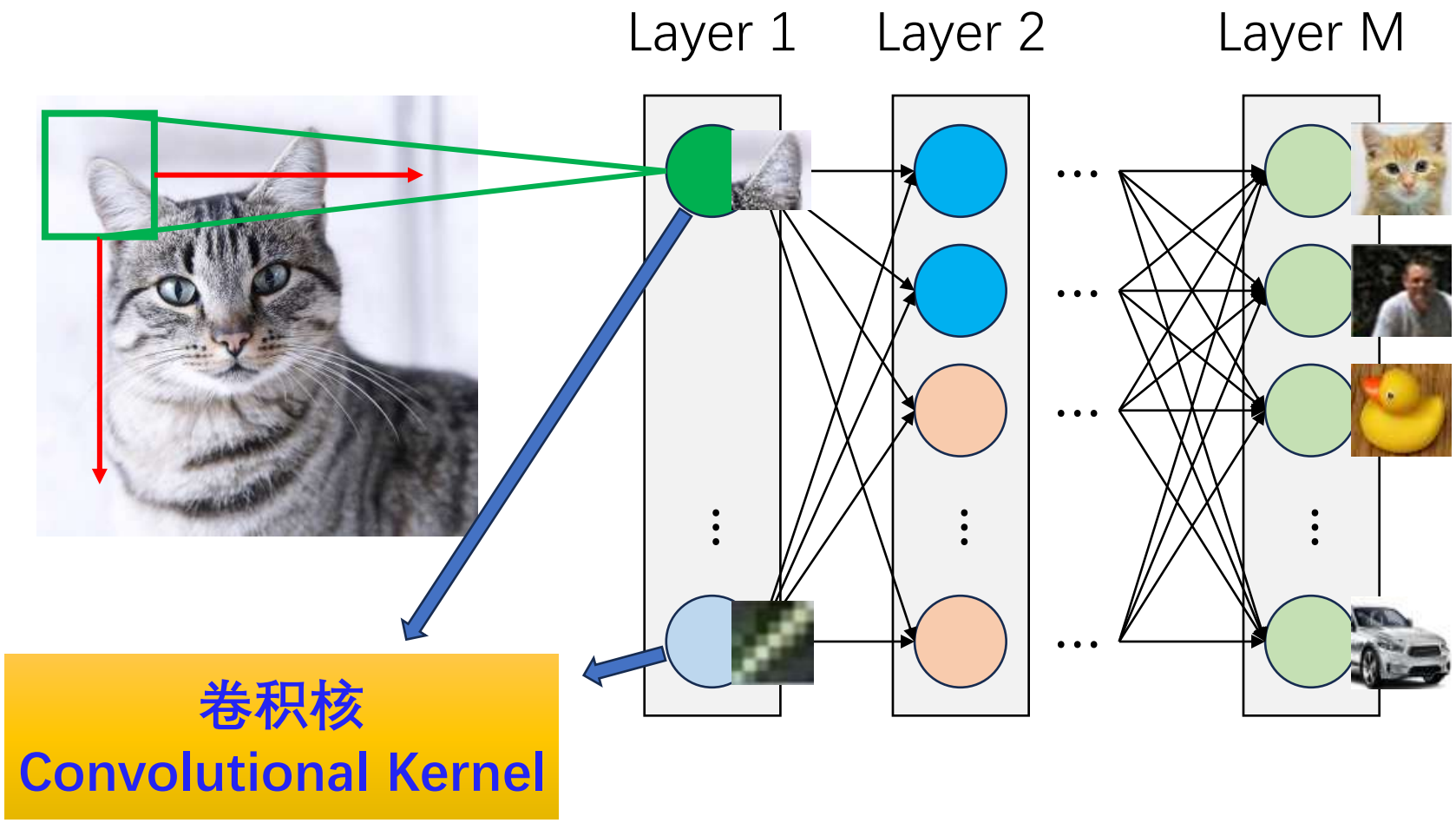
看看边缘检测怎么做

定义检测块

卷积操作：遍历整幅图，使用卷积核处理

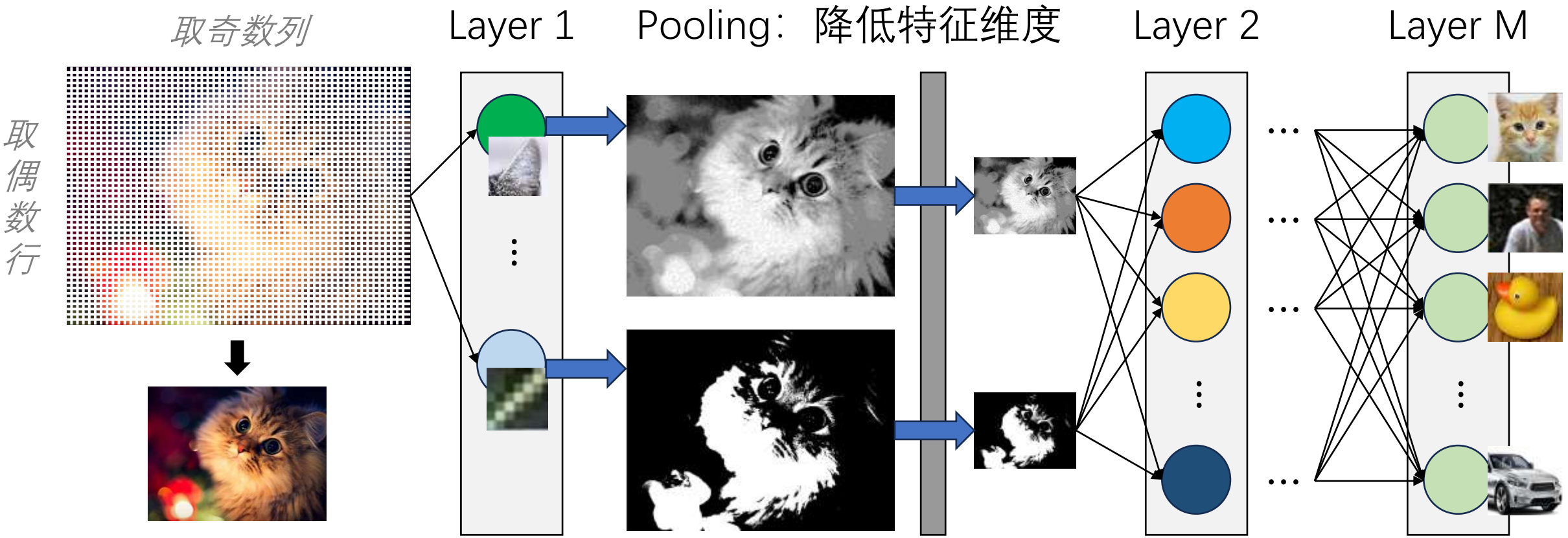
4.3 深度神经网络的设计

借鉴一下滑动机制



4.3 深度神经网络的设计

关于图片特征的另一个观察：**合理的下采样并不会影响语义的表达**

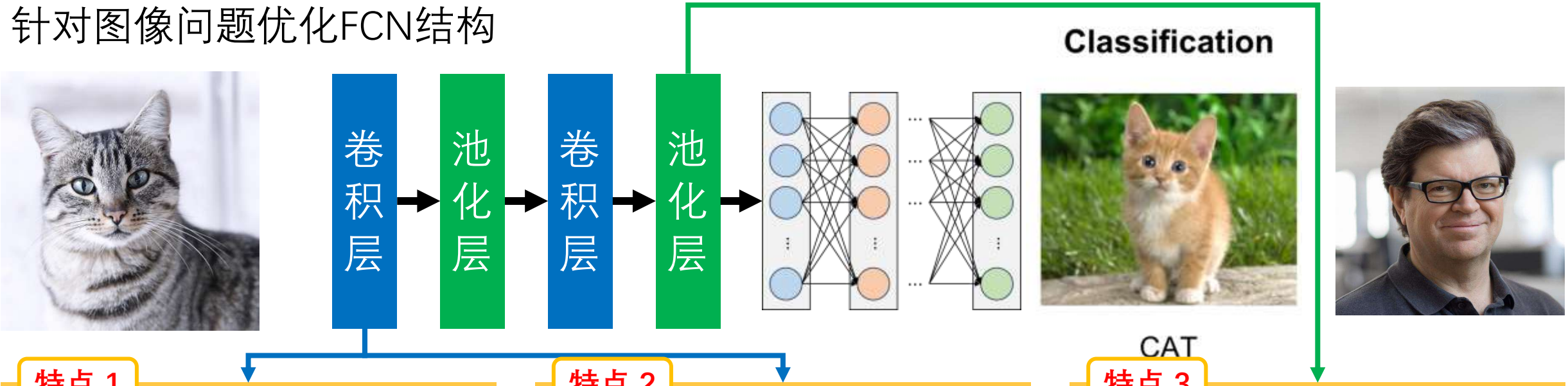


下采样后的图片特征更少，但并不影响判断结果

增加一个池化层（Pooling）

4.3 深度神经网络的设计

针对图像问题优化FCN结构



特点 1

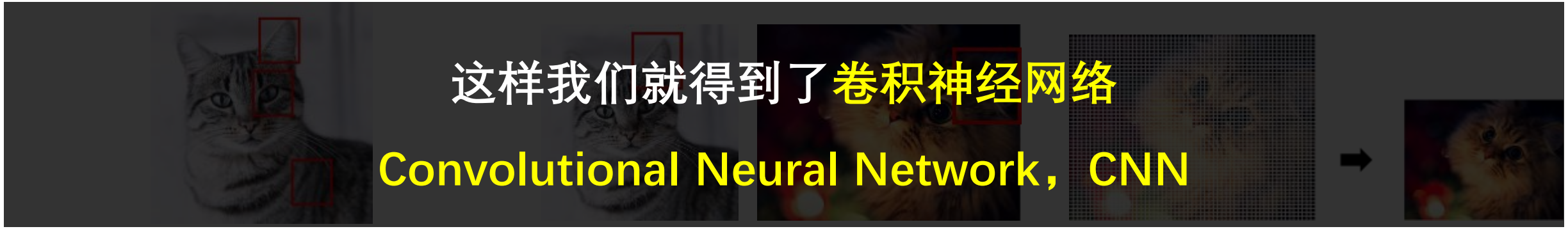
局部特征远小于整幅图像

特点 2

相同模式出现在不同位置

特点 3

下采样不影响结果



这样我们就得到了**卷积神经网络**

Convolutional Neural Network，CNN

回到房价问题

Year: **1978**

x_1：犯罪率（%）	0.00632	……
x_2：大住宅用地占比（%）	18.00	……
x_3：非零售商业用地占比（%）	2.31	……
x_4：景观房（0/1）	0	……
x_5：氮氧化物浓度（ppm）	0.538	……
x_6：平均房间数（个）	6.575	……
……	……	……
y：房价中位数（千美元）	**24.00**	……

Year: **1979**　Year: **1980**　Year: ……

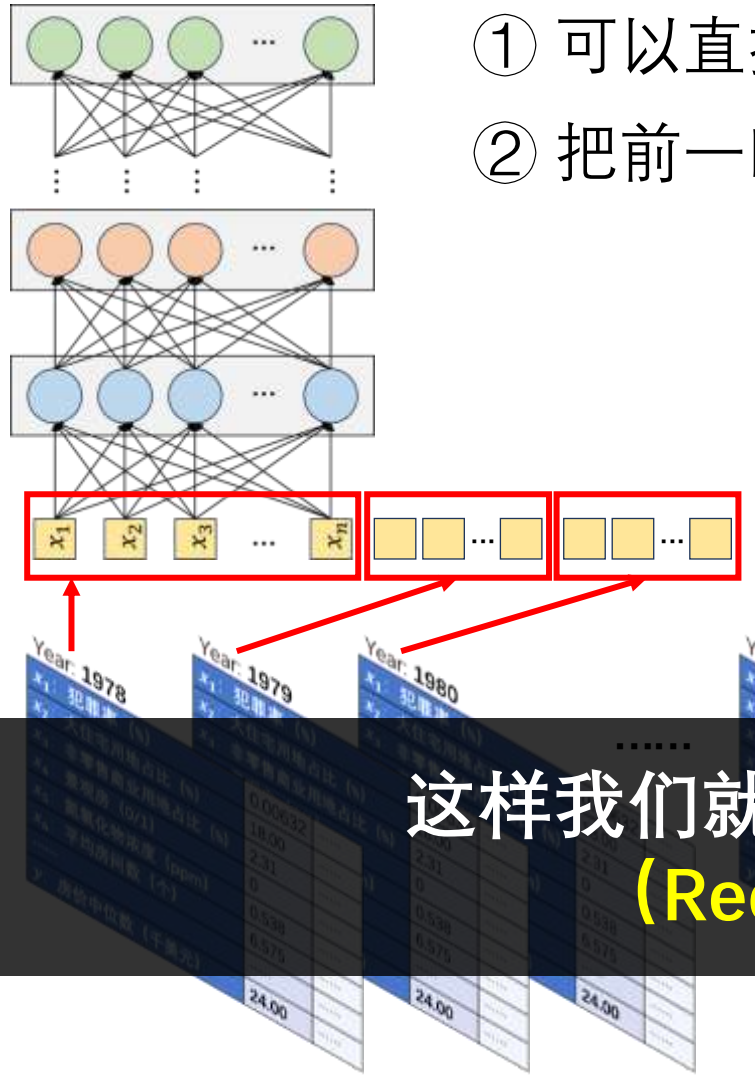
x_1：犯罪率（%）	0.00732	……
x_2：大住宅用地占比（%）	18.00	……
x_3：非零售商业用地占比（%）	2.31	……
x_4：景观房（0/1）	0	……
x_5：氮氧化物浓度（ppm）	0.638	……
x_6：平均房间数（个）	6.375	……
……	……	……
y：房价中位数（千美元）	**26.00**	……

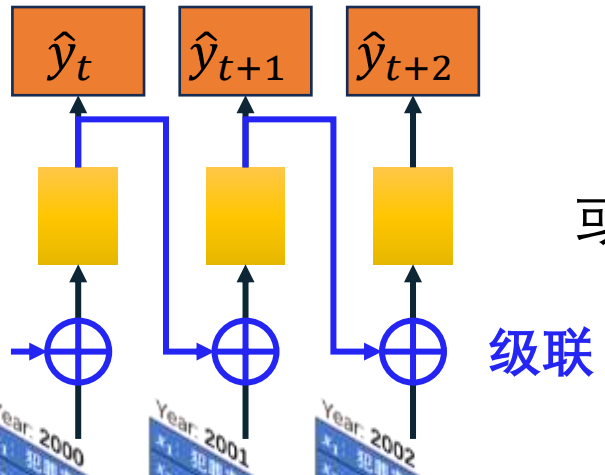
如何以**时间维度**的信息作为参考?

4.3 深度神经网络的设计

① 可以直接将时间窗口内的特征作为输入　**“看不见”窗口之外的信息**

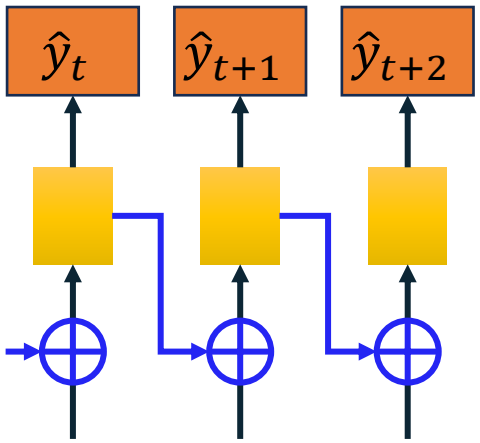
② 把前一时刻的信息作为输入





级联

或者

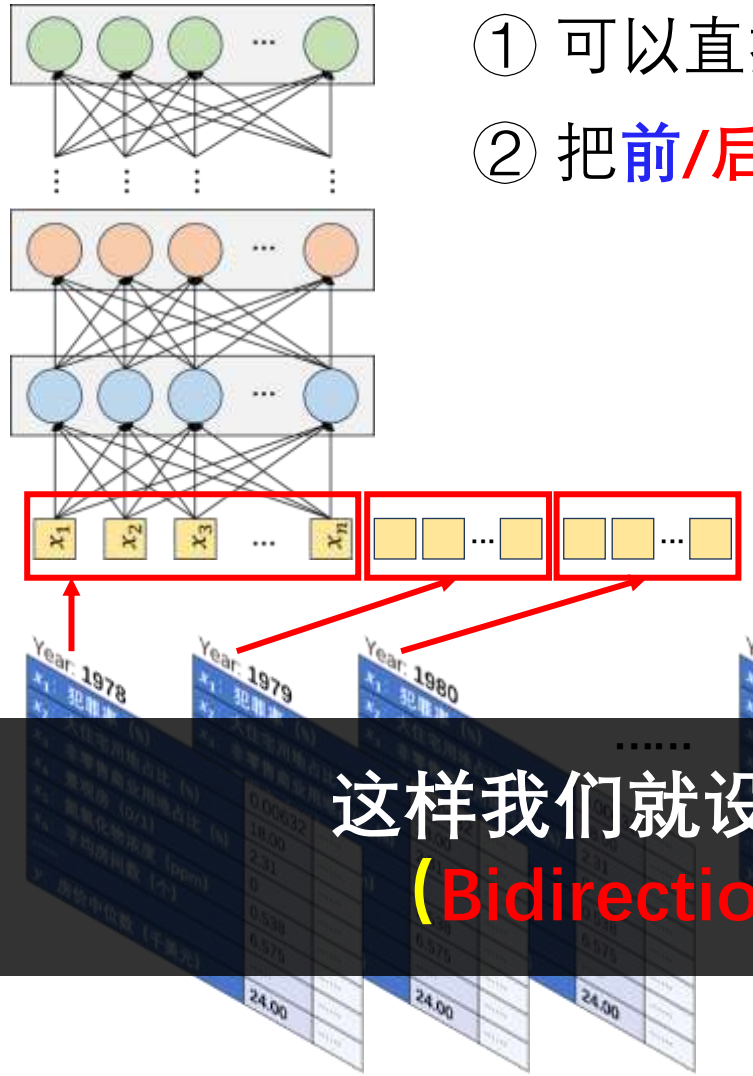


将前一时刻的特征作为当前的输入

这样我们就设计出了**具有记忆能力的循环神经网络**（Recurrent Neural Network，RNN）

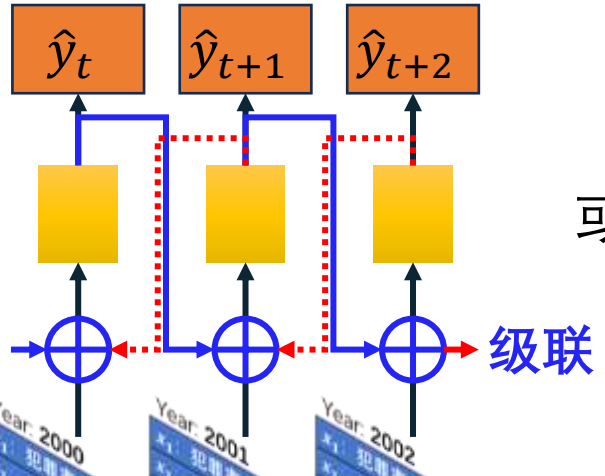
如何以**时间维度**的信息作为参考？

4.3 深度神经网络的设计

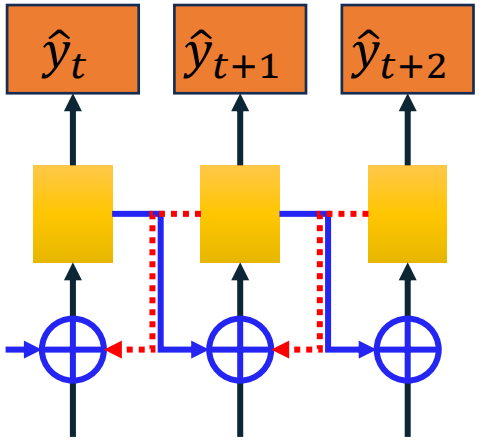


① 可以直接将时间窗口内的特征作为输入　“看不见”窗口之外的信息

② 把前/后时刻的信息作为输入



或者



将前一时刻的特征作为当前的输入

这样我们就设计出了具有记忆能力的双向循环神经网络（Bidirectional Recurrent Neural Network，RNN）

如何以时间维度的信息作为参考？

4.3 深度神经网络的设计

问题 1：DNN是“回归”还是“分类”？

DNN是一种模型结构
并不局限于具体问题

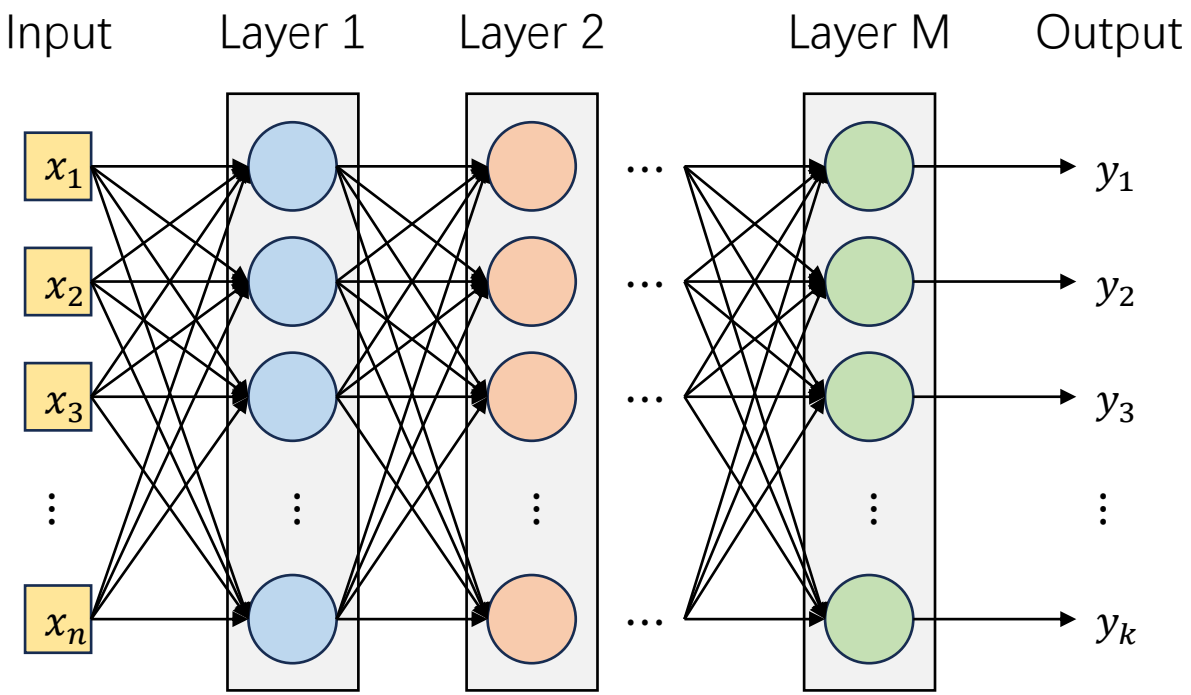
问题 2：模型结构对任务表现有影响吗？

肯定的

问题 3：应该如何设计层数和节点数？

试错 + 直觉

这也是“炼丹”的一环



接下来的问题……

如何利用数据确定最优的 θ 呢?

步骤3

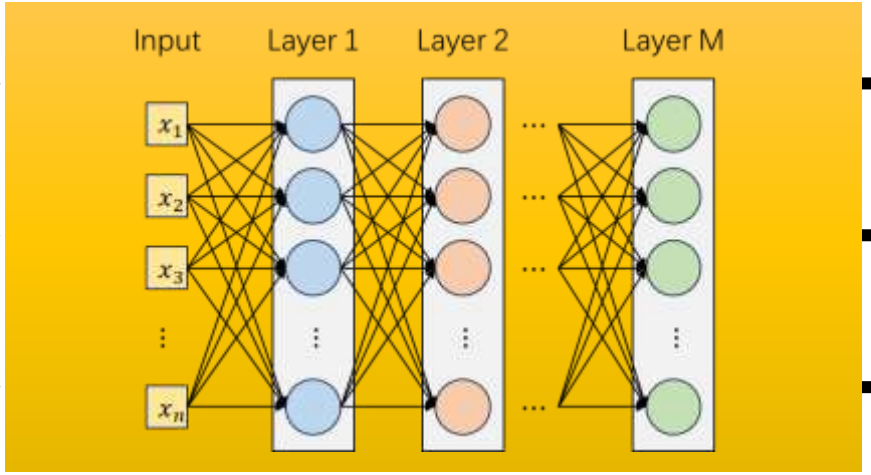
选择最优模型

损失函数
梯度下降

4.4 训练深度神经网络——反向传播

Softmax回归 → 深度神经网络

x



$\hat{y}$

$\hat{y}_1$

$\hat{y}_2$

$\hat{y}_3$

交叉熵
Cross Entropy Loss

y

y_1

y_2

y_3

$$C = -\sum_{i=1}^{3} y_i ln\hat{y}_i$$

求偏导

$$\frac{\partial C}{\partial w_i}, \frac{\partial C}{\partial b_i}$$

反向传播
Backpropagation

更新参数

$\Delta w_1,\quad \Delta b_1$

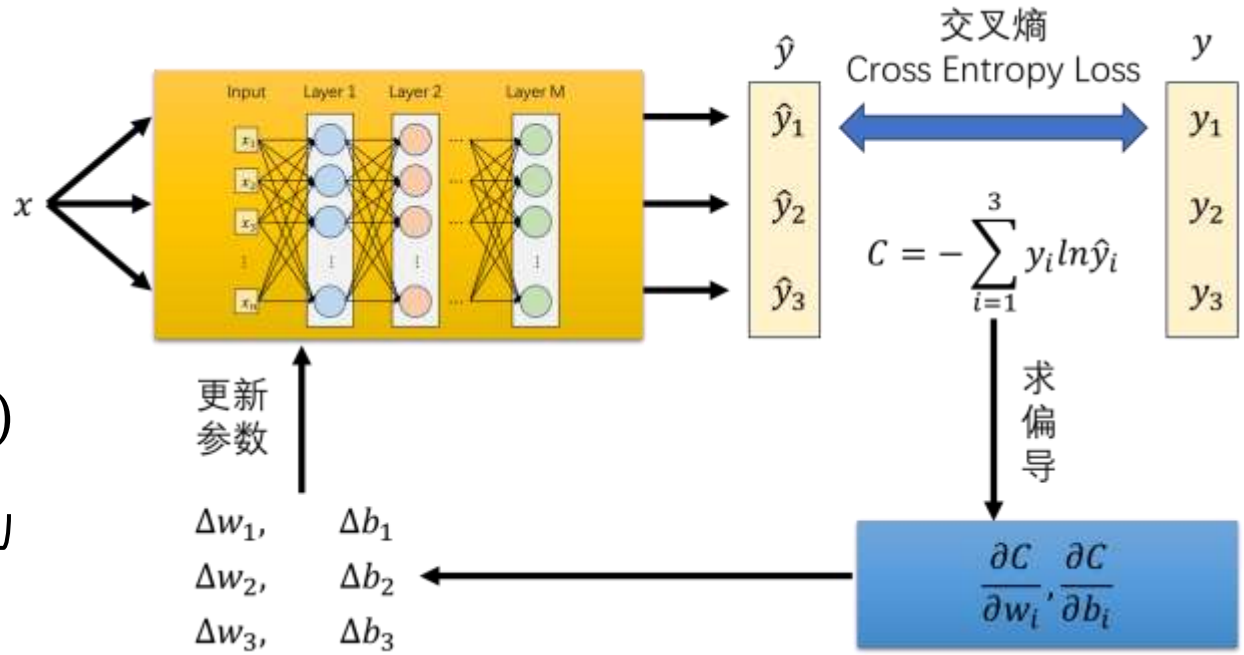
$\Delta w_2,\quad \Delta b_2$

$\Delta w_3,\quad \Delta b_3$

4.4 训练深度神经网络——反向传播

$$\theta = (w_1, w_2, \dots, b_1, b_2, \dots)$$

$$\Delta\theta = \underbrace{\left(\frac{\partial C}{\partial w_1}, \frac{\partial C}{\partial w_2}, \dots, \frac{\partial C}{\partial b_1}, \frac{\partial C}{\partial b_2}, \dots\right)}_{\text{维度极高}}$$



反向传播研究的问题：如何高效的计算 $\Delta\theta$ ？

4.4 训练深度神经网络——反向传播

案例①： $y = g(x) \quad z = h(y)$

$$\Delta x \to \Delta y \to \Delta z \quad \Rightarrow \quad \frac{dz}{dx} = \frac{dz}{dy}\frac{dy}{dx}$$

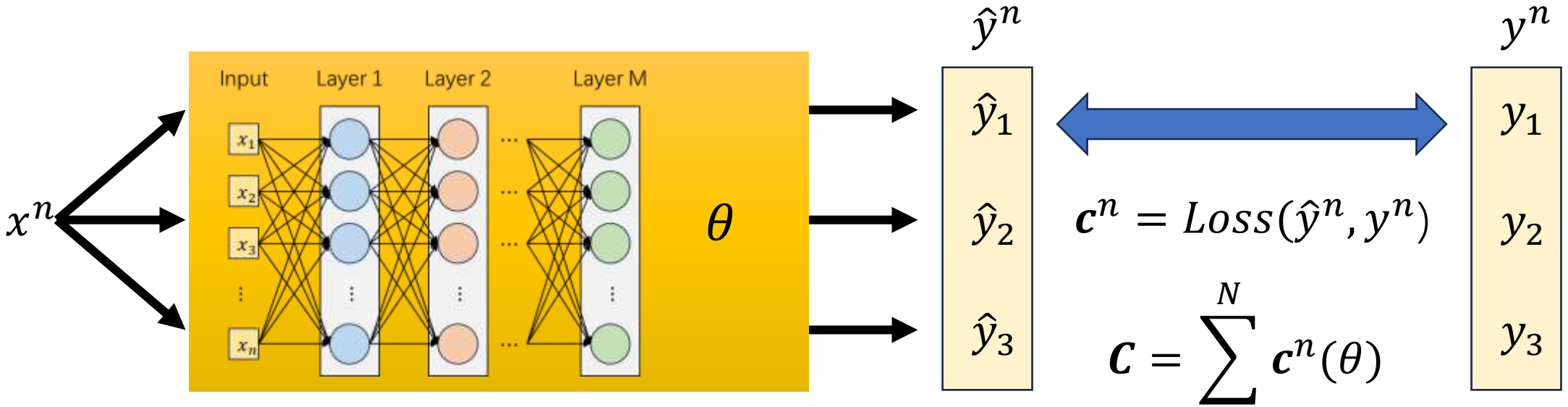
案例②： $x = g(s) \quad y = h(s) \quad z = f(x, y)$

$$\begin{tikzcd} & \Delta x \arrow[dr] & \\ \Delta s \arrow[ur] \arrow[dr] & & \Delta z \\ & \Delta y \arrow[ur] & \end{tikzcd} \quad \Rightarrow \quad \frac{dz}{ds} = \frac{\partial z}{\partial x}\frac{dx}{ds} + \frac{\partial z}{\partial y}\frac{dy}{ds}$$

案例③： $x = g(s) \quad y = h(x) \quad z = f(y)$

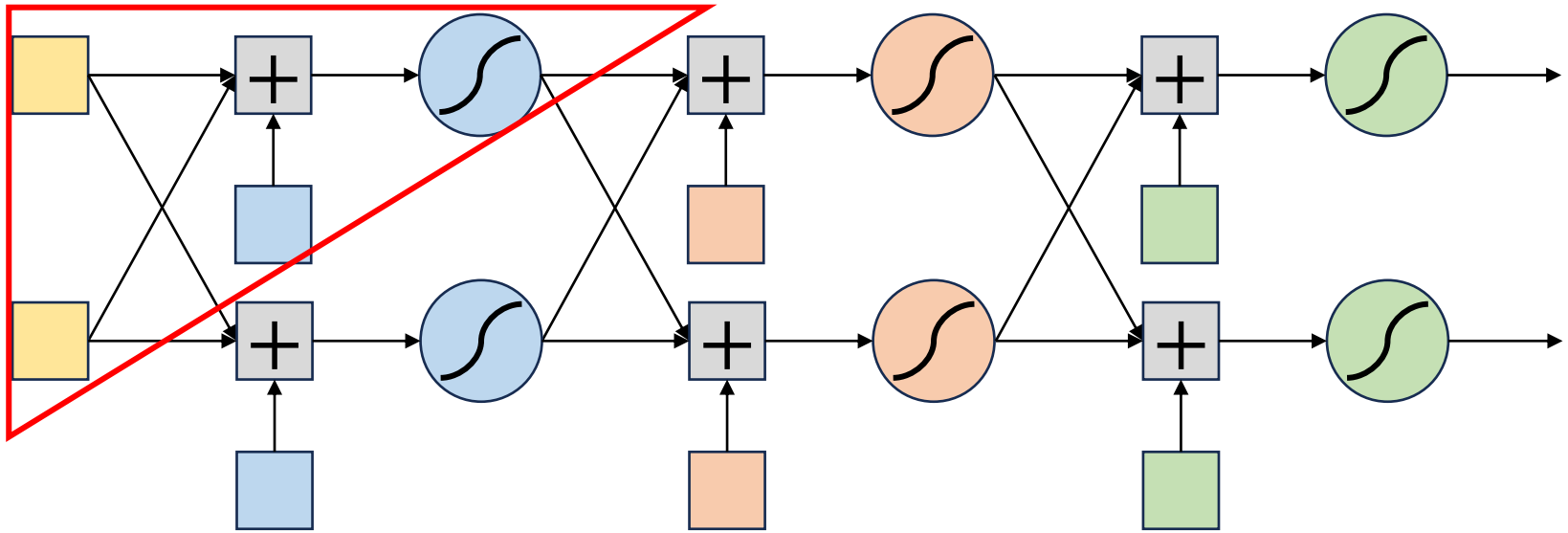
$$\Delta s \to \Delta x \to \Delta y \to \Delta z \quad \Rightarrow \quad \frac{dz}{ds} = \frac{dz}{dx}\frac{dx}{ds} = \frac{dz}{dy}\frac{dy}{dx}\frac{dx}{ds}$$

4.4 训练深度神经网络——反向传播



$$\boldsymbol{c}^n = Loss(\hat{y}^n, y^n)$$

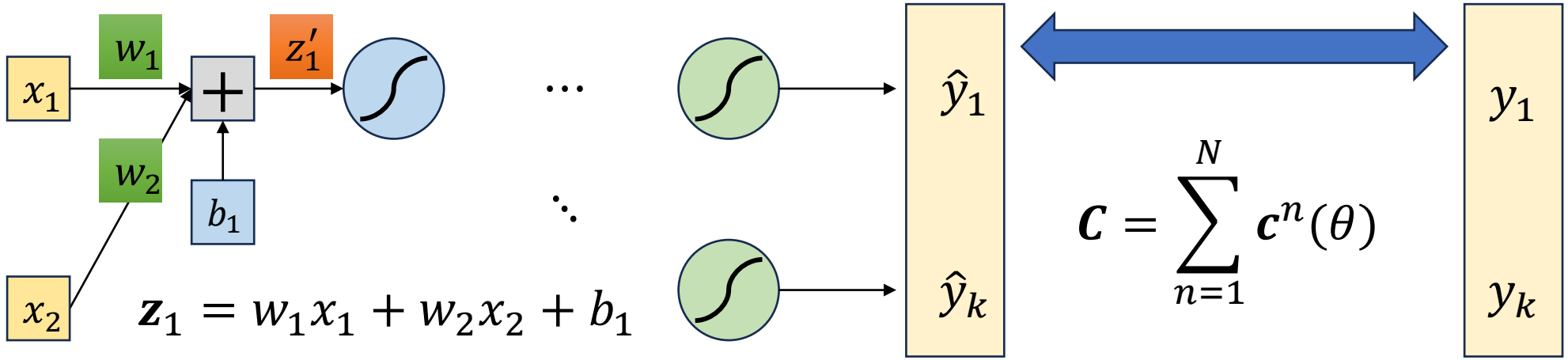
$$\boldsymbol{C} = \sum_{n=1}^{N} \boldsymbol{c}^n(\theta)$$



计算参数梯度

$$\frac{\partial \boldsymbol{C}}{\partial w} = \sum_{n=1}^{N} \frac{\partial \boldsymbol{c}^n(\theta)}{\partial w}$$

计算一个就能计算所有



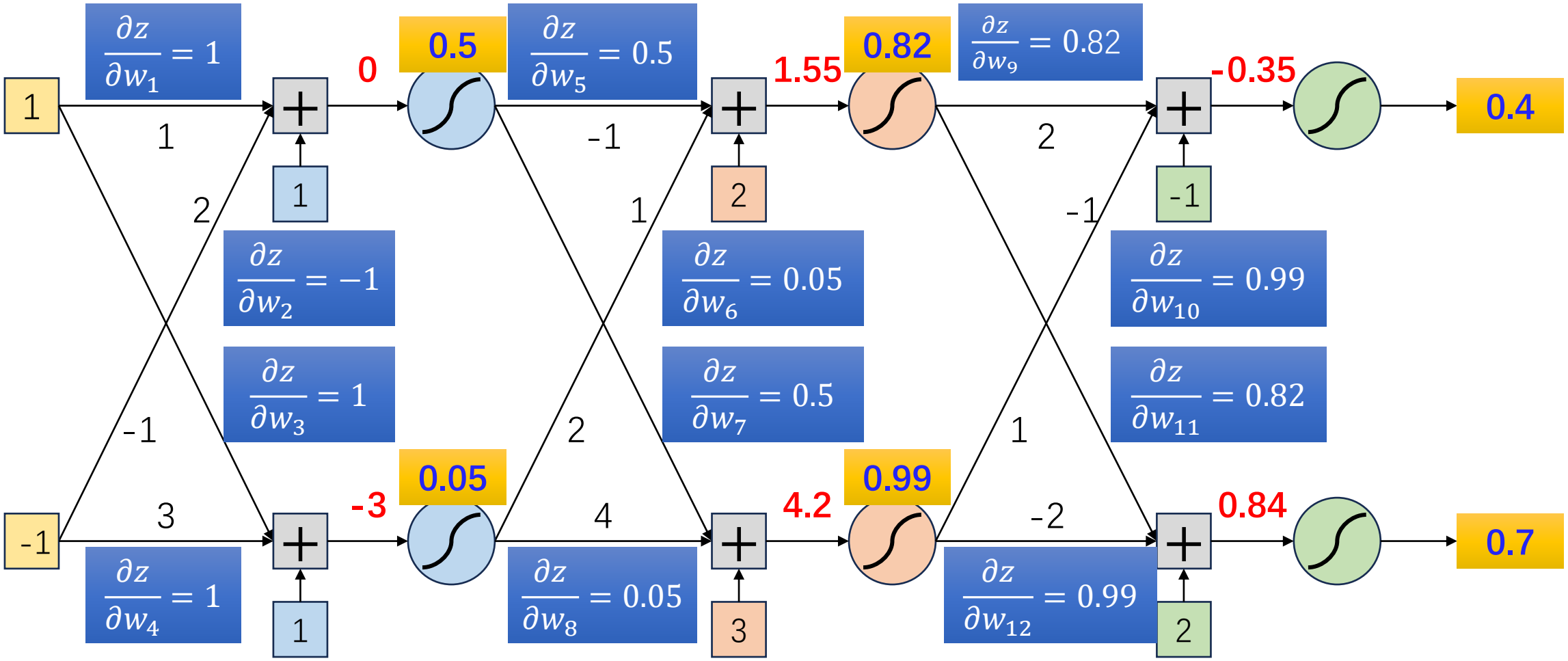
$$z_1 = w_1 x_1 + w_2 x_2 + b_1$$

$$C = \sum_{n=1}^{N} c^n(\theta)$$

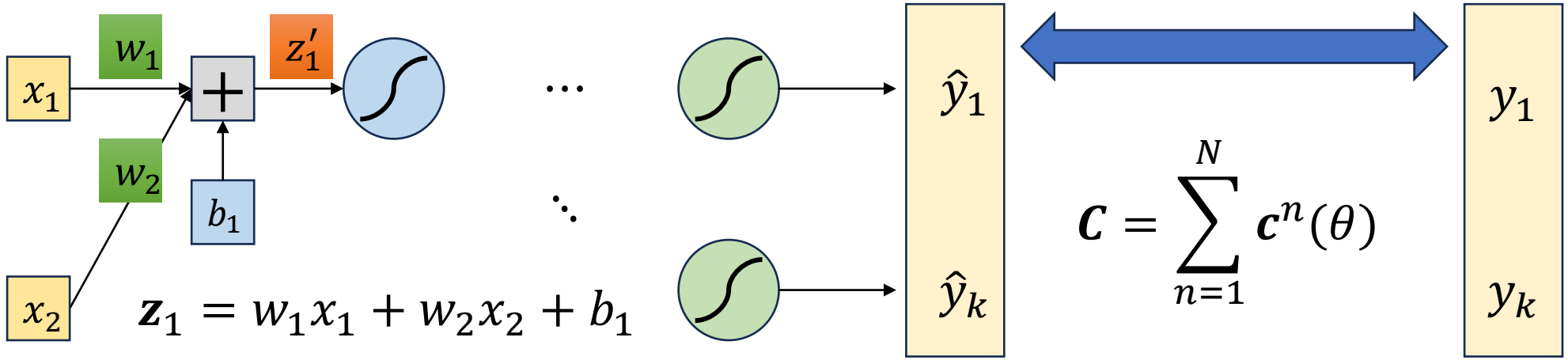
$$\frac{\partial C}{\partial w_1} = \boxed{\frac{\partial z_1}{\partial w_1}} \frac{\partial C}{\partial z_1} \qquad \frac{\partial C}{\partial w_2} = \boxed{\frac{\partial z_1}{\partial w_2}} \frac{\partial C}{\partial z_1} \qquad \frac{\partial C}{\partial b_1} = \boxed{\frac{\partial z_1}{\partial b_1}} \frac{\partial C}{\partial z_1}$$

x_1 $\qquad$ x_2 $\qquad$ 1

规律： $\frac{\partial z}{\partial w}$ 的值为输入的数值， $\frac{\partial z}{\partial b}$ 的值为1



规律： $\frac{\partial z}{\partial w}$ 的值可在预测时获得，即正向传播



$$\boldsymbol{z}_1 = w_1 x_1 + w_2 x_2 + b_1$$

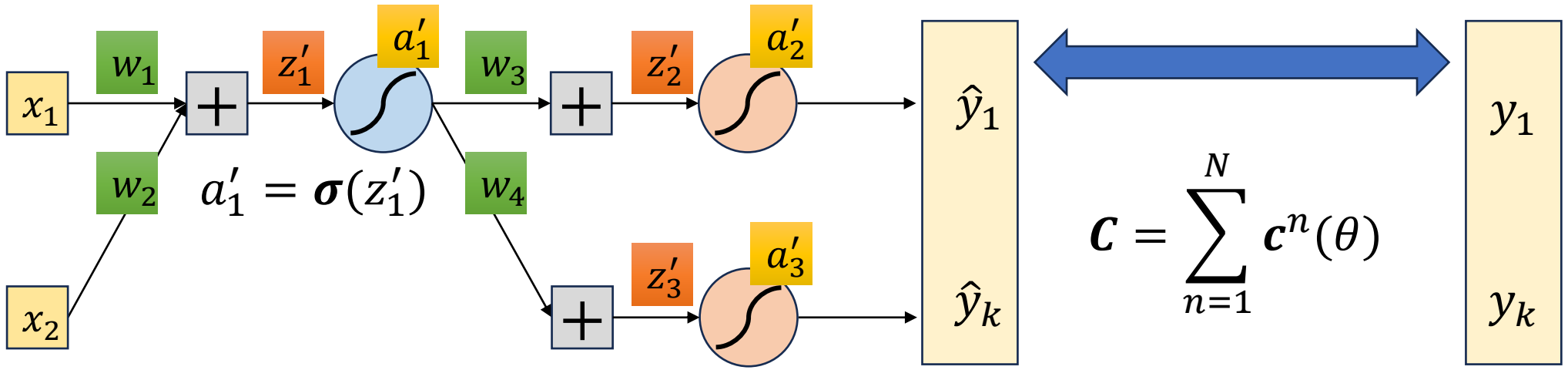
$$\boldsymbol{C} = \sum_{n=1}^{N} \boldsymbol{c}^n(\theta)$$

$$\frac{\partial \boldsymbol{C}}{\partial w_1} = \boxed{\frac{\partial \boldsymbol{z}_1}{\partial w_1}} \frac{\partial \boldsymbol{C}}{\partial \boldsymbol{z}_1} \qquad \frac{\partial \boldsymbol{C}}{\partial w_2} = \boxed{\frac{\partial \boldsymbol{z}_1}{\partial w_2}} \frac{\partial \boldsymbol{C}}{\partial \boldsymbol{z}_1} \qquad \frac{\partial \boldsymbol{C}}{\partial b_1} = \boxed{\frac{\partial \boldsymbol{z}_1}{\partial b_1}} \frac{\partial \boldsymbol{C}}{\partial \boldsymbol{z}_1}$$

正向传播

该如何计算 $\dfrac{\partial \boldsymbol{C}}{\partial \boldsymbol{z}_1}$ ？ **反向传播**

已达Output Layer

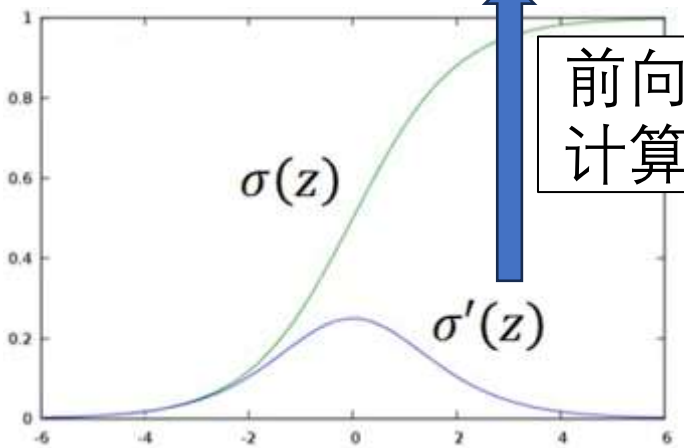


$$a_1' = \boldsymbol{\sigma}(z_1')$$

$$\boldsymbol{C} = \sum_{n=1}^{N} \boldsymbol{c}^n(\theta)$$

$$\frac{\partial \boldsymbol{C}}{\partial \boldsymbol{z}_1} = \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{z}_1}\frac{\partial \boldsymbol{C}}{\partial \boldsymbol{\sigma}} \rightarrow \frac{\partial \boldsymbol{C}}{\partial \boldsymbol{\sigma}} = \frac{\partial \boldsymbol{z}_2}{\partial \boldsymbol{\sigma}}\frac{\partial \boldsymbol{C}}{\partial \boldsymbol{z}_2} + \frac{\partial \boldsymbol{z}_3}{\partial \boldsymbol{\sigma}}\frac{\partial \boldsymbol{C}}{\partial \boldsymbol{z}_3} \rightarrow \frac{\partial \boldsymbol{C}}{\partial \boldsymbol{z}_3} = \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{z}_3}\frac{\partial \boldsymbol{C}}{\partial \boldsymbol{\sigma}}$$

w_3 w_4

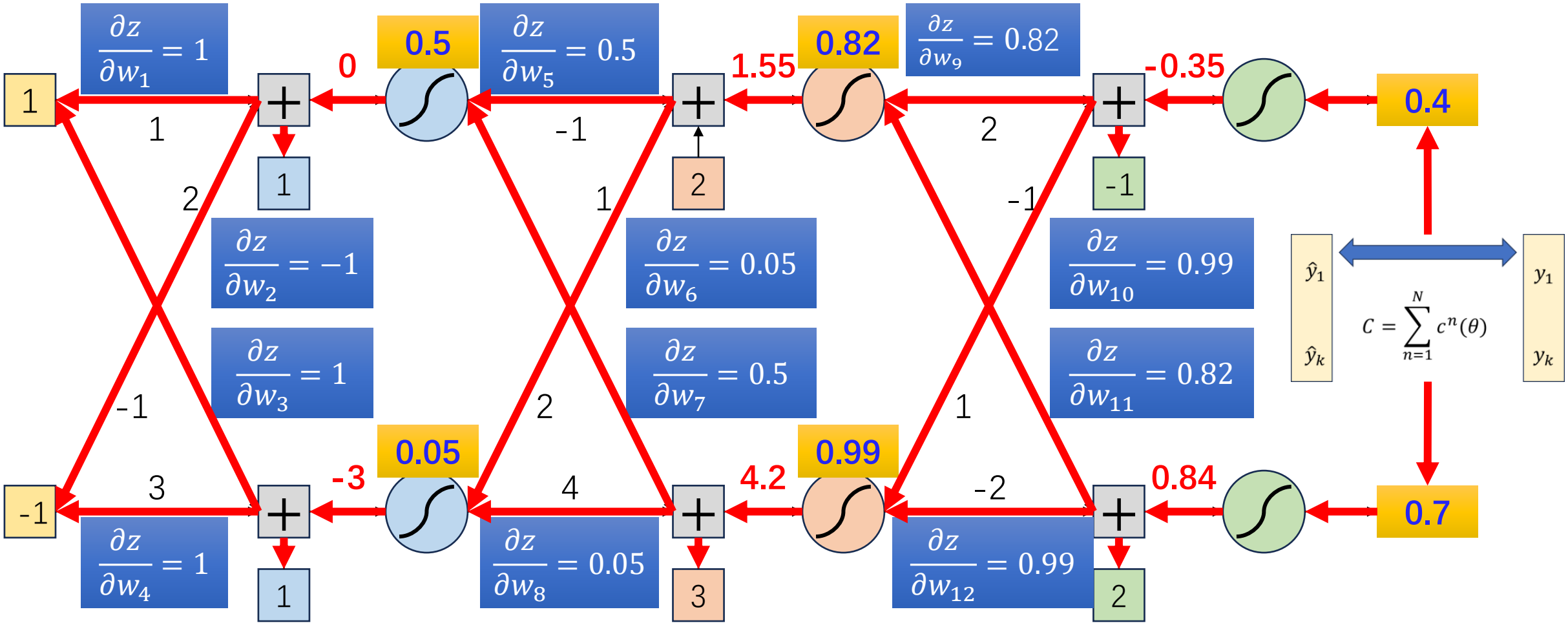


前向传播
计算可得

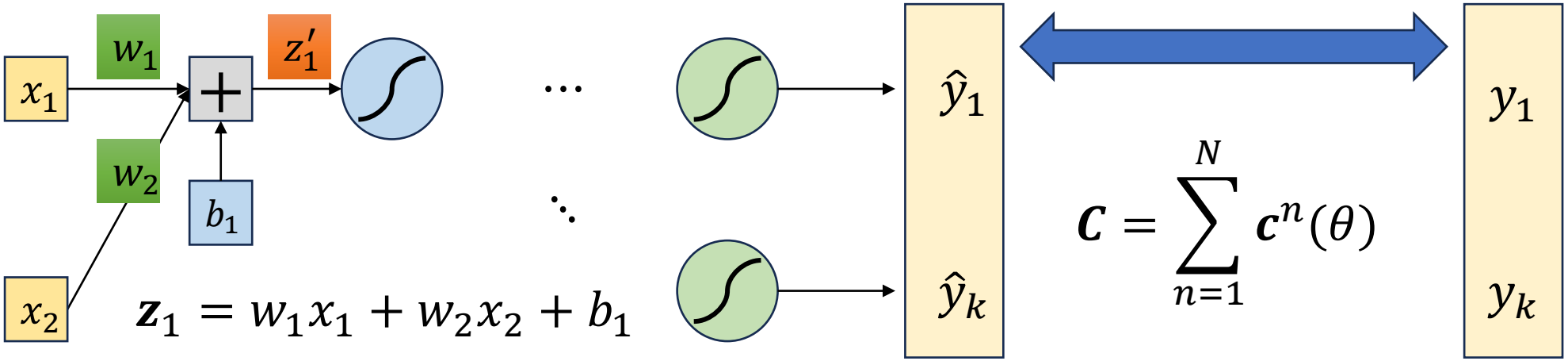
$$\frac{\partial \boldsymbol{C}}{\partial \boldsymbol{\sigma}} \rightarrow \frac{\partial \boldsymbol{c}}{\partial \boldsymbol{\sigma}} \rightarrow \frac{\partial \boldsymbol{l}(\boldsymbol{y}, \hat{\boldsymbol{y}})}{\partial \boldsymbol{\sigma}}$$

$$\rightarrow \frac{\partial \boldsymbol{l}(\boldsymbol{y}, \boldsymbol{\sigma})}{\partial \boldsymbol{\sigma}} \rightarrow \frac{\partial \boldsymbol{l}(\boldsymbol{\sigma})}{\partial \boldsymbol{\sigma}}$$

无法直接得出结果的$\frac{\partial C}{\partial z_1}$越来越多，但形式是重复的



规律：$\frac{\partial C}{\partial z}$ 的值需在一次预测之后，反向计算而得，即反向传播



$$\boldsymbol{z}_1 = w_1 x_1 + w_2 x_2 + b_1$$

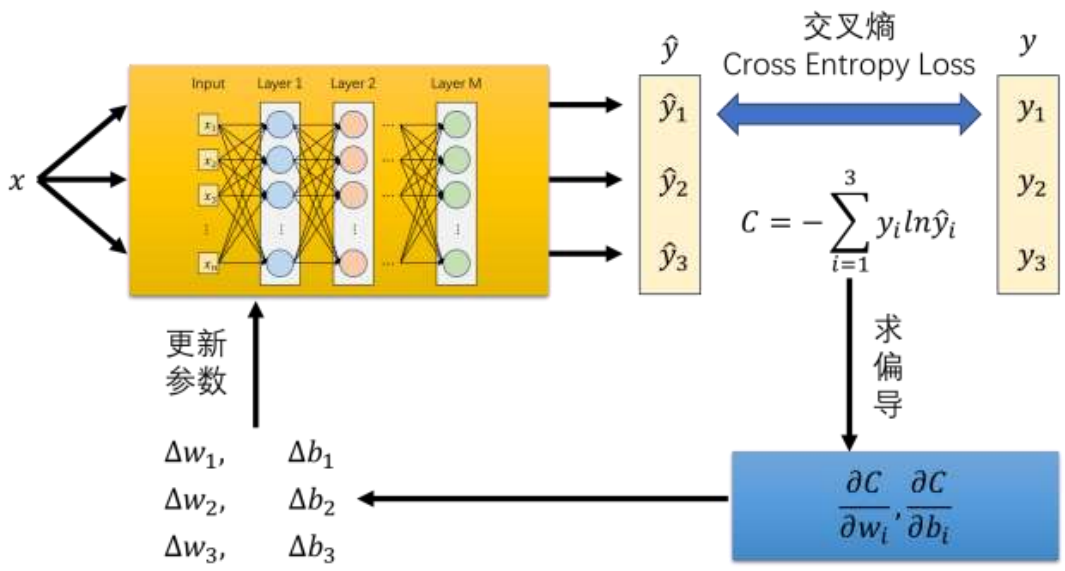
$$\boldsymbol{C} = \sum_{n=1}^{N} \boldsymbol{c}^n(\theta)$$

$$\frac{\partial \boldsymbol{C}}{\partial w_1} = \boxed{\frac{\partial \boldsymbol{z}_1}{\partial w_1}} \frac{\partial \boldsymbol{C}}{\partial \boldsymbol{z}_1} \qquad \frac{\partial \boldsymbol{C}}{\partial w_2} = \boxed{\frac{\partial \boldsymbol{z}_1}{\partial w_2}} \frac{\partial \boldsymbol{C}}{\partial \boldsymbol{z}_1} \qquad \frac{\partial \boldsymbol{C}}{\partial b_1} = \boxed{\frac{\partial \boldsymbol{z}_1}{\partial b_1}} \frac{\partial \boldsymbol{C}}{\partial \boldsymbol{z}_1}$$

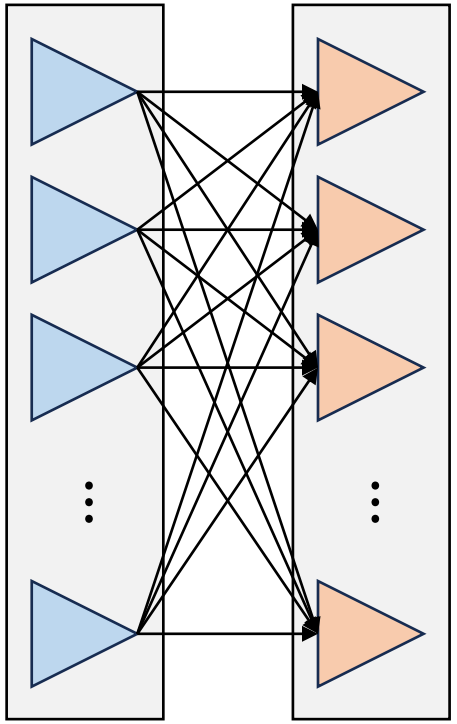
正向传播

该如何计算 $\dfrac{\partial \boldsymbol{C}}{\partial \boldsymbol{z}_1}$? **反向传播**

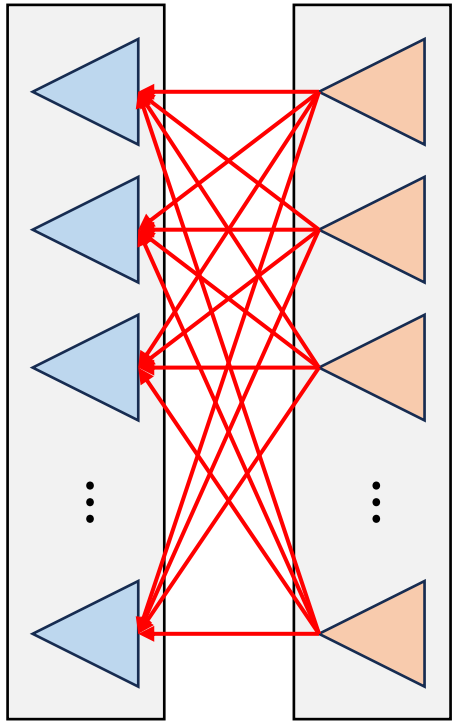
4.4 训练深度神经网络——反向传播



正向传播



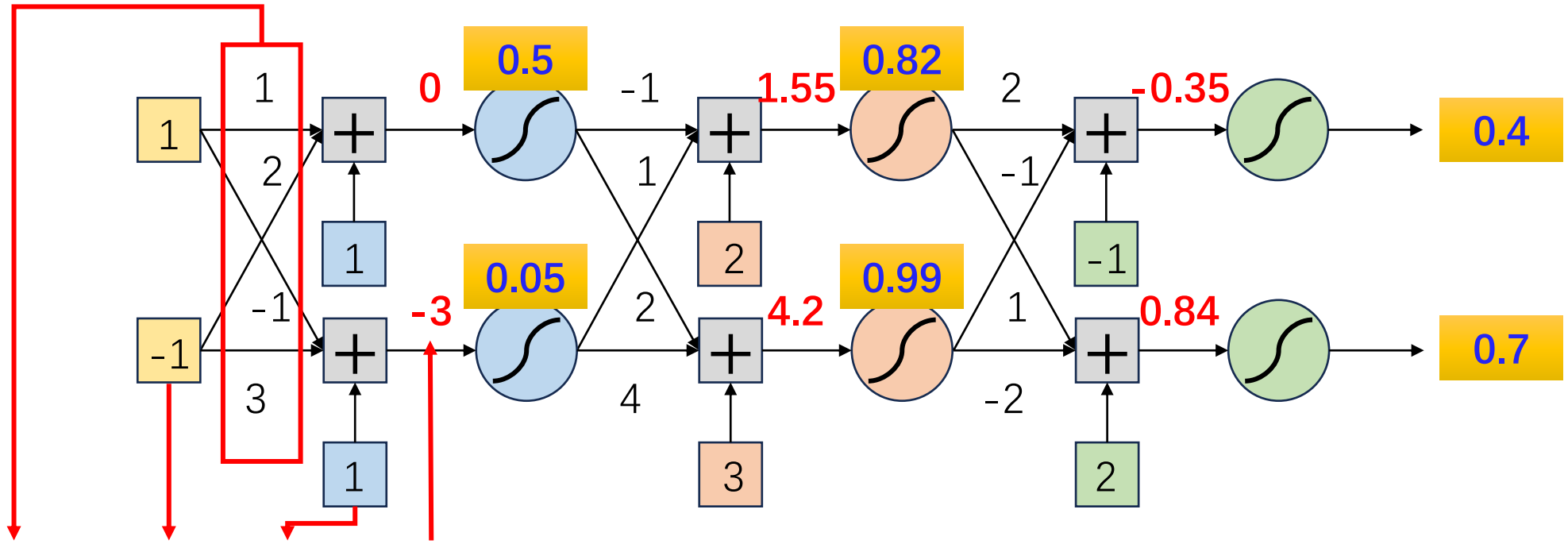
反向传播



研究如何高效的计算 $\Delta\theta$ ？

$$\frac{\partial C}{\partial w} = \frac{\partial z}{\partial w} \times \frac{\partial C}{\partial z}$$

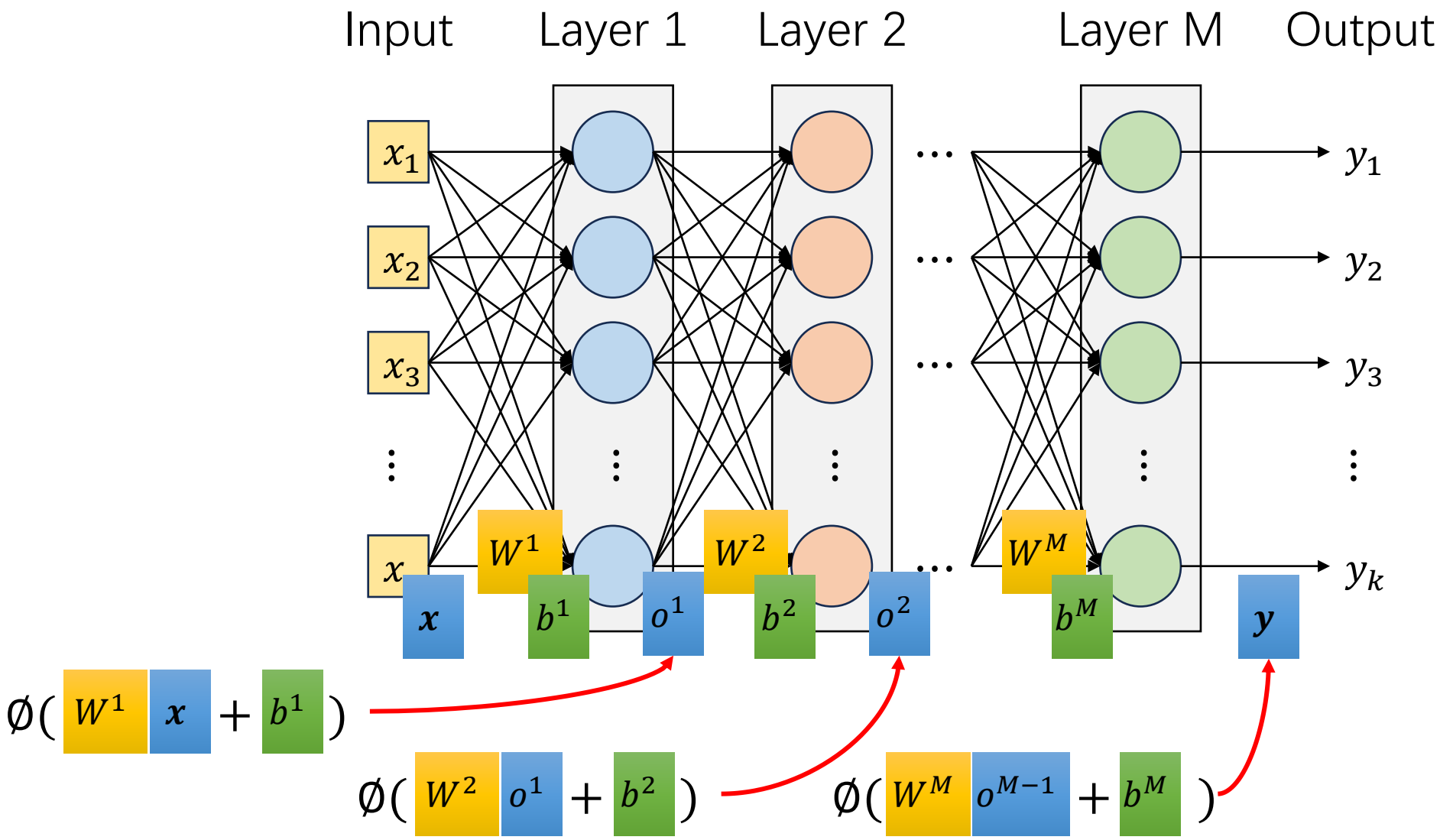
4.4 训练深度神经网络——矩阵运算



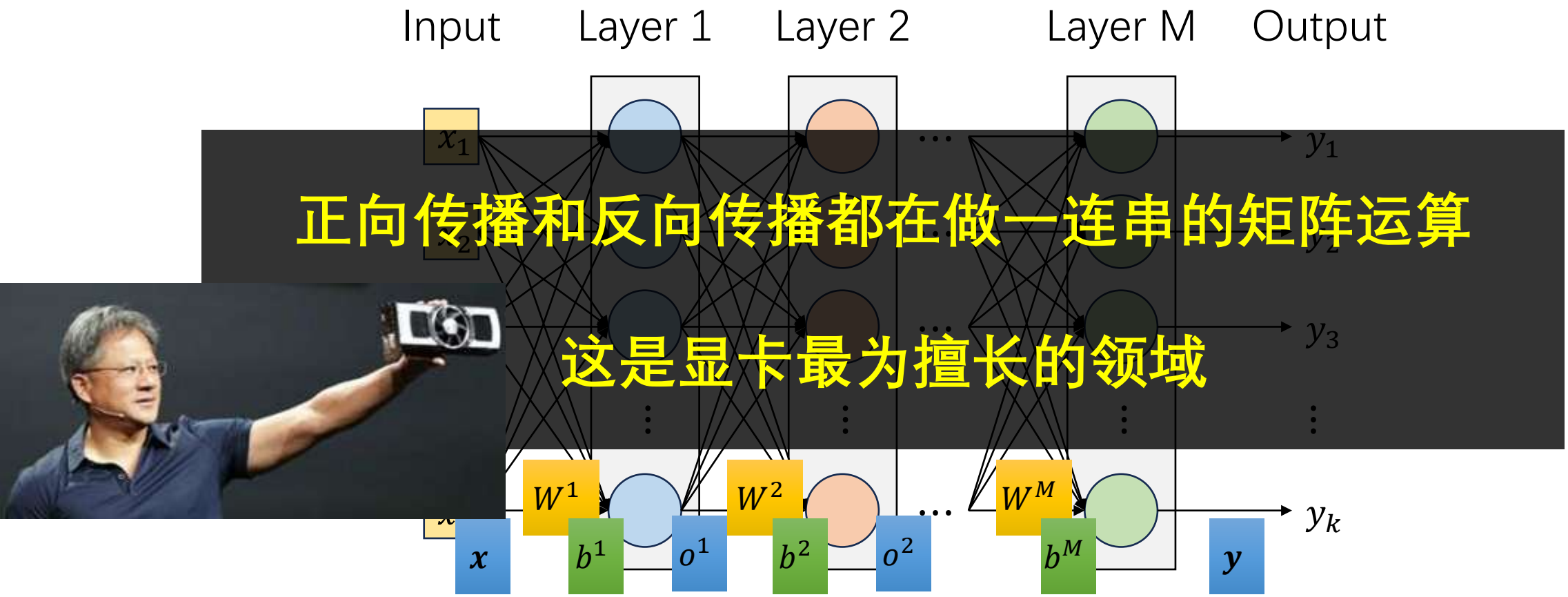
$$\begin{bmatrix}1 & 2\\ -1 & 3\end{bmatrix}\begin{bmatrix}1\\ -1\end{bmatrix}+\begin{bmatrix}1\\ 1\end{bmatrix}=\begin{bmatrix}0\\ -3\end{bmatrix}\quad \begin{bmatrix}-1 & 1\\ 2 & 4\end{bmatrix}\begin{bmatrix}0.5\\ 0.05\end{bmatrix}+\begin{bmatrix}2\\ 3\end{bmatrix}=\begin{bmatrix}1.55\\ 4.2\end{bmatrix}\quad \begin{bmatrix}2 & -1\\ 1 & -2\end{bmatrix}\begin{bmatrix}0.82\\ 0.99\end{bmatrix}+\begin{bmatrix}-1\\ 2\end{bmatrix}=\begin{bmatrix}-0.35\\ 0.84\end{bmatrix}$$

$$\phi\left(\begin{bmatrix}0\\ -3\end{bmatrix}\right)=\begin{bmatrix}0.5\\ 0.05\end{bmatrix}\qquad \phi\left(\begin{bmatrix}1.55\\ 4.2\end{bmatrix}\right)=\begin{bmatrix}0.82\\ 0.99\end{bmatrix}\qquad \phi\left(\begin{bmatrix}-0.35\\ 0.84\end{bmatrix}\right)=\begin{bmatrix}0.4\\ 0.7\end{bmatrix}$$

4.4 训练深度神经网络——矩阵运算



4.4 训练深度神经网络——矩阵运算

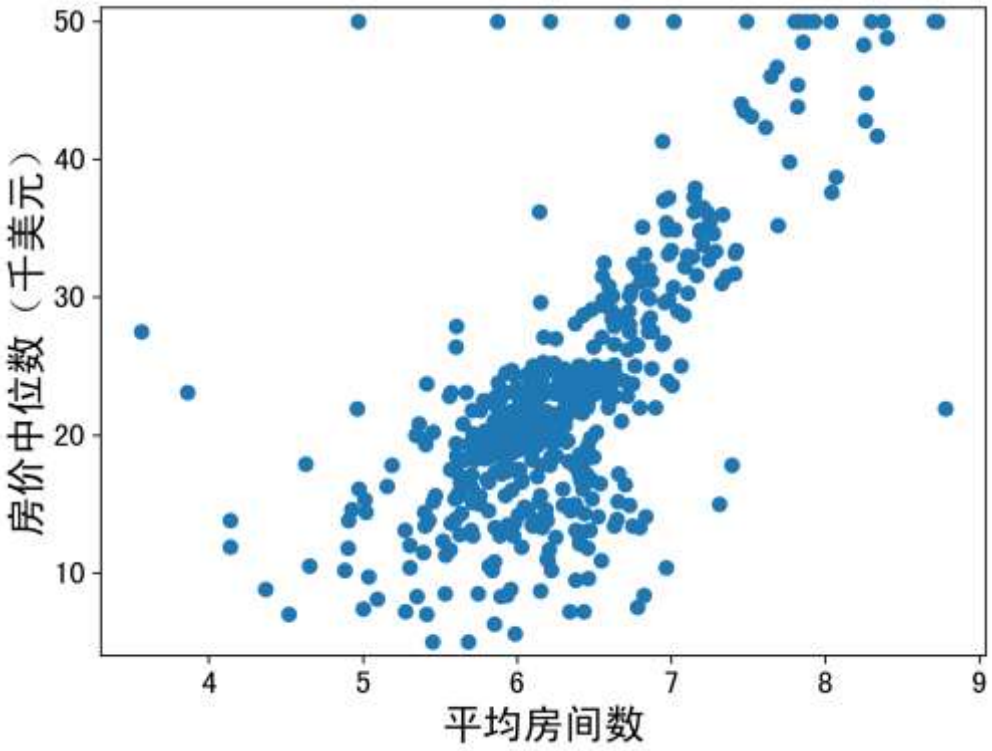


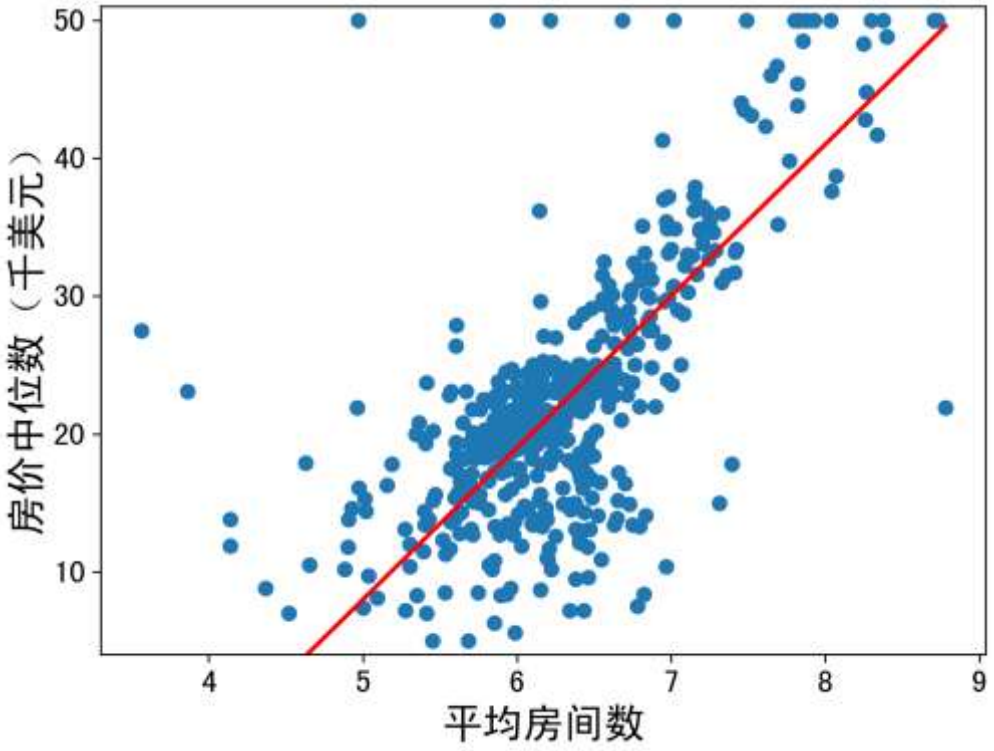
正向传播和反向传播都在做一连串的矩阵运算

这是显卡最为擅长的领域

$$y = \emptyset(W^M \cdots \emptyset(W^2 \emptyset(W^1 x + b^1) + b^2) + b^M)$$

思考一下：训练完的模型能直接用吗？





泛 化
Generalization

这是模型的**关键能力**

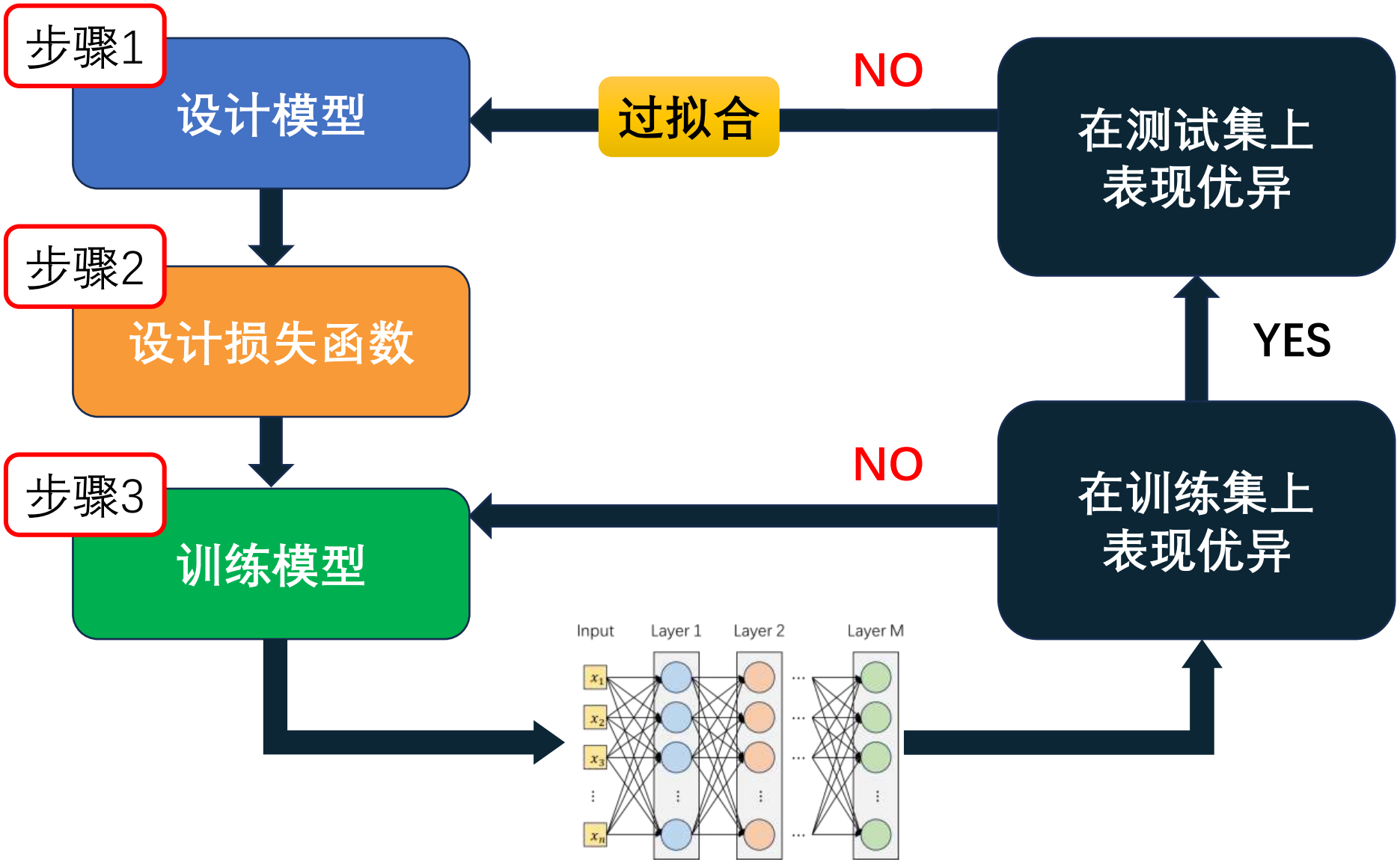
机器学习中的**泛化能力**（generalization ability）指的是模型在训练数据之外、在未见过的测试数据或真实世界数据上表现的能力。换句话说，泛化能力衡量模型从有限的训练数据中学到的知识是否能够有效地应用到新数据上。

在机器学习中，泛化能力（Generalization）指的是模型对未见过的数据进行预测的能力。一个具有良好泛化能力的模型能够从训练数据中学习到通用的规律，并且能够将这些规律应用到新的、未见过的数据上，从而做出准确的预测。

泛化能力是机器学习模型性能的关键指标之一，它与模型的过拟合（Overfitting）和欠拟合（Underfitting）密切相关：

泛化主要关注模型面对训练过程中**未见过的数据**时的能力

训练深度神经网络一点经验



如果我没有猜错的话

训练深度神经网络一点经验

早停法

正则化

Dropout

新的激活函数

更优的学习算法

NO

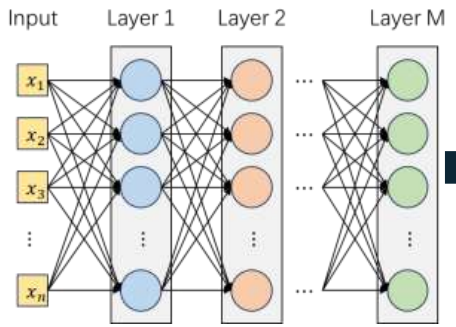
在测试集上表现优异

对症下药

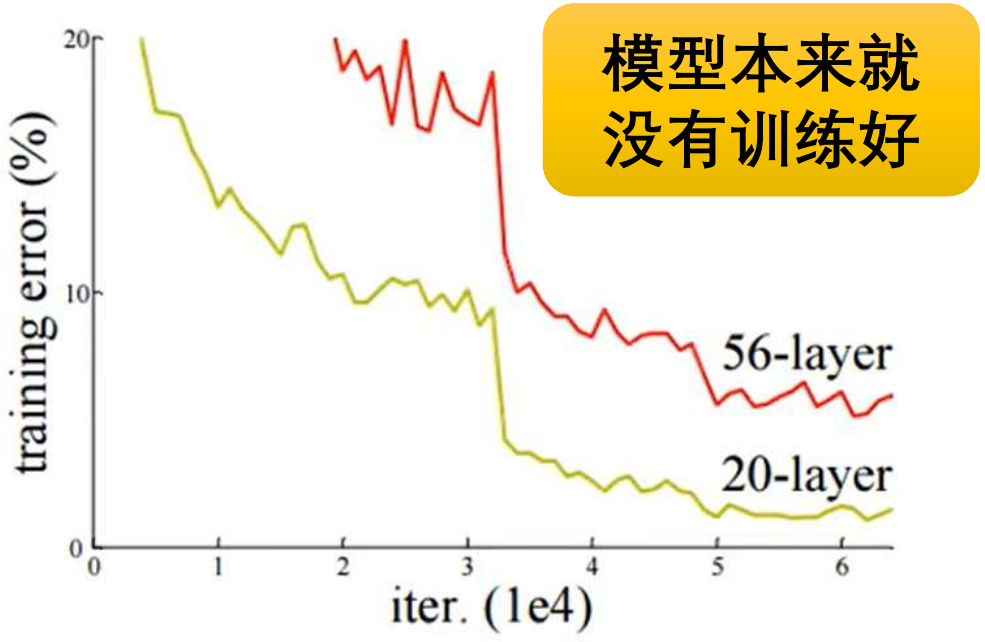
YES

NO

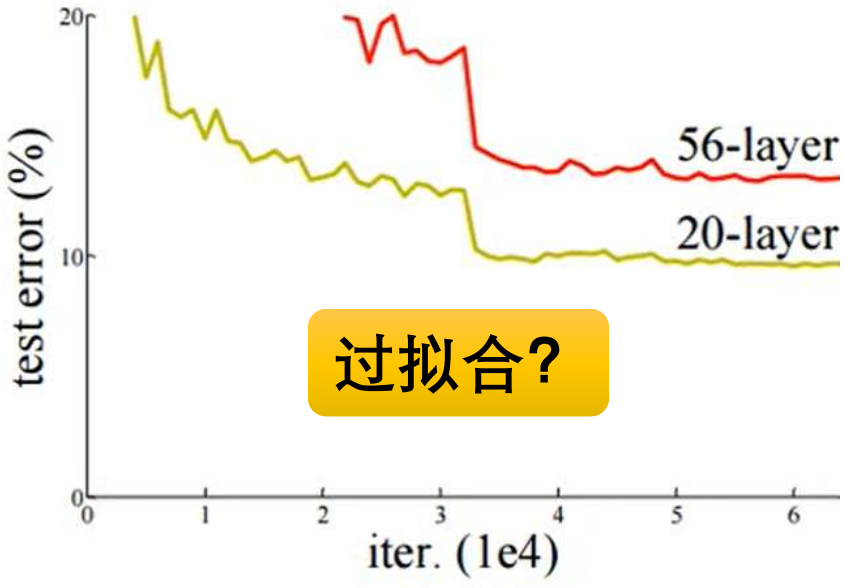
在训练集上表现优异



4.5 梯度消失和激活函数

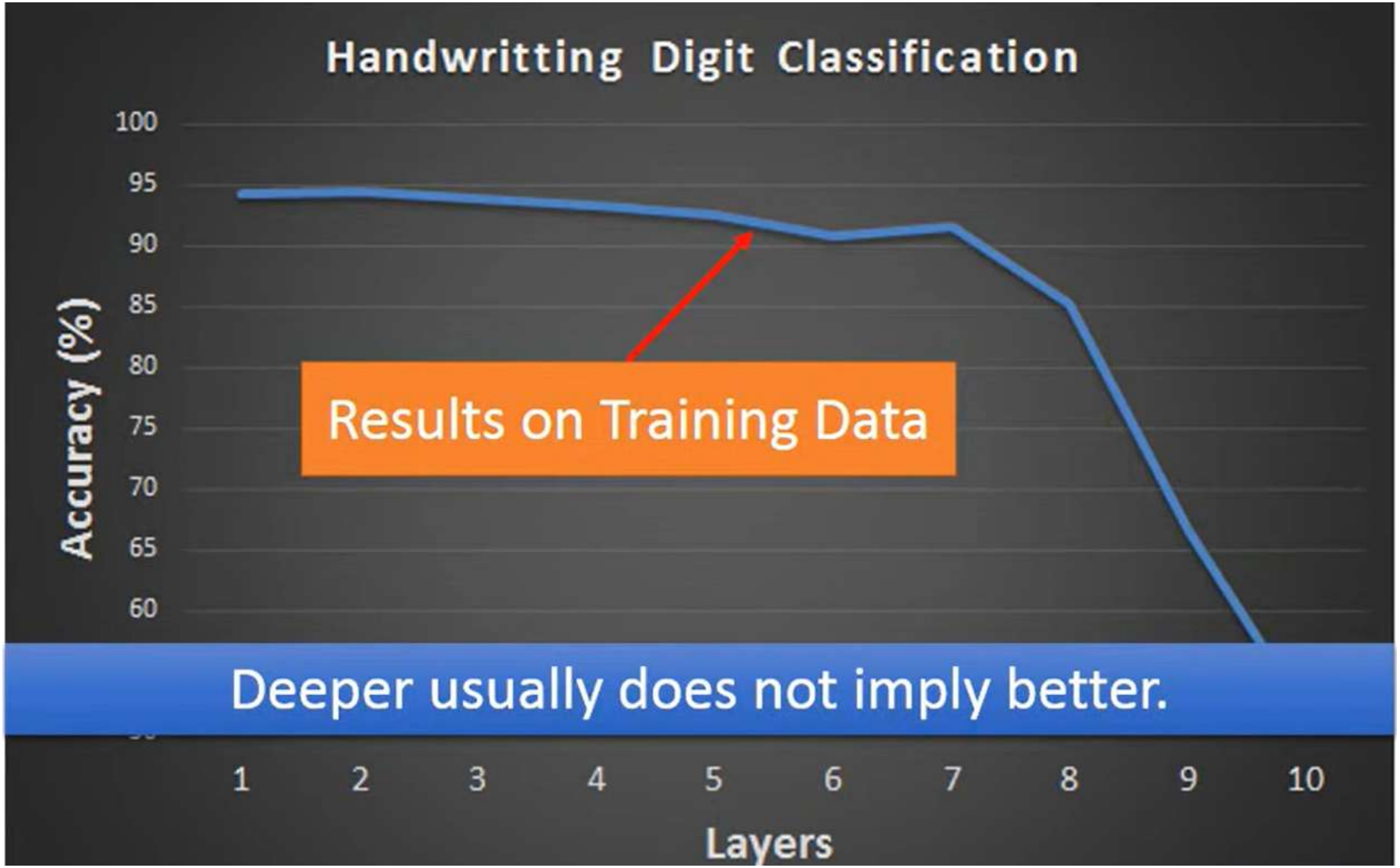


Training Data



Testing Data

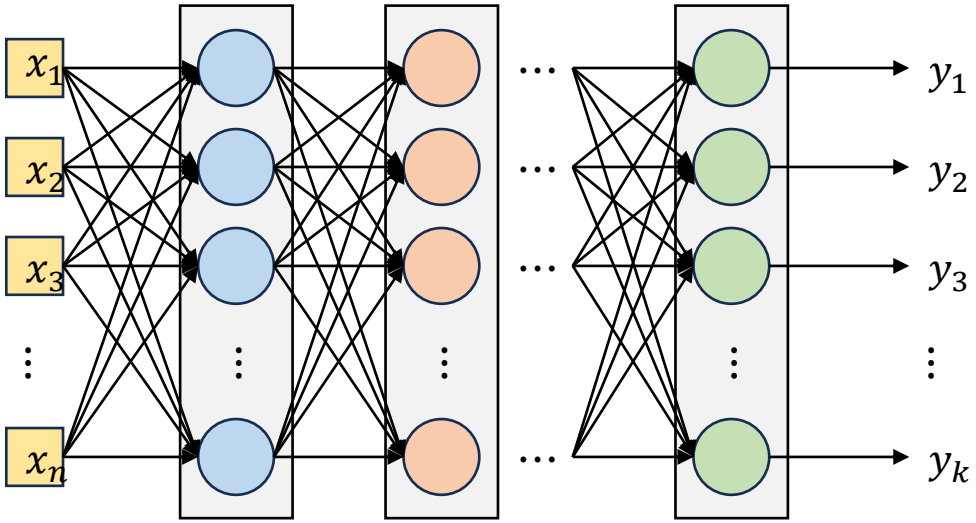
4.5 梯度消失和激活函数



https://youtu.be/xki61j7z-30?si=Hd6awEwQb7SsSAvV

4.5 梯度消失和激活函数

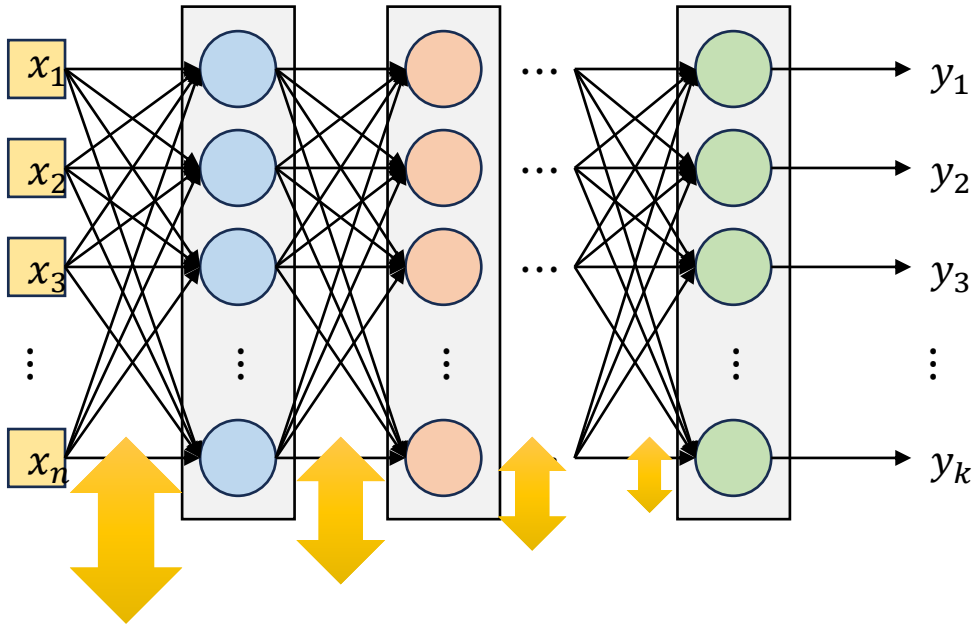
梯度消失

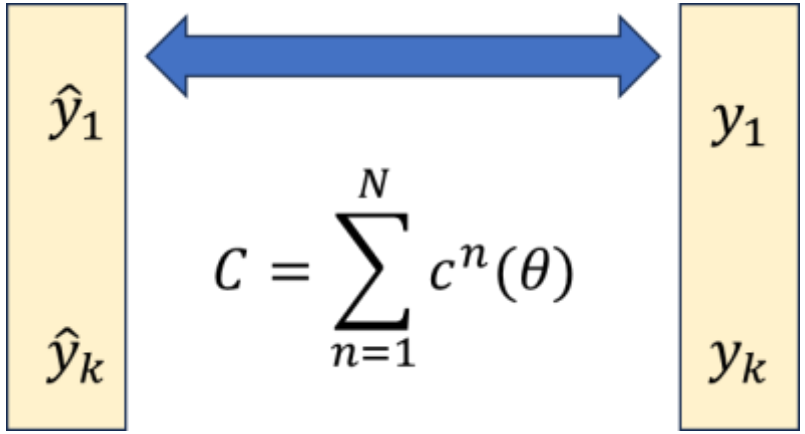


梯度较小	梯度较大
参数更新慢	参数更新快
参数处于随机状态	参数已经完成收敛

4.5 梯度消失和激活函数

梯度消失



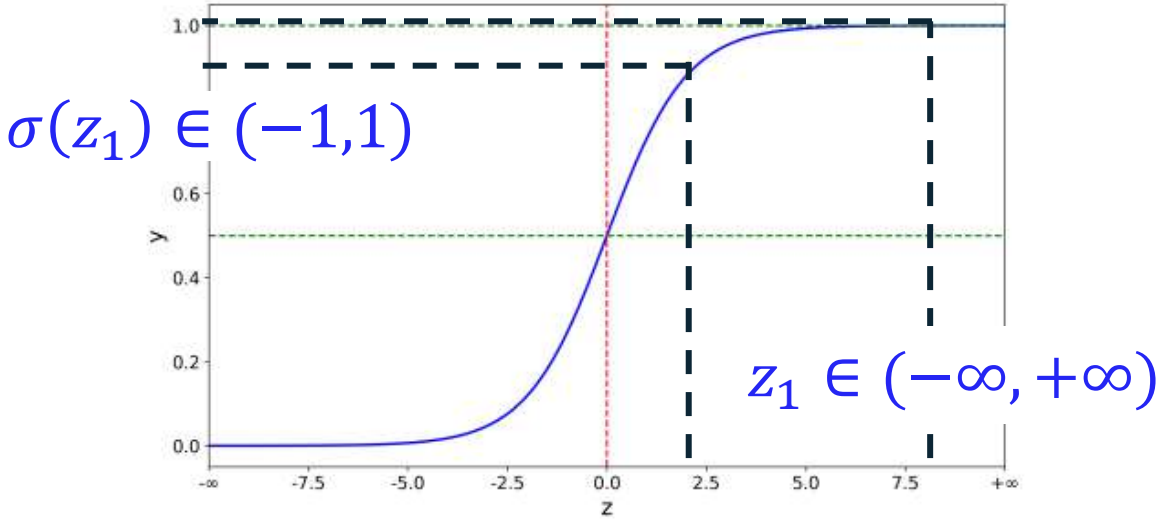


$$C = \sum_{n=1}^{N} c^n(\theta)$$

$$\frac{\partial C}{\partial w} \rightarrow \frac{\Delta C}{\Delta w}$$

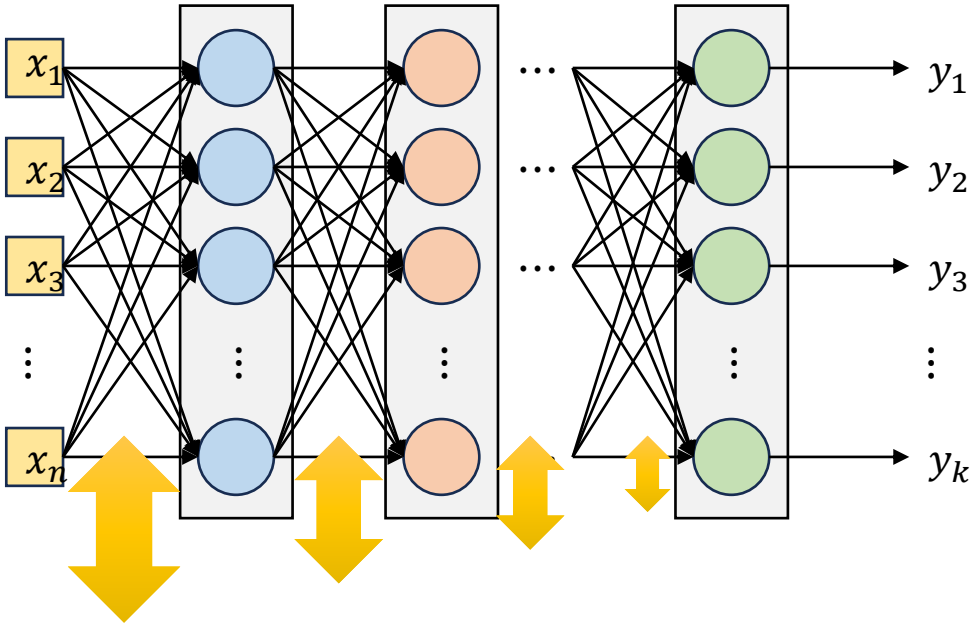
随着层数增高，低层参数对C的影响逐渐降低

低层还很混乱，高层已经收敛



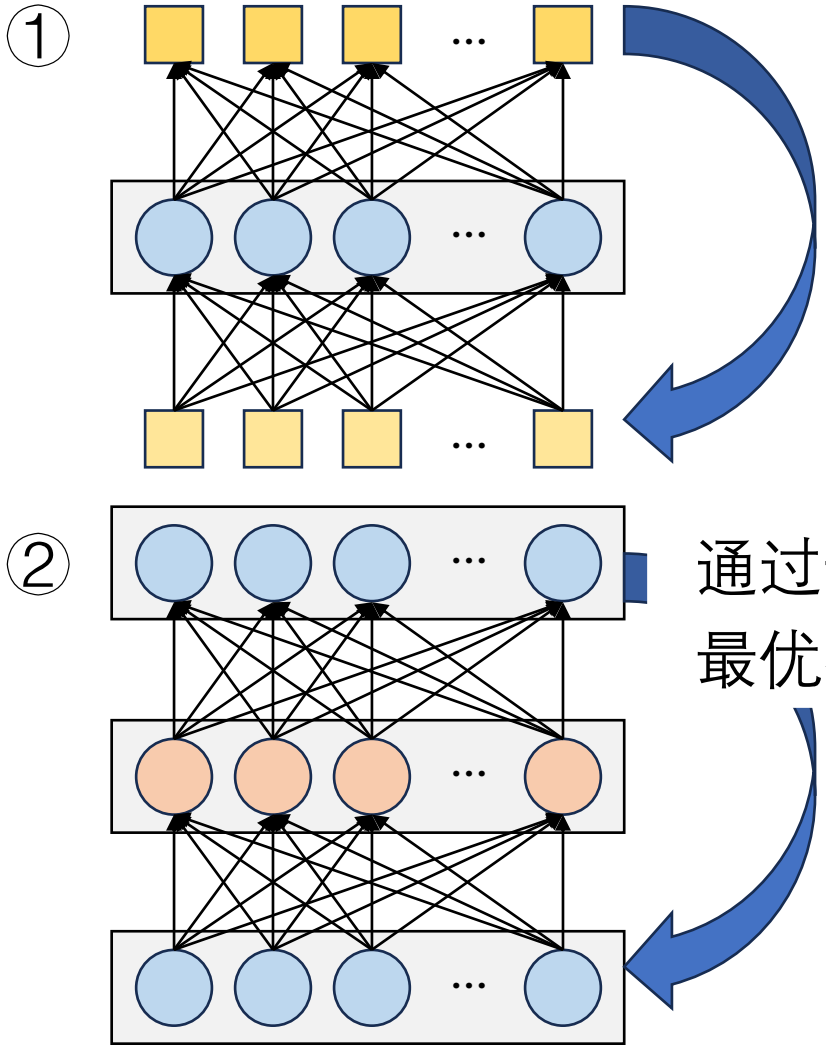
4.5 梯度消失和激活函数

梯度消失



随着层数增高，低层参数
对C的影响逐渐降低

如果一开始就找到合适的初始化参数呢？



$\hat{\boldsymbol{x}} = \boldsymbol{x}$

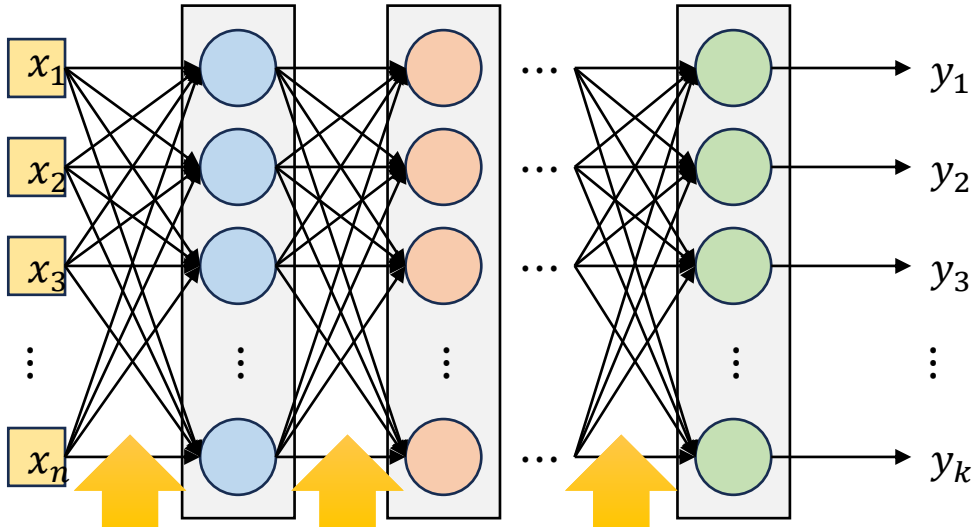
通过预训练找到
最优初始化参数

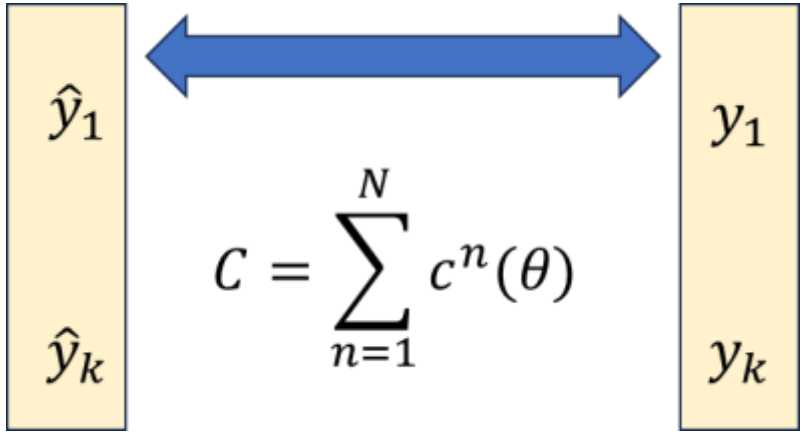
$\hat{\boldsymbol{a}} = \boldsymbol{a}$

有限玻尔兹曼机

4.5 梯度消失和激活函数

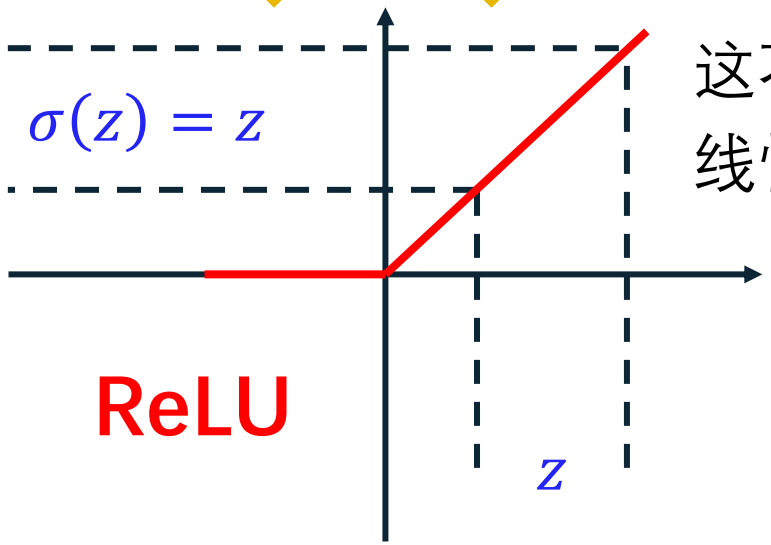
梯度消失





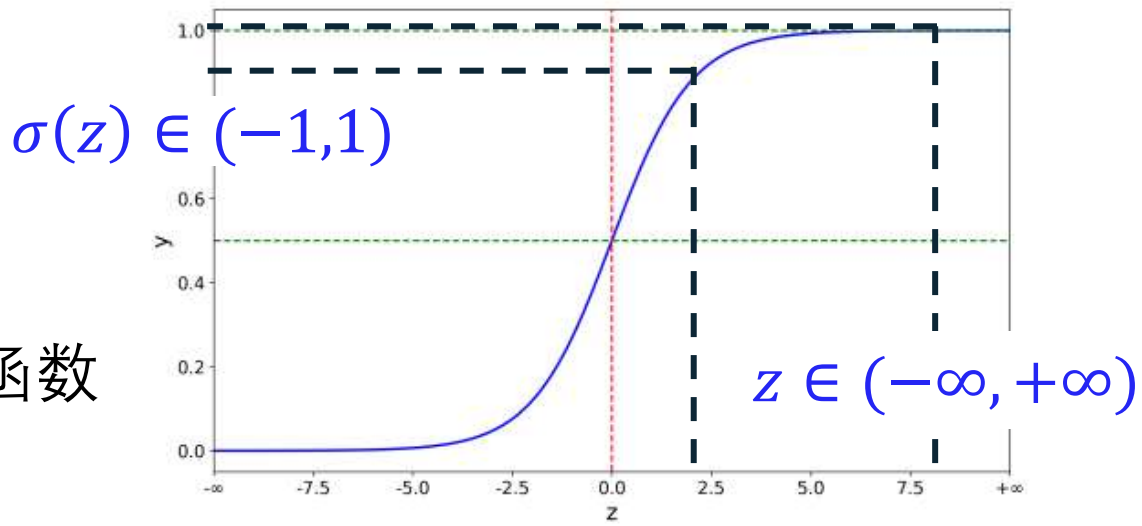
$$C = \sum_{n=1}^{N} c^n(\theta)$$

$$\frac{\partial C}{\partial w} \rightarrow \frac{\Delta C}{\Delta w}$$

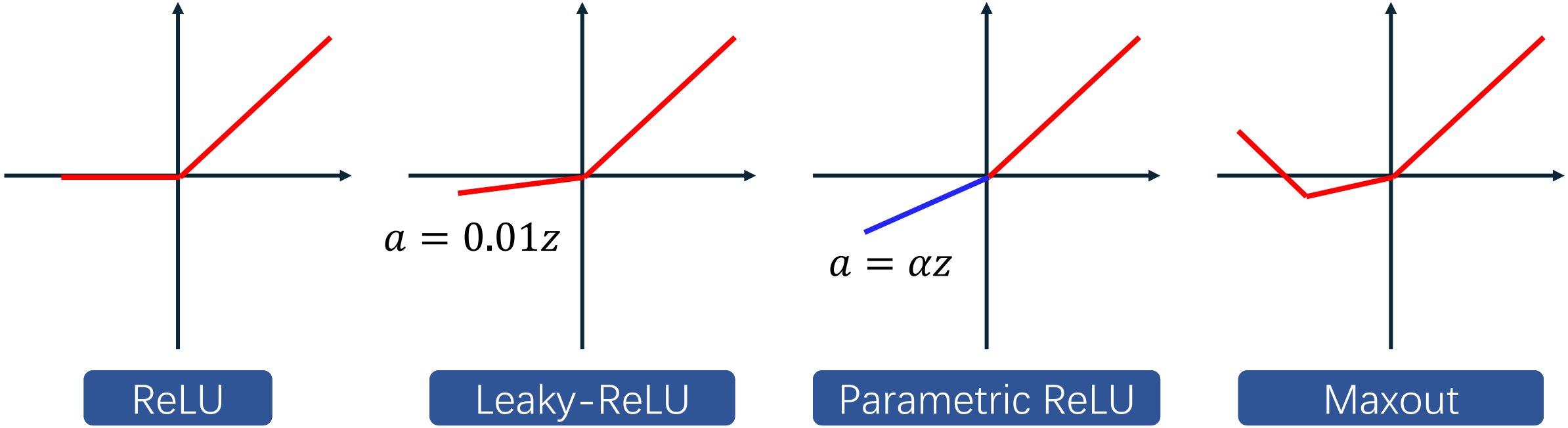


这不就退化成线性方程了？

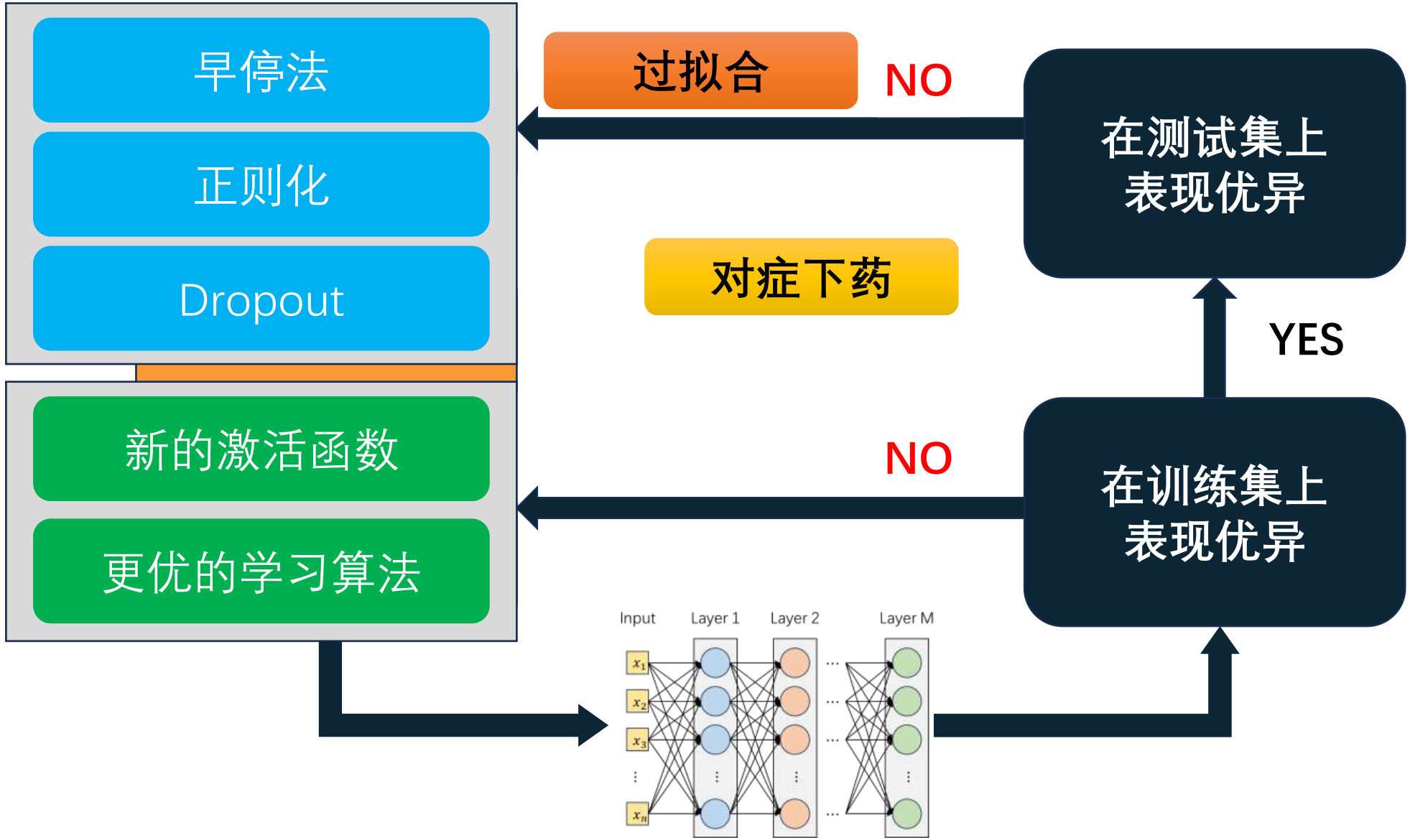
改变激活函数



ReLU和它的好兄弟们



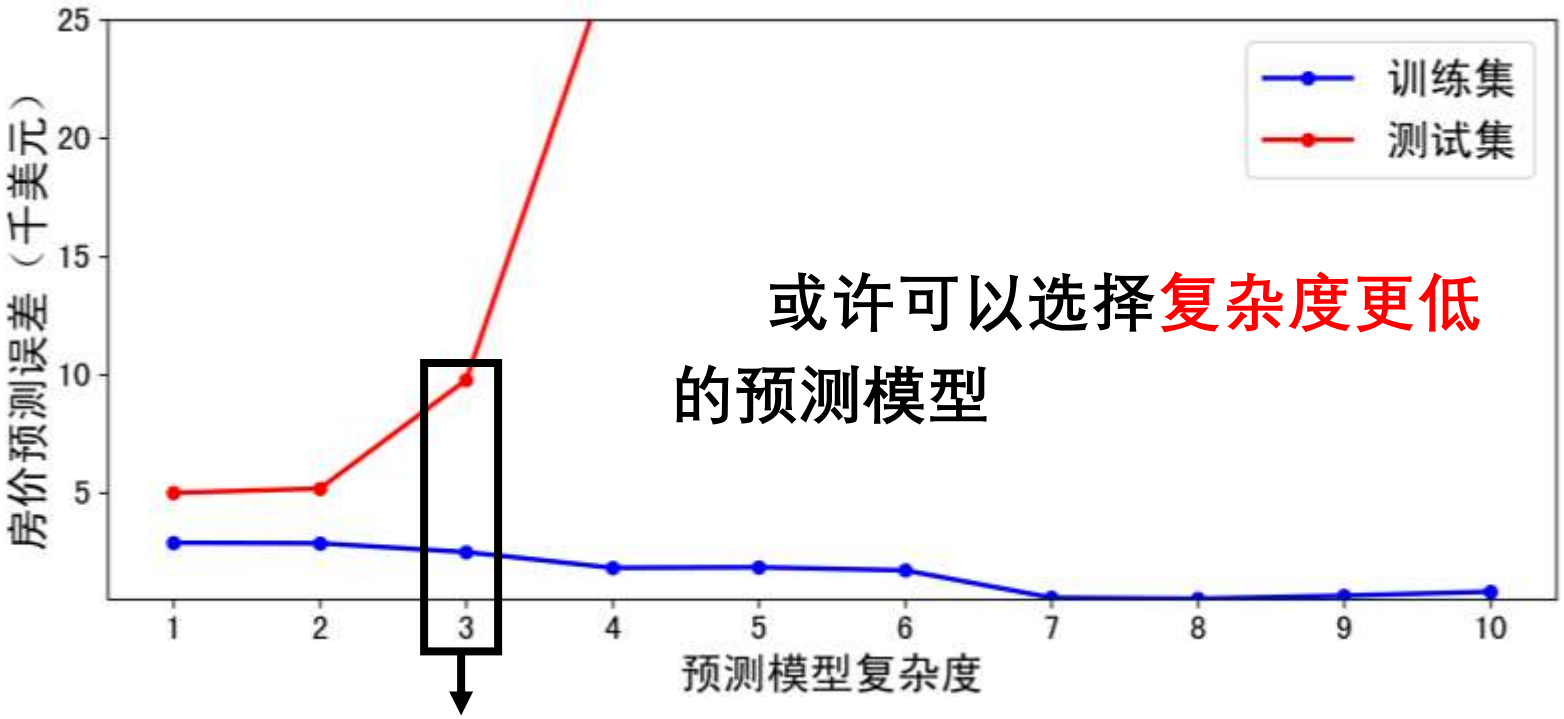
训练深度神经网络一点经验



4.6 过拟合及解决思路①：降低模型复杂度

不同复杂度模型的训练和测试误差

复杂度	训练误差	测试误差
1	2.8898	4.9989
2	2.8688	2.1824
3	2.4948	9.7763
4	1.8300	27.6936
5	1.8525	26.4811
6	1.7230	142.1823
7	0.5671	3563.2944
8	0.5238	6518.7184
9	0.6230	26062.6516
10	0.8185	91591.8384



不出意外，这里应该是出意外了

训练集表现良好，测试集表现很差。

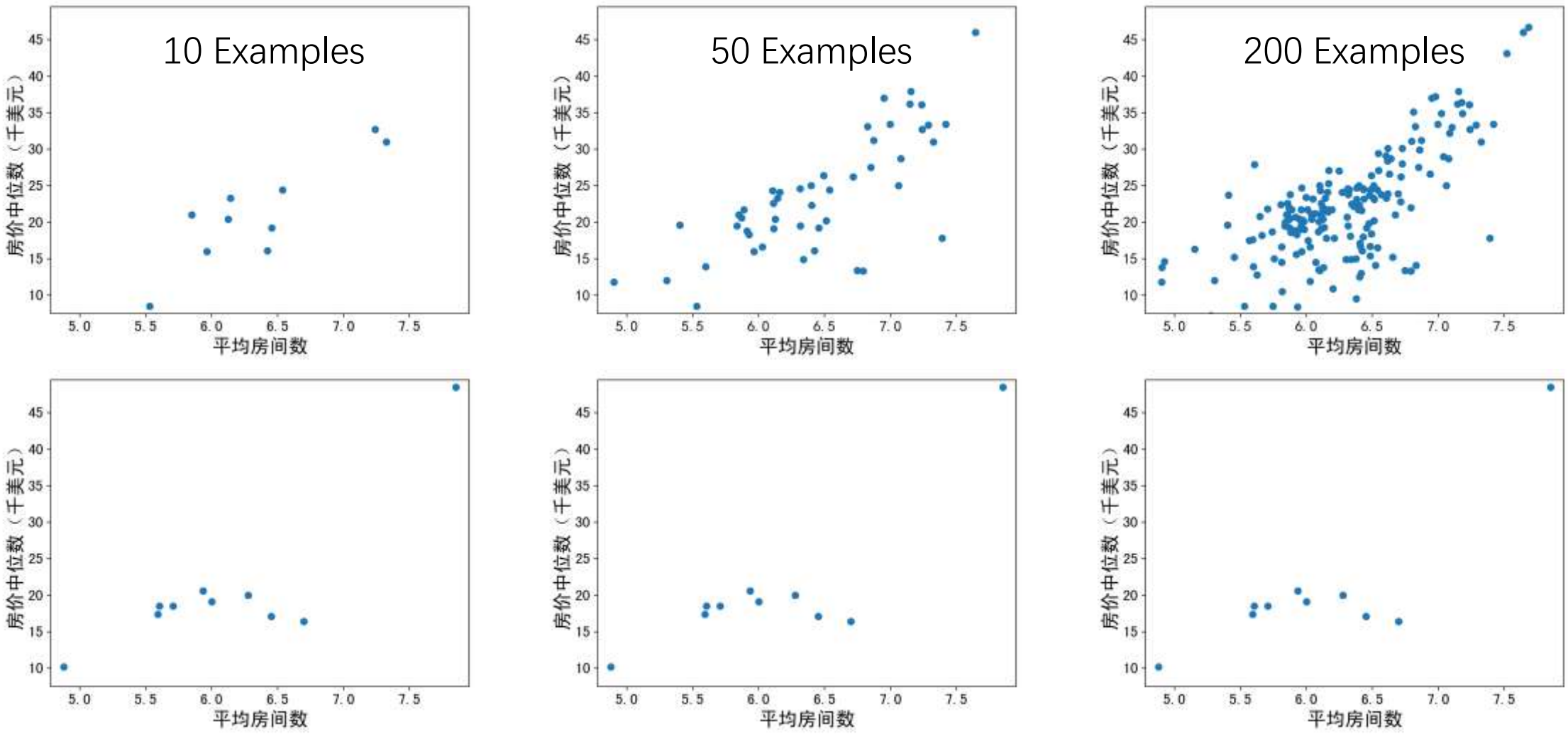
大概率发生了**过拟合**

4.6 过拟合及解决思路②：数据增强（扩展真实数据体量）

例： $y = b + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 + w_6x^6 + w_7x^7 + w_8x^8 + w_9x^9 + w_{10}x^{10}$

训练集

测试集



4.6 过拟合及解决思路②：数据增强（扩展真实数据体量）

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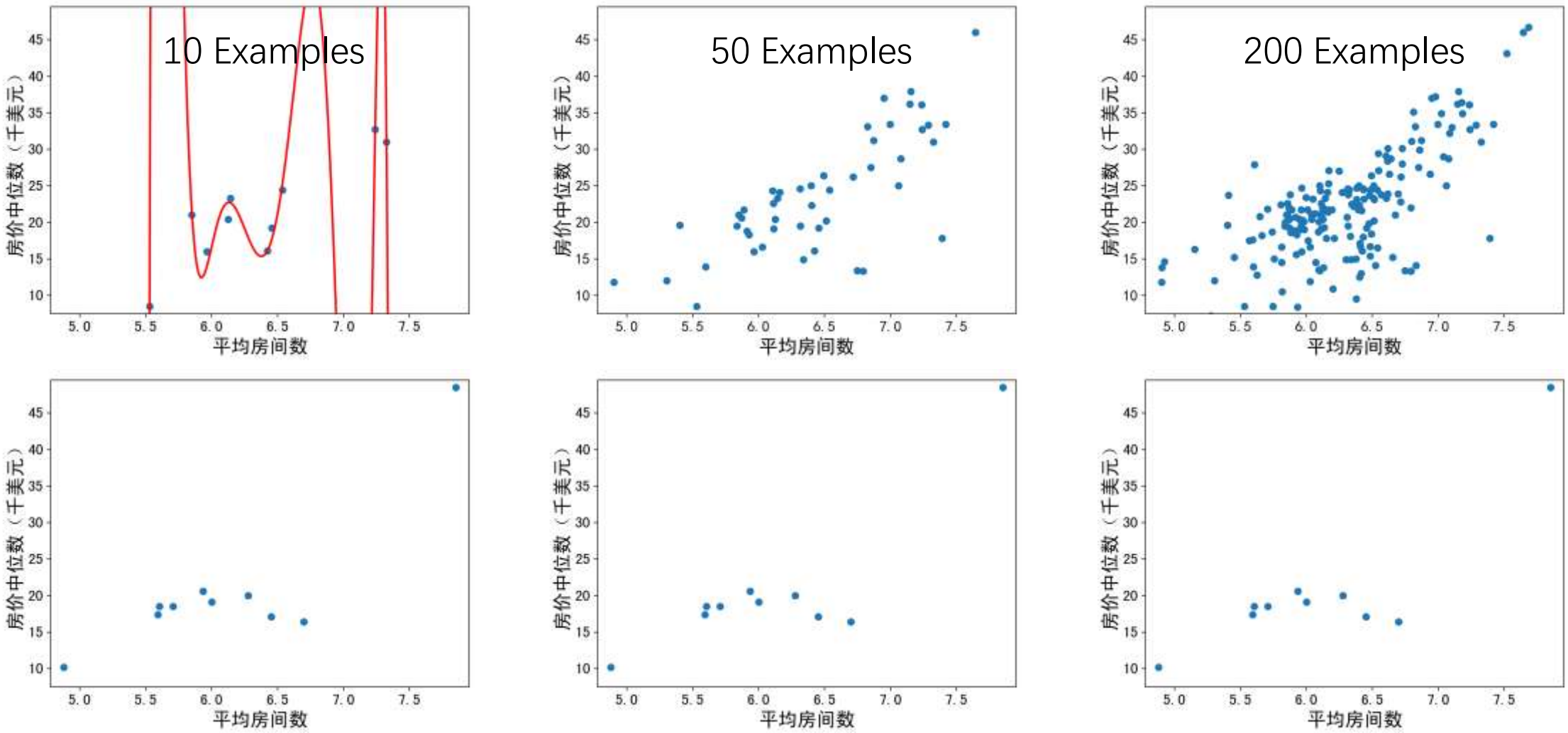
训练集

测试集



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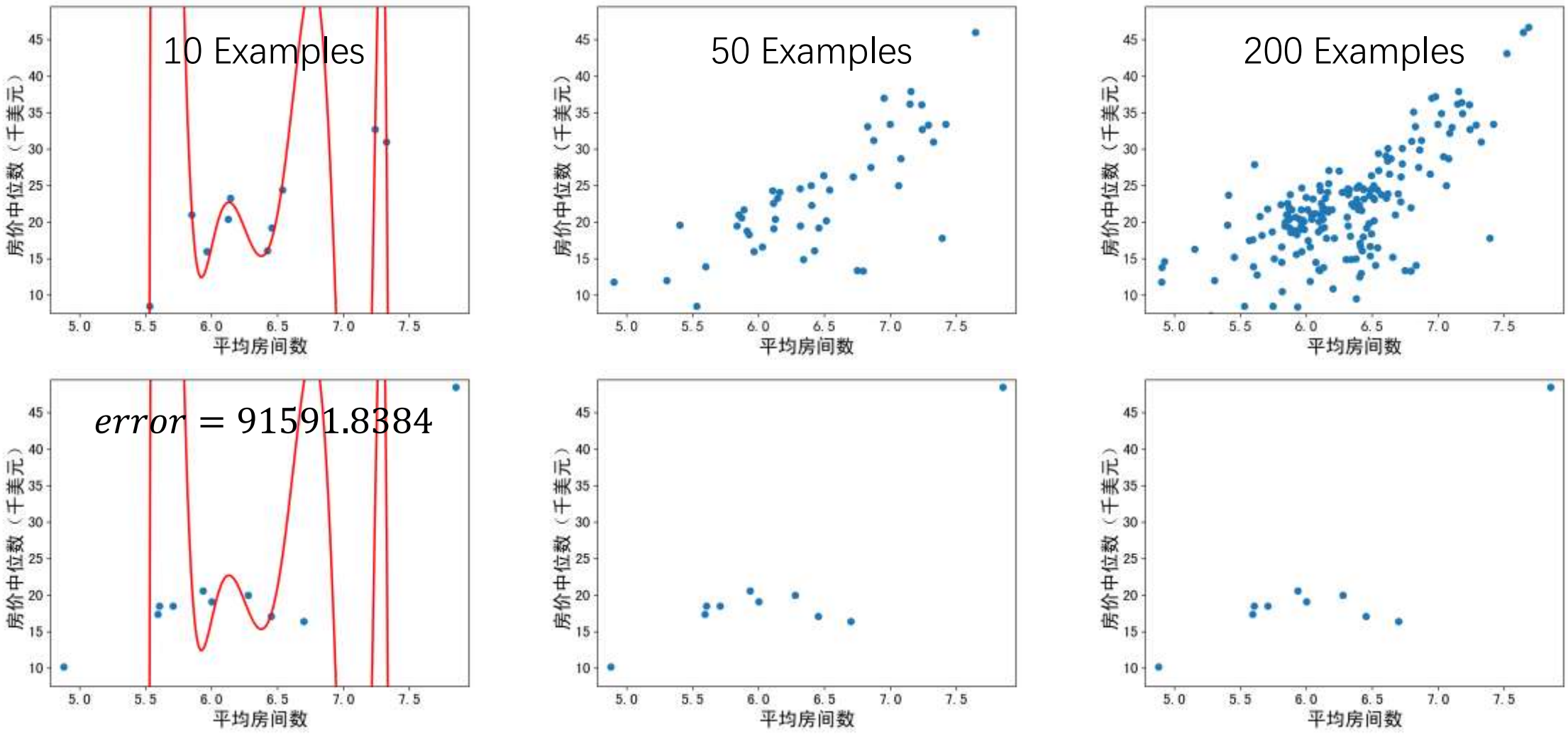
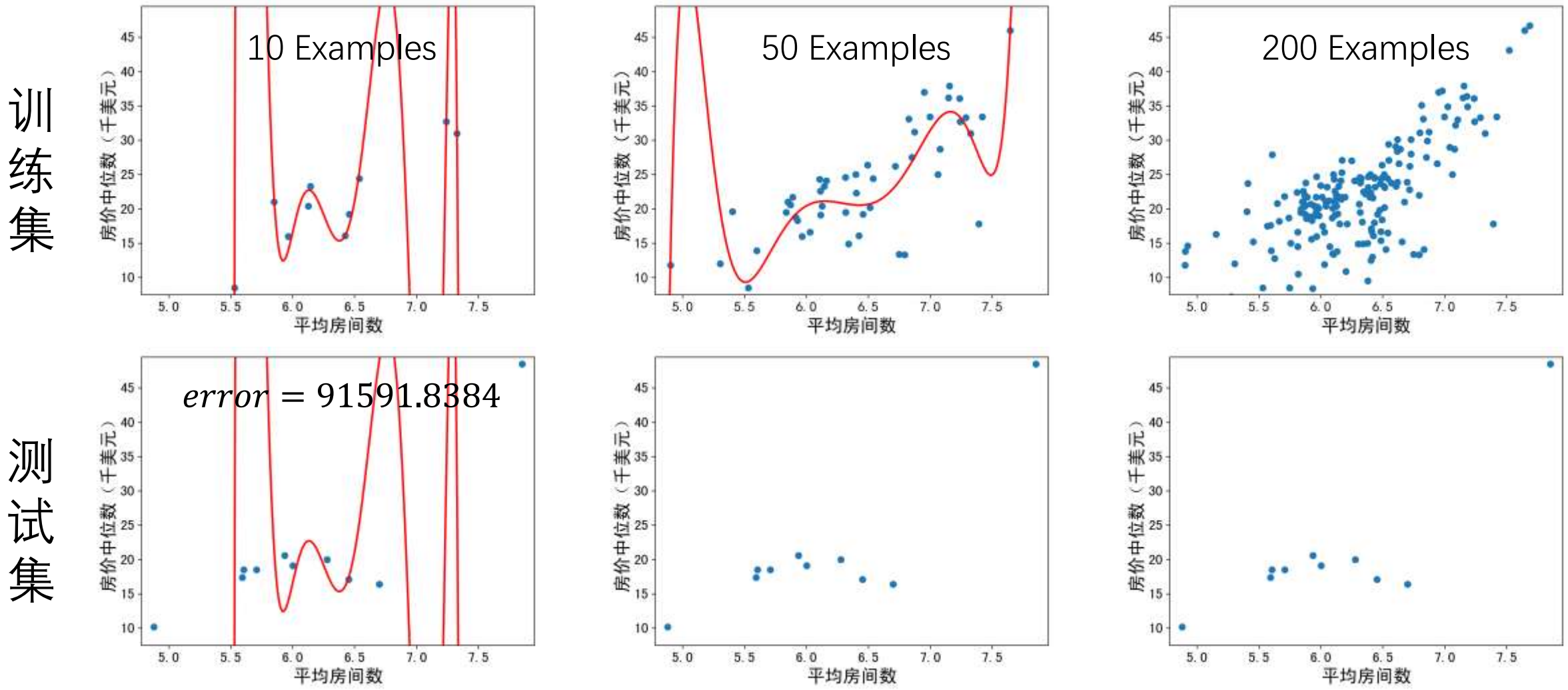
训练集

测试集



4.6 过拟合及解决思路②：数据增强（扩展真实数据体量）

例： $y = b + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 + w_6x^6 + w_7x^7 + w_8x^8 + w_9x^9 + w_{10}x^{10}$

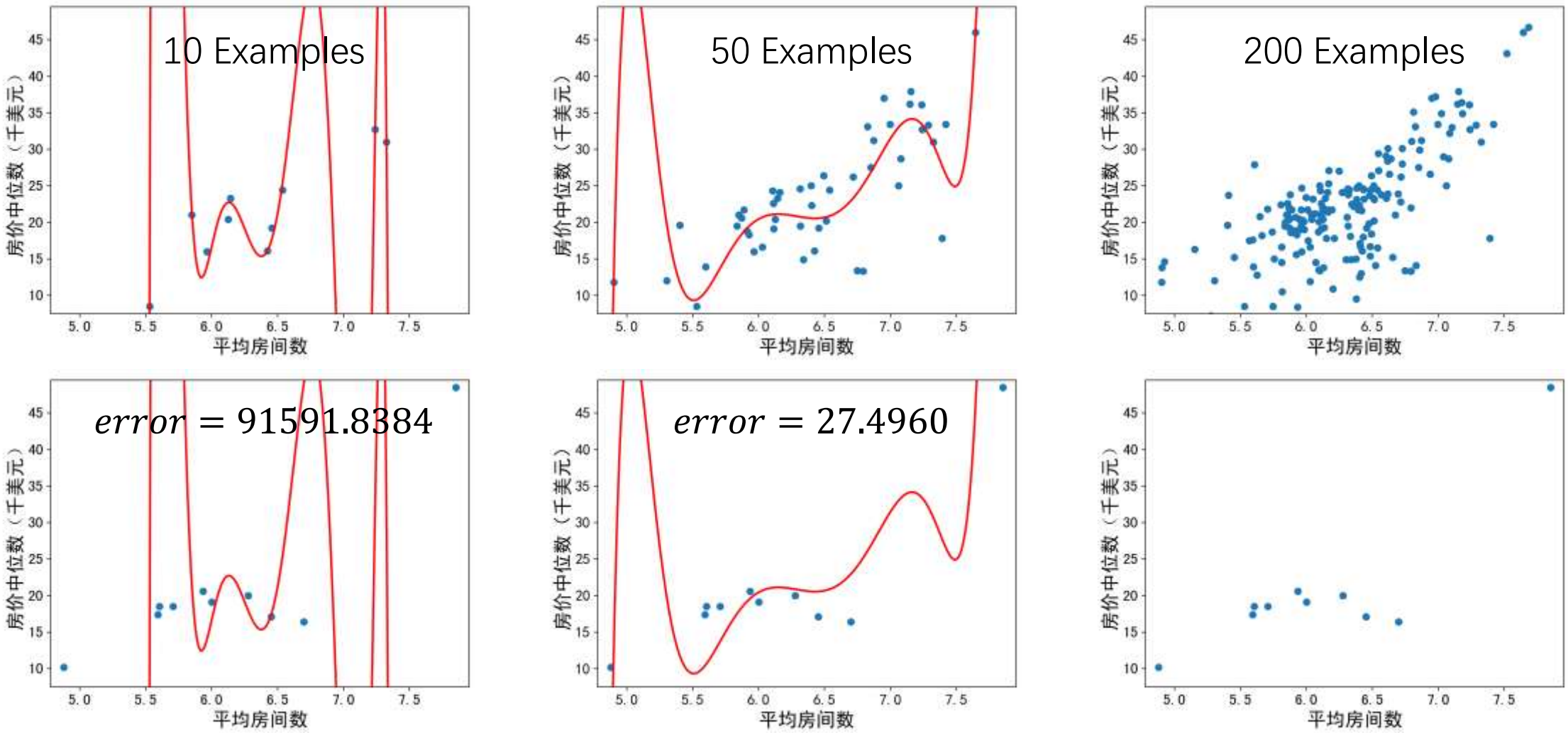


4.6 过拟合及解决思路②：数据增强（扩展真实数据体量）

例： $y = b + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 + w_6x^6 + w_7x^7 + w_8x^8 + w_9x^9 + w_{10}x^{10}$

训练集

测试集

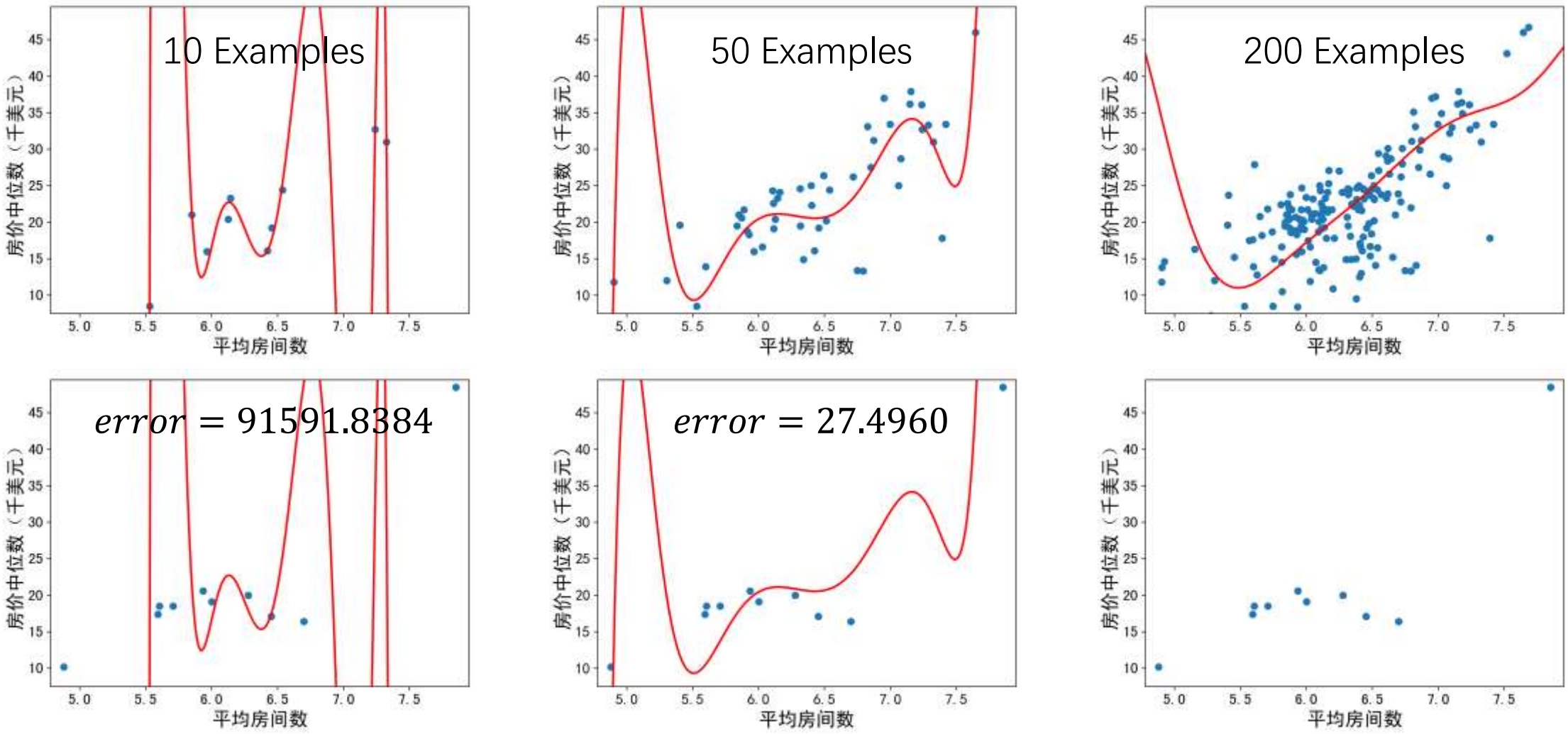


4.6 过拟合及解决思路②：数据增强（扩展真实数据体量）

例： $y = b + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 + w_6x^6 + w_7x^7 + w_8x^8 + w_9x^9 + w_{10}x^{10}$

训练集

测试集

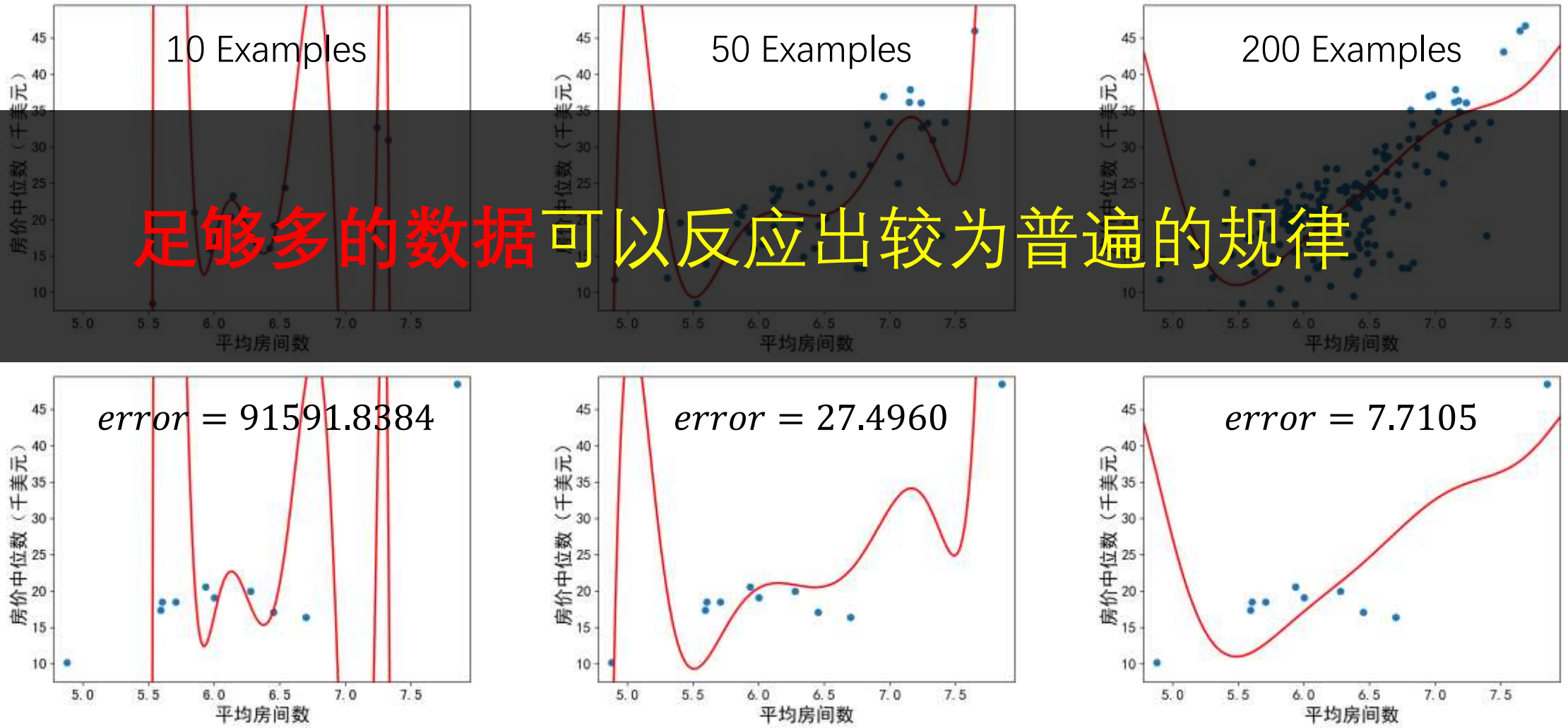


4.6 过拟合及解决思路②：数据增强（扩展真实数据体量）

例： $y = b + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 + w_6x^6 + w_7x^7 + w_8x^8 + w_9x^9 + w_{10}x^{10}$

训练集

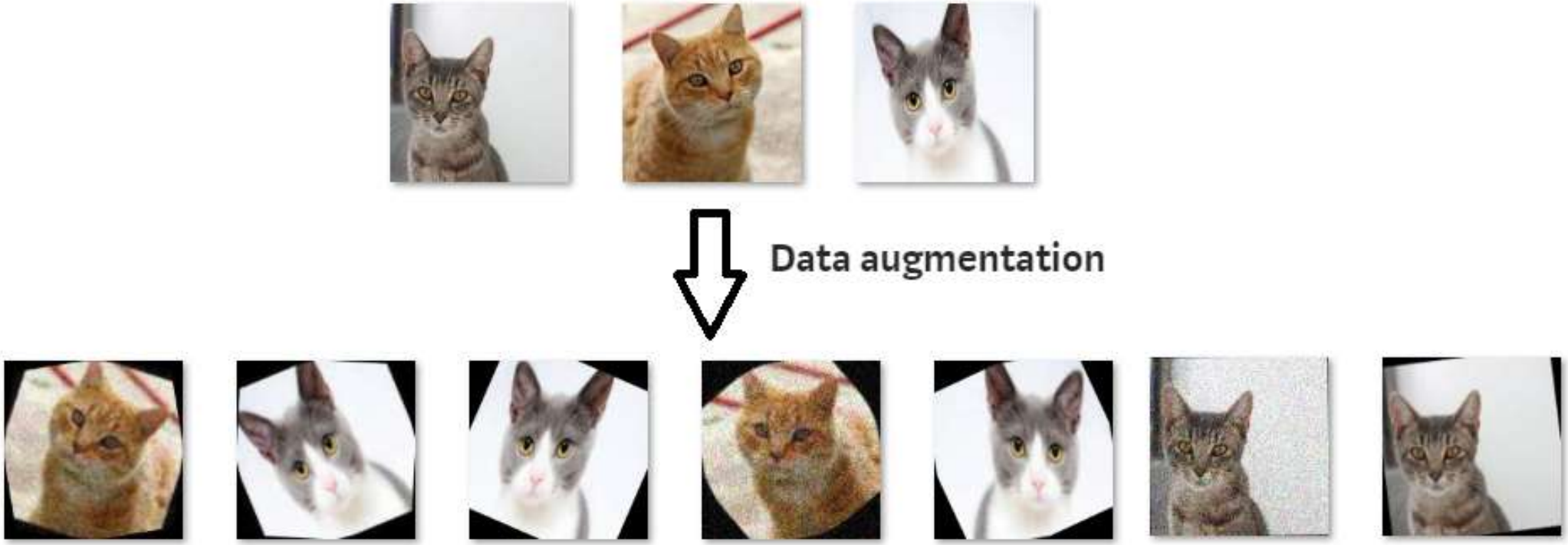
测试集



足够多的数据可以反应出较为普遍的规律

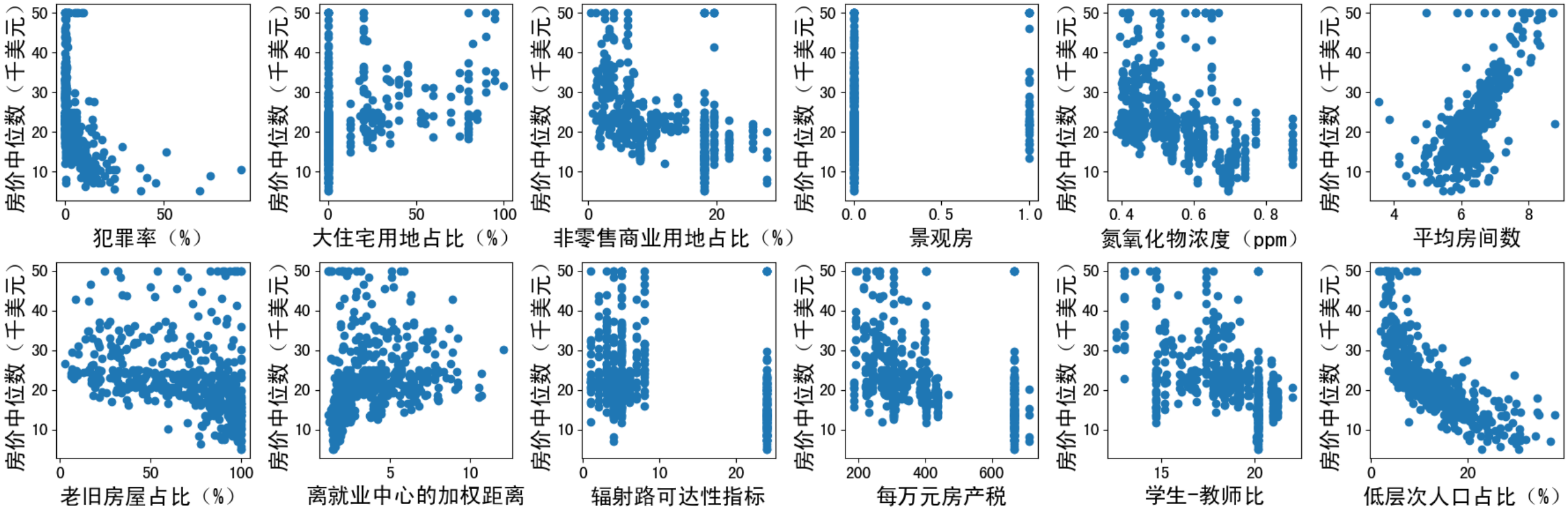
4.6 过拟合及解决思路②：数据增强（扩展虚拟数据体量）

使用虚拟数据可以模拟特征多样性，避免模型过于关注细节



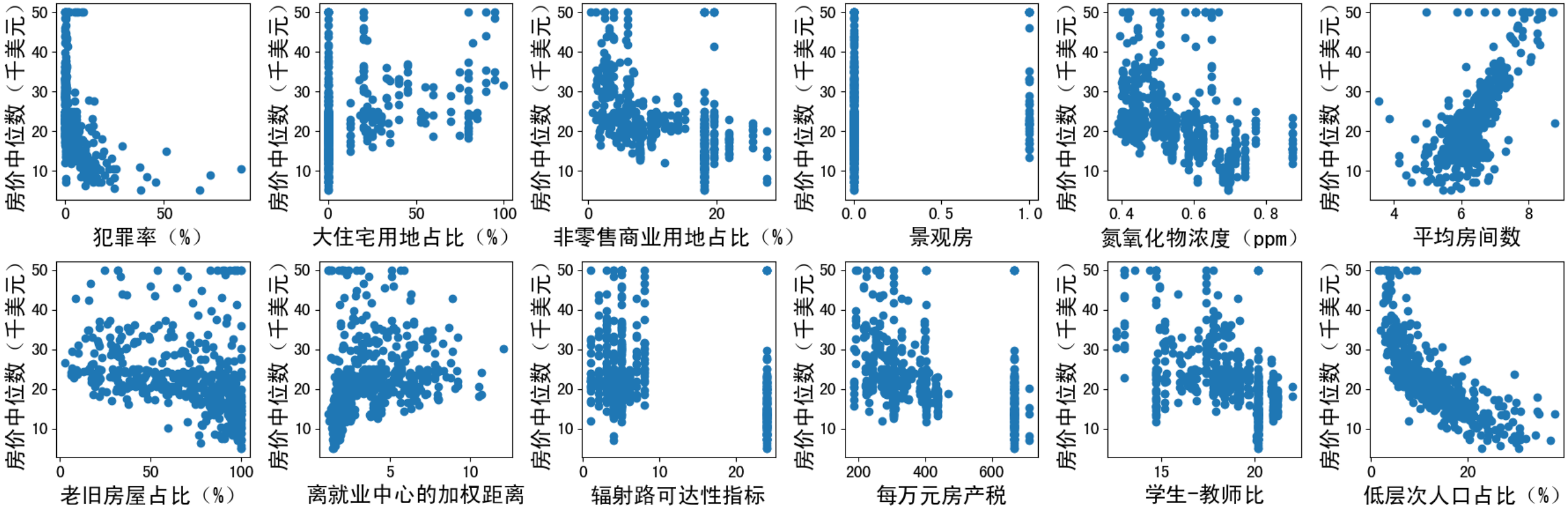
4.6 过拟合及解决思路③：更丰富的特征&正则化

例： $y = b + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 + w_6x^6 + w_7x^7 + w_8x^8 + w_9x^9 + w_{10}x^{10}$



4.6 过拟合及解决思路③：更丰富的特征&正则化

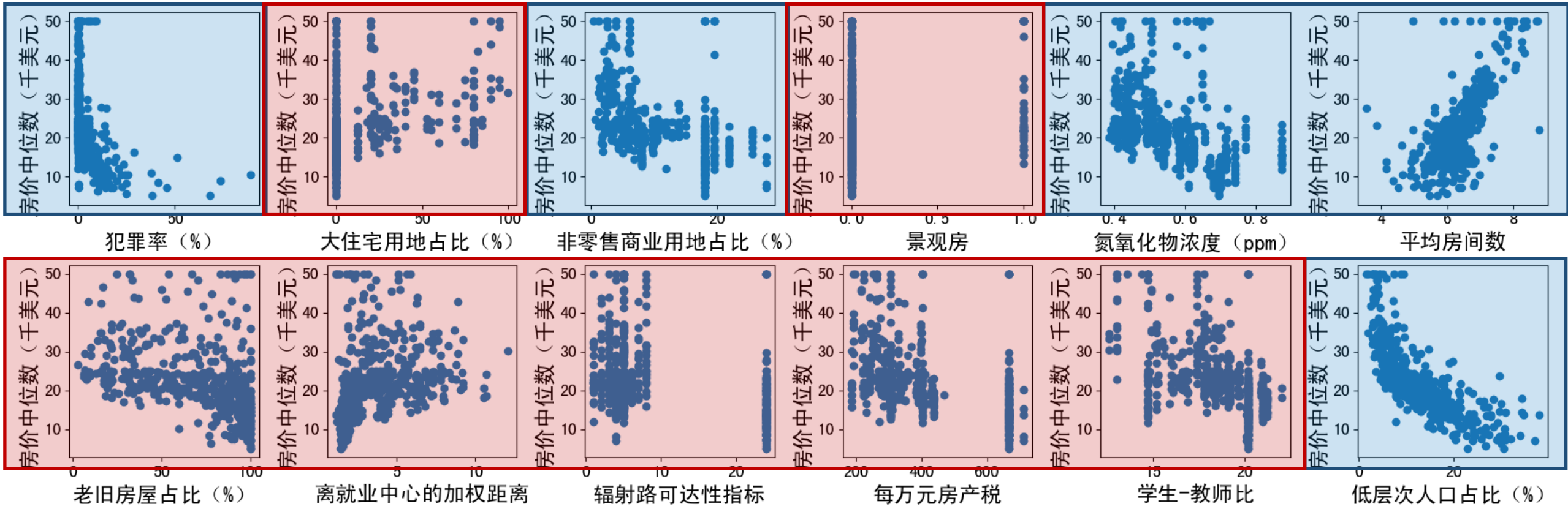
例：$y = b + w_1x_{cr} + w_2x_{zn} + w_3x_{nr} + w_4x_{ch} + w_5x_{ppm} + w_6x_{rn} + w_7x_{ag} + w_8x_{cd} + w_9x_{ra}$
$+w_{10}x_{rt} + w_{11}x_{st} + w_{12}x_{lr}$



4.6 过拟合及解决思路③：更丰富的特征&正则化

哪些特征更有用咧？

肉眼可辨的规律　　需要深耕多年的从业经验



4.6 过拟合及解决思路③：更丰富的特征&正则化（L2范式）

在Loss函数中引入一个奇怪的部分 $\eta \cdot \sum w^2$

$$\hat{y} = \sum_{i=1}^{M} w_i x_i + b$$

这样就能确定有意义的特征吗？

$$Loss = \frac{1}{2N} \sum_{k=1}^{N} (\widehat{y_k} - y_k)^2 + \boxed{\eta \cdot \sum_{i=1}^{M} w_i^2}$$

4.6 过拟合及解决思路③：更丰富的特征&正则化（L2范式）

在Loss函数中引入一个奇怪的部分 $\eta \cdot \sum w^2$

$$\hat{y} = \sum_{i=1}^{M} w_i x_i + b$$

$$Loss = \frac{1}{2N} \sum_{k=1}^{N} (\widehat{y_k} - y_k)^2 + \eta \cdot \sum_{i=1}^{M} w_i^2$$

需明确的关键一点：

不论Loss的形式如何，训练的时候只会机械寻找参数，使得Loss最低。

4.6 过拟合及解决思路③：更丰富的特征&正则化（L2范式）

在Loss函数中引入一个奇怪的部分 $\eta \cdot \sum w^2$

$$\hat{y} = \sum_{i=1}^{M} w_i x_i + b$$

$$Loss = \frac{1}{2N}\sum_{k=1}^{N}(\widehat{y_k} - y_k)^2 + \eta \cdot \sum_{i=1}^{M} w_i^2$$

预测的准确度

此部分的值降低有什么意义？

正则化是为了让参数变小，让函数变得更加

平 滑（Smooth)

这两个部分的值在训练的时候都会降低

4.6 过拟合及解决思路③：更丰富的特征&正则化（L2范式）

在Loss函数中引入一个奇怪的部分 $\eta \cdot \sum w^2$

$$\hat{y} = \sum_{i=1}^{M} w_i x_i + b$$

为什么参数小就能使函数变得平滑？

$$Loss = \frac{1}{2N} \sum_{k=1}^{N} (\widehat{y_k} - y_k)^2 + \eta \cdot \sum_{i=1}^{M} w_i^2$$

所谓平滑是指，预测函数的输出对于输入的变化不敏感，不容易受到噪声影响。

4.6 过拟合及解决思路③：更丰富的特征&正则化（L2范式）

在Loss函数中引入一个奇怪的部分 $\eta \cdot \sum w^2$

$$\hat{y} = \sum_{i=1}^{M} w_i x_i + b$$

x_i: $+\Delta$

$\hat{y}$: $+w_i(x_i + \Delta)$

$$Loss = \frac{1}{2N}\sum_{k=1}^{N}(\widehat{y_k} - y_k)^2 + \eta \cdot \sum_{i=1}^{M} w_i^2$$

- 如果w_i不够小，Δ较大，$\hat{y}$变化大
- 如果w_i足够小，Δ较大，$\hat{y}$变化小
- 如果特征x_i与$\hat{y}$无关，正则化可使模型学到较小的w_i（理想$w_i \to 0$）

⬇

确定有意义的特征

所谓平滑是指，预测函数的输出对于输入的变化不敏感，不容易受到噪声影响。

4.6 过拟合及解决思路③：更丰富的特征&正则化（L2范式）

η 的作用又是什么？

$$Loss = \frac{1}{2N}\sum_{k=1}^{N}(\widehat{y_k} - y_k)^2 + \eta \cdot \sum_{i=1}^{M} w_i^2$$

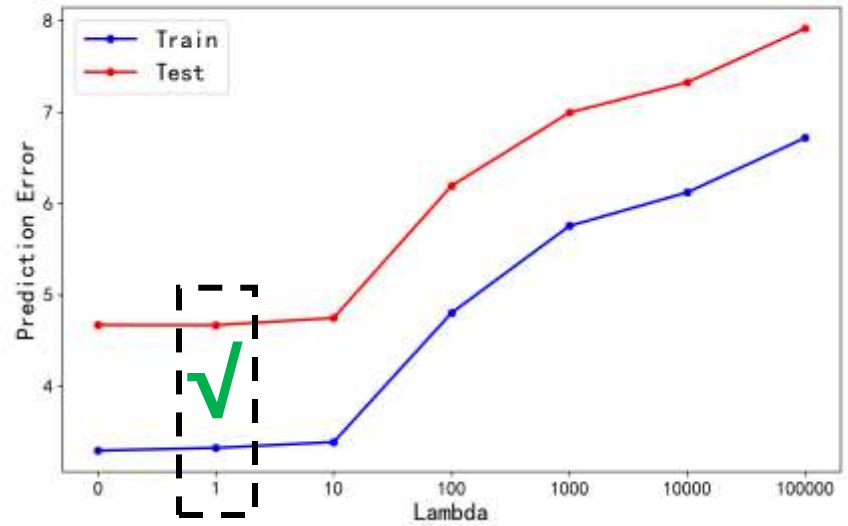
4.6 过拟合及解决思路③：更丰富的特征&正则化（L2范式）

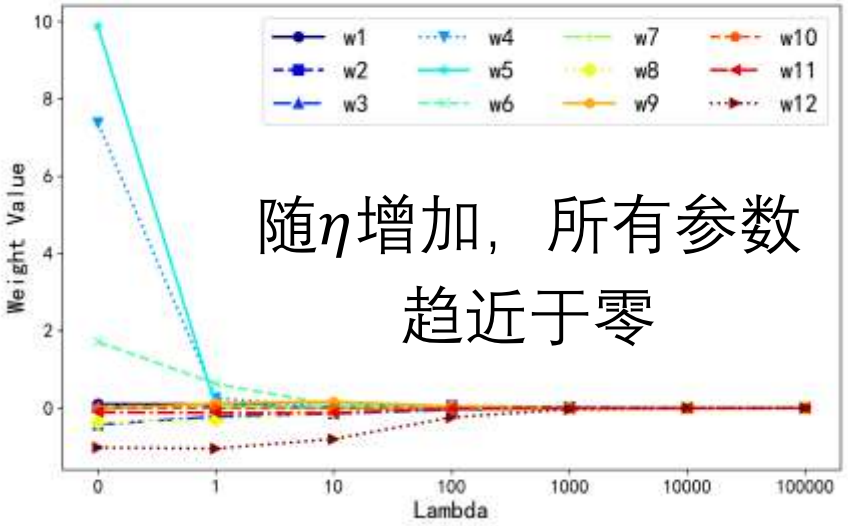
η 的作用又是什么？ 是人工预定义的值，描述正则化倾向（重要性）

$$Loss = \boxed{\frac{1}{2N}\sum_{k=1}^{N}(\widehat{y_k} - y_k)^2} + \eta \cdot \boxed{\sum_{i=1}^{M} w_i^2}$$

预测的准确度　　平滑度

η	训练误差	测试误差
0	3.2953	4.6701
1	3.3242	4.6683
10	3.3877	4.7461
100	4.8020	6.1936
1000	5.7546	6.9953
10000	6.1221	7.3265
100000	6.7197	7.9166





随η增加，所有参数趋近于零

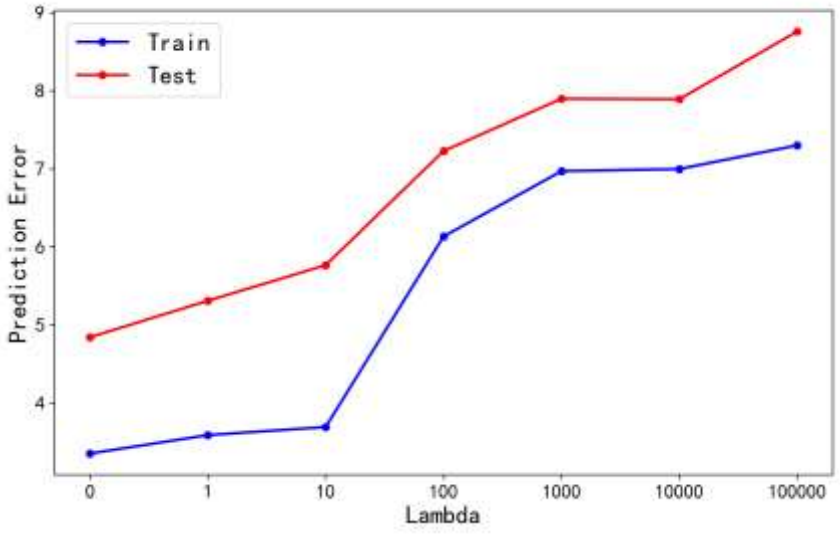
4.6 过拟合及解决思路③：更丰富的特征&正则化（L1范式）

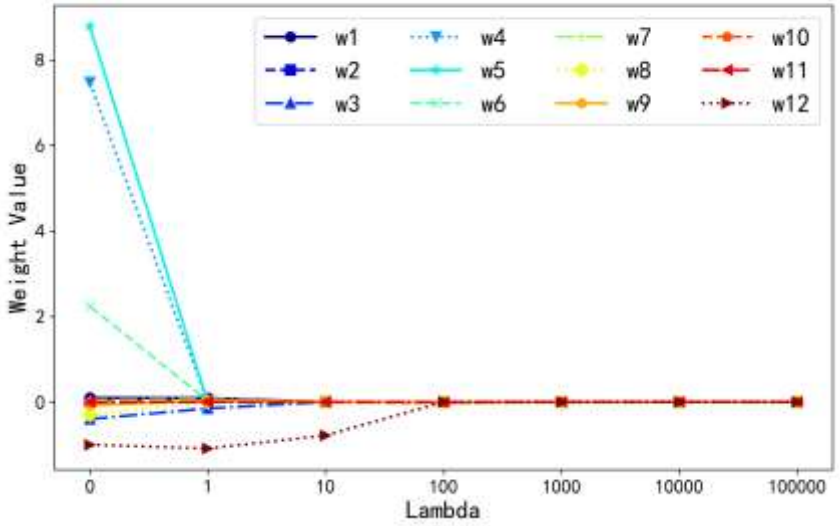
在Loss函数中引入另一个奇怪的部分 $\eta \cdot \sum|w|$

$$Loss = \frac{1}{2N}\sum_{k=1}^{N}(\widehat{y_k} - y_k)^2 + \eta \cdot \sum_{i=1}^{M}|w_i|$$

- L1范式适用于样本较大、特征较多的场景
- 能使w为0，降低存储开销
- 在0处不可微分，一般使用正弦函数近似

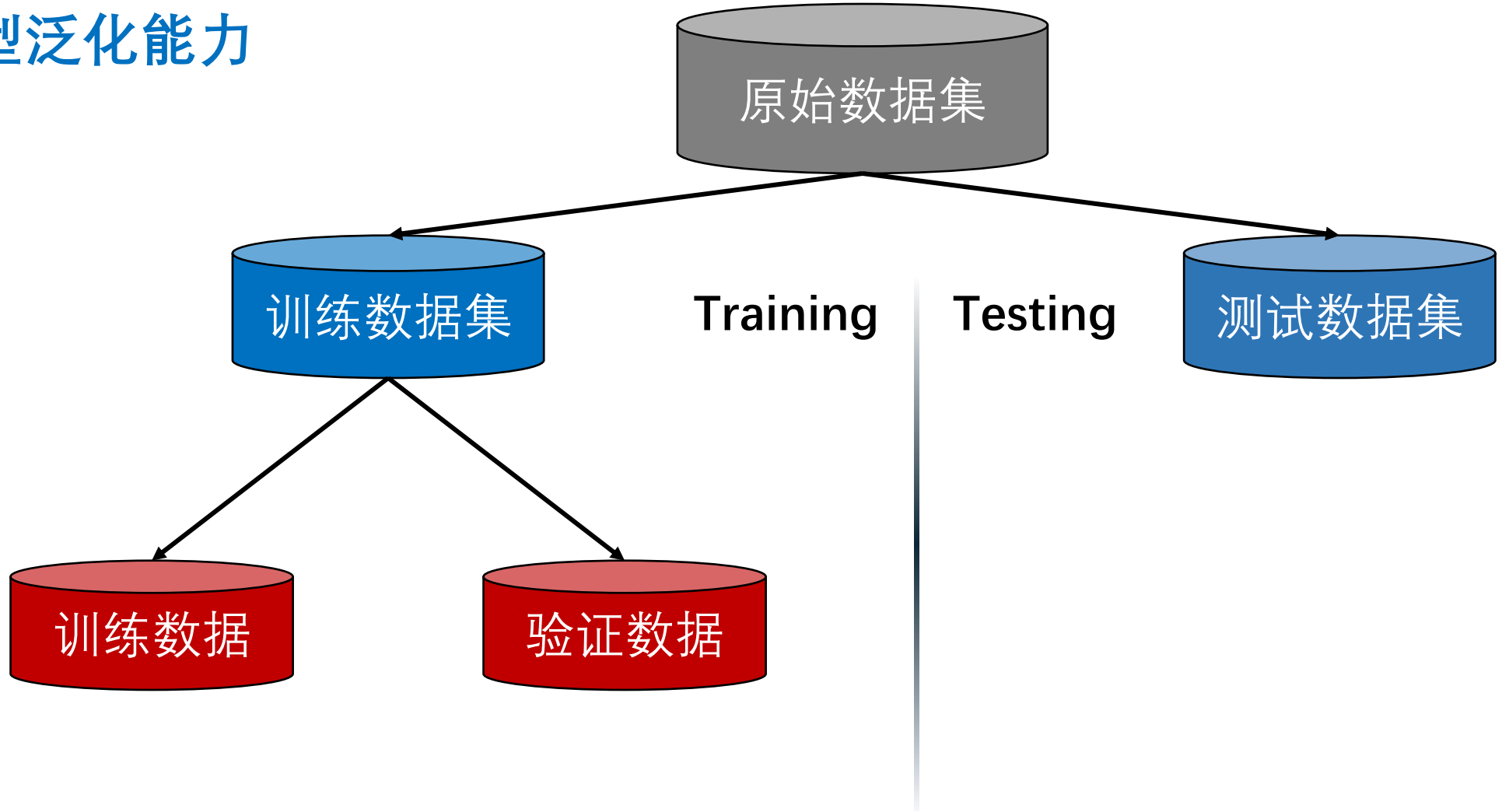
$$\frac{d_{Loss}}{d_{w_i}} = \frac{1}{N}\sum_{k=1}^{N}(\widehat{y_k} - y_k)x_k + \eta \cdot \sin(w_i)$$





4.6 过拟合及解决思路④：交叉验证增强评估能力

在训练时评估
模型泛化能力



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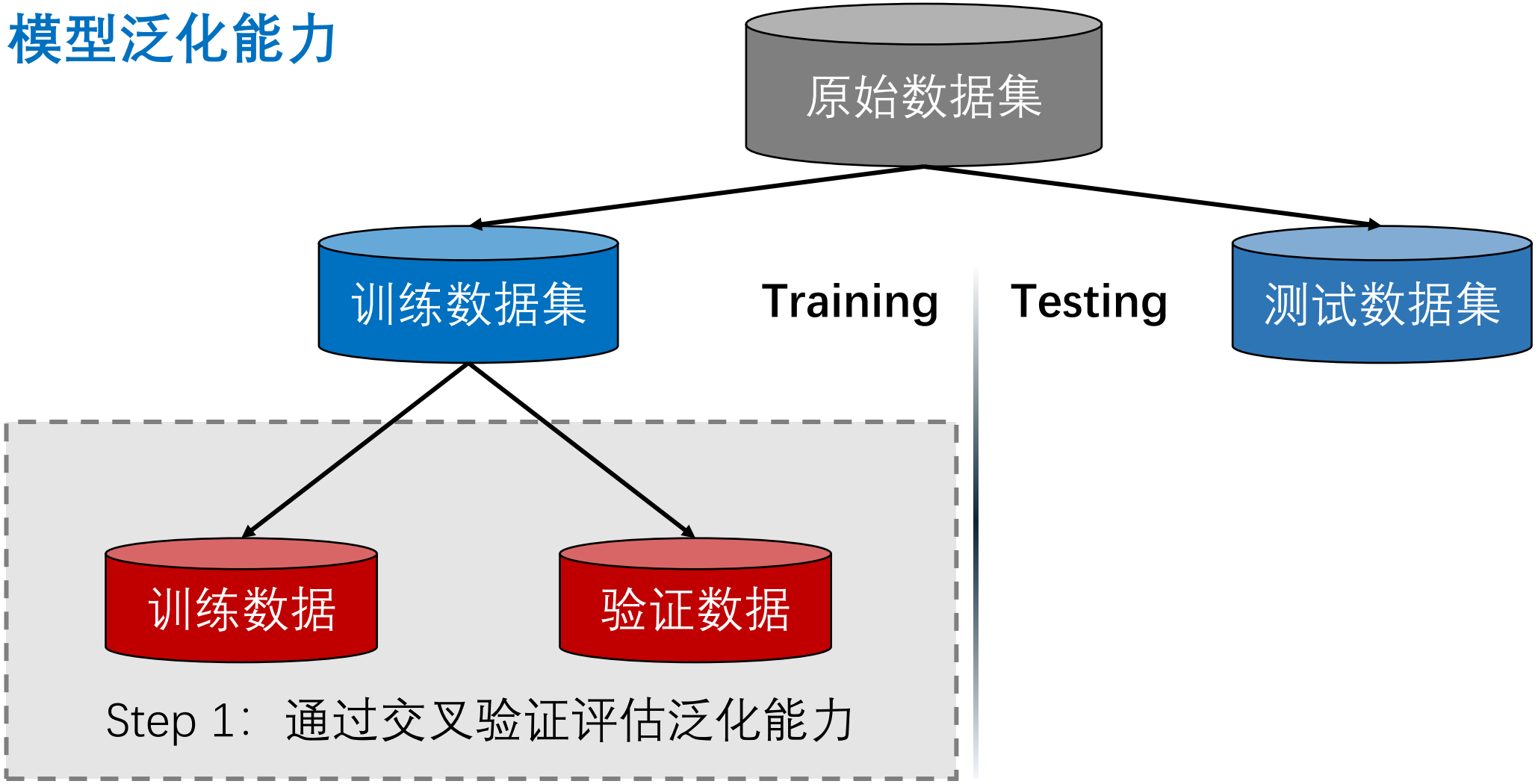

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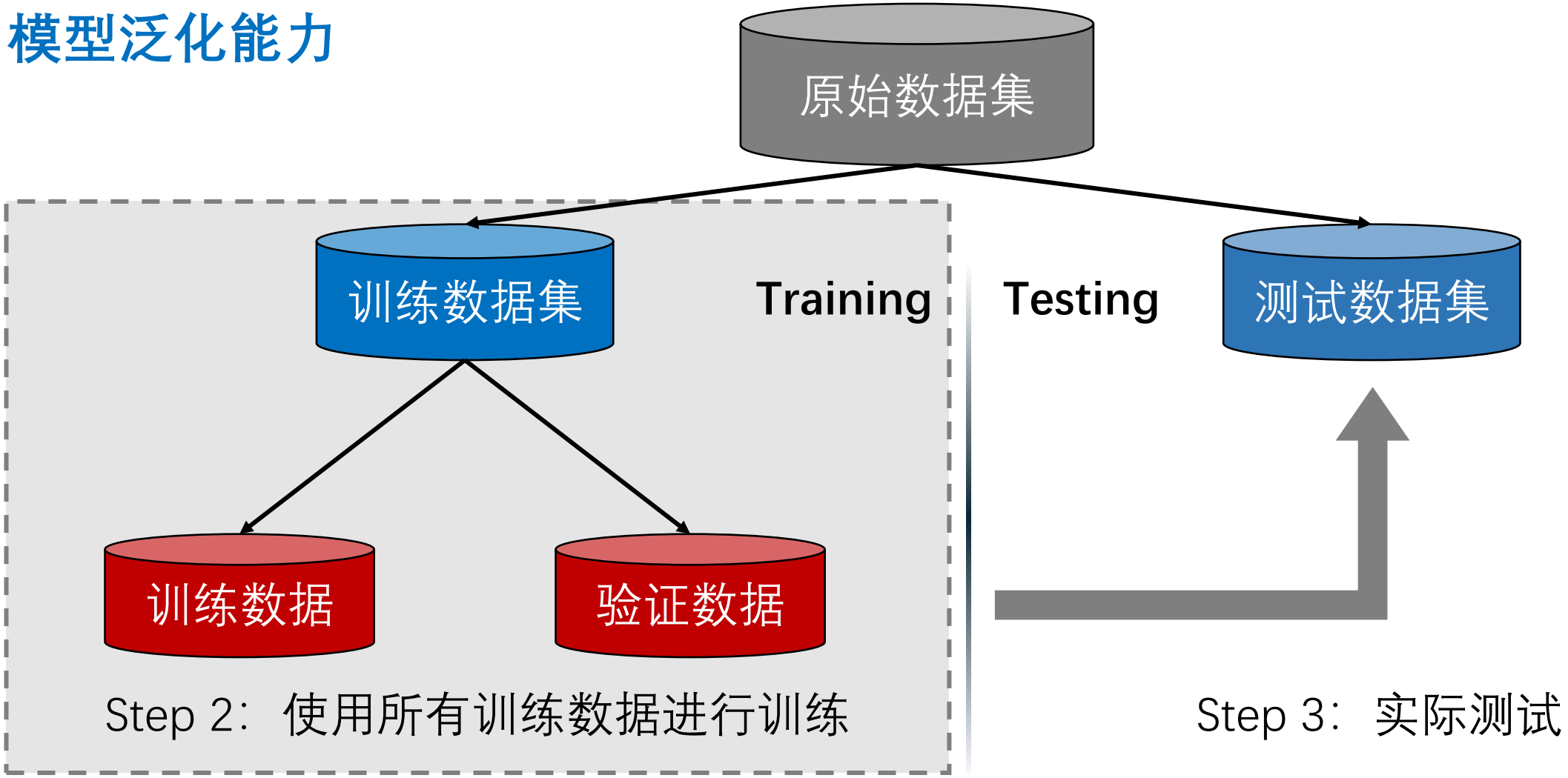
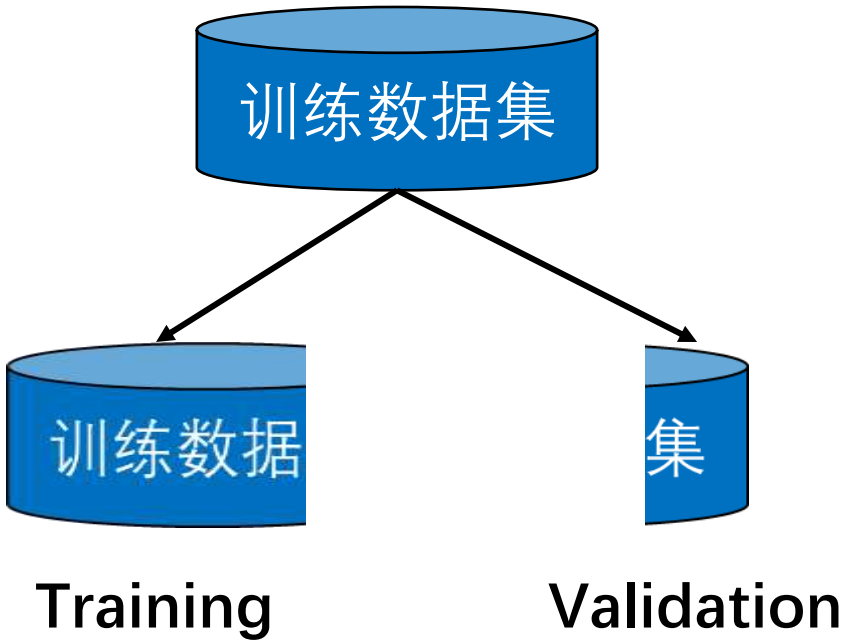


4.6 过拟合及解决思路④：交叉验证增强评估能力

常用的交叉验证方法有哪些咧?

- 留出法 Leave-P-Out



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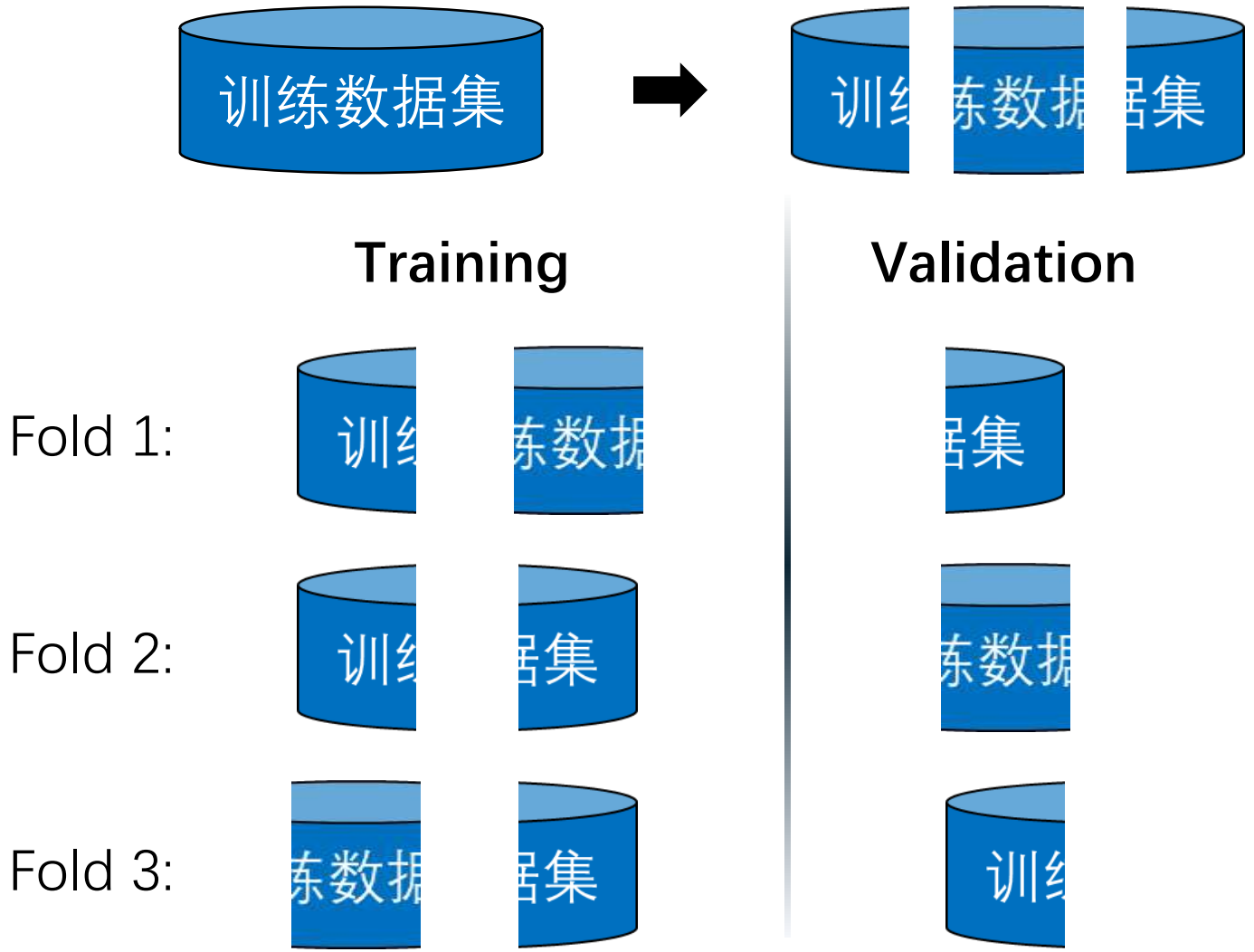
- K折法 K-fold: k = 3



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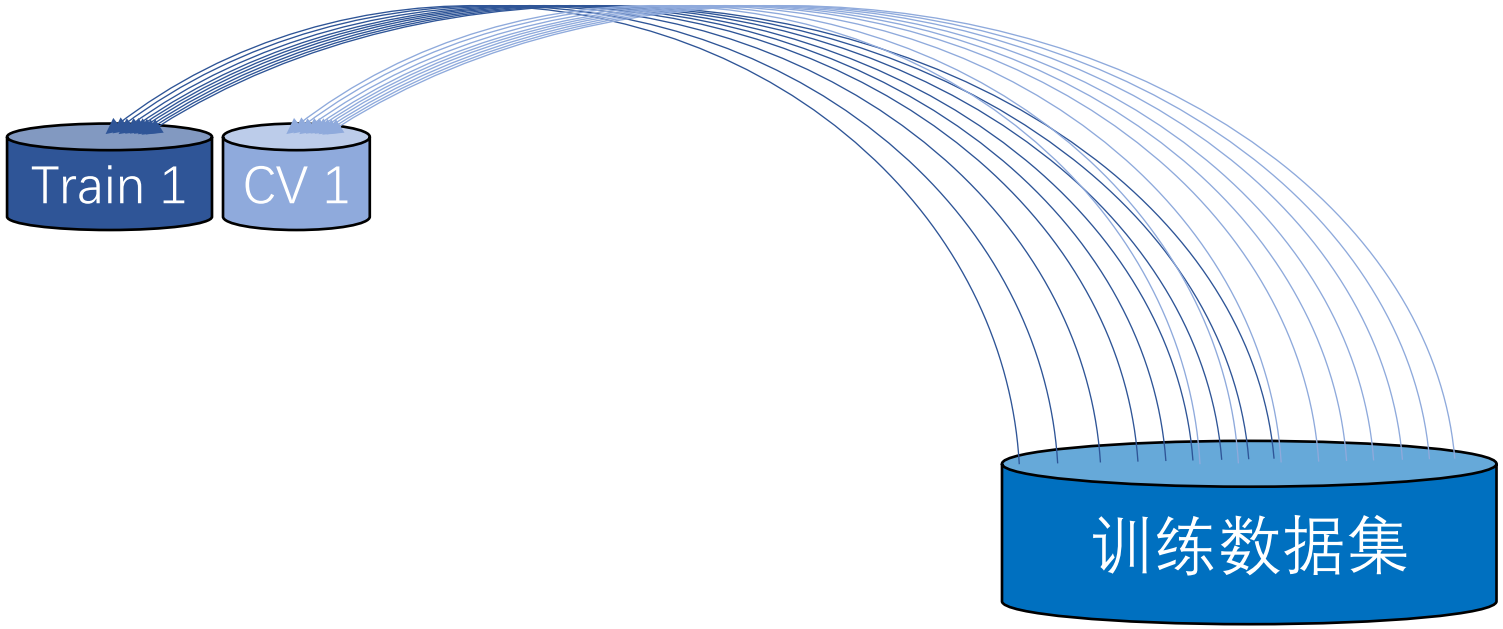
- 自助法 Boostrapping



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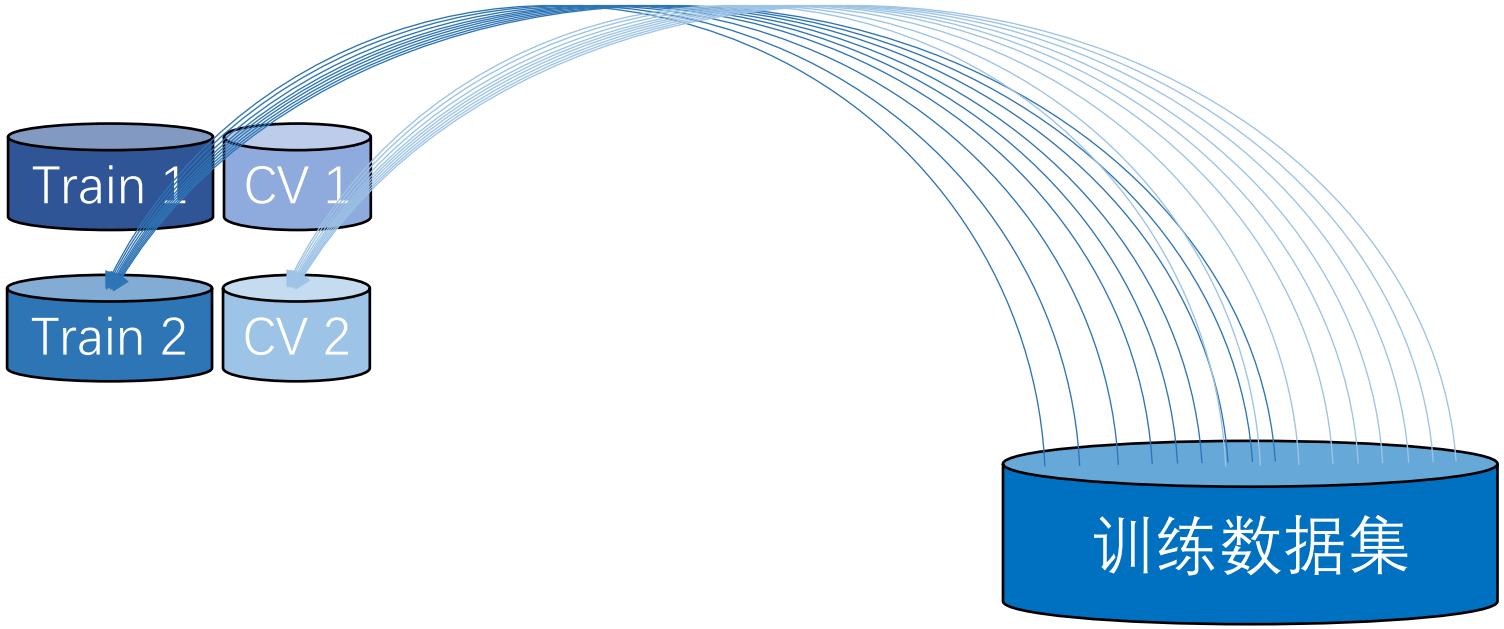

4.6 过拟合及解决思路④：交叉验证增强评估能力

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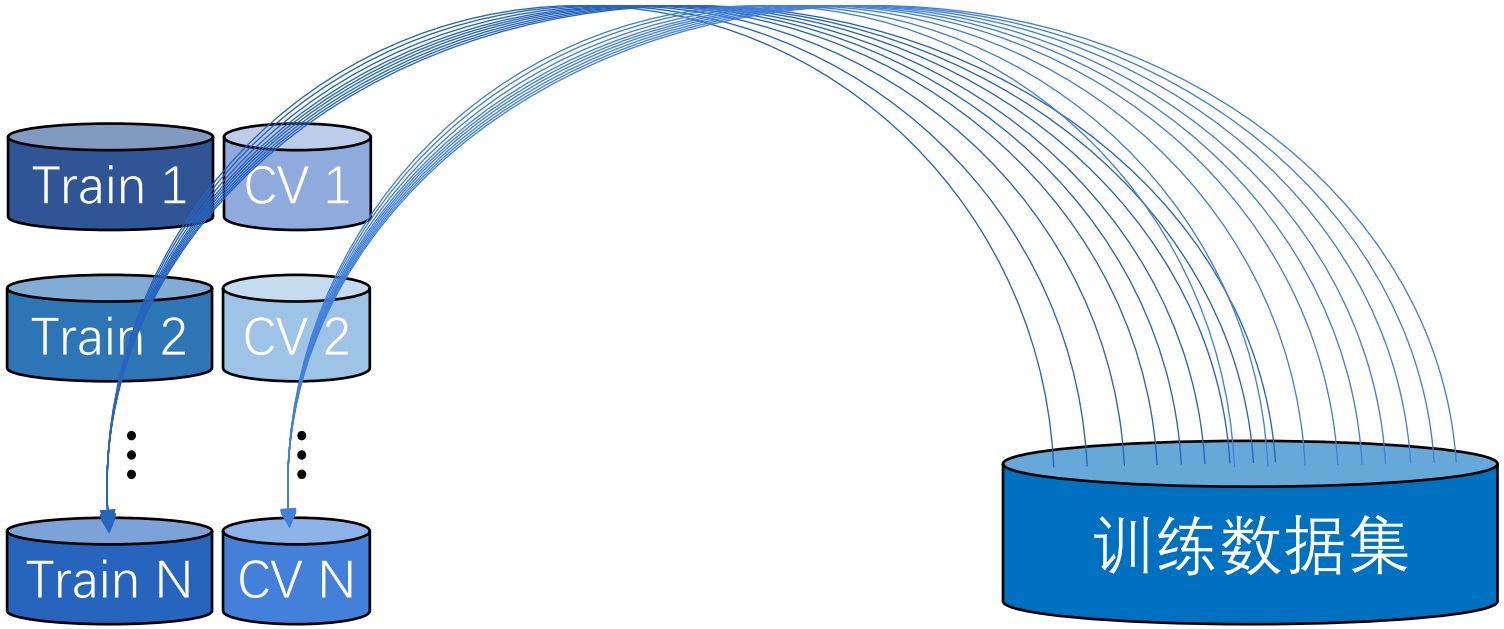
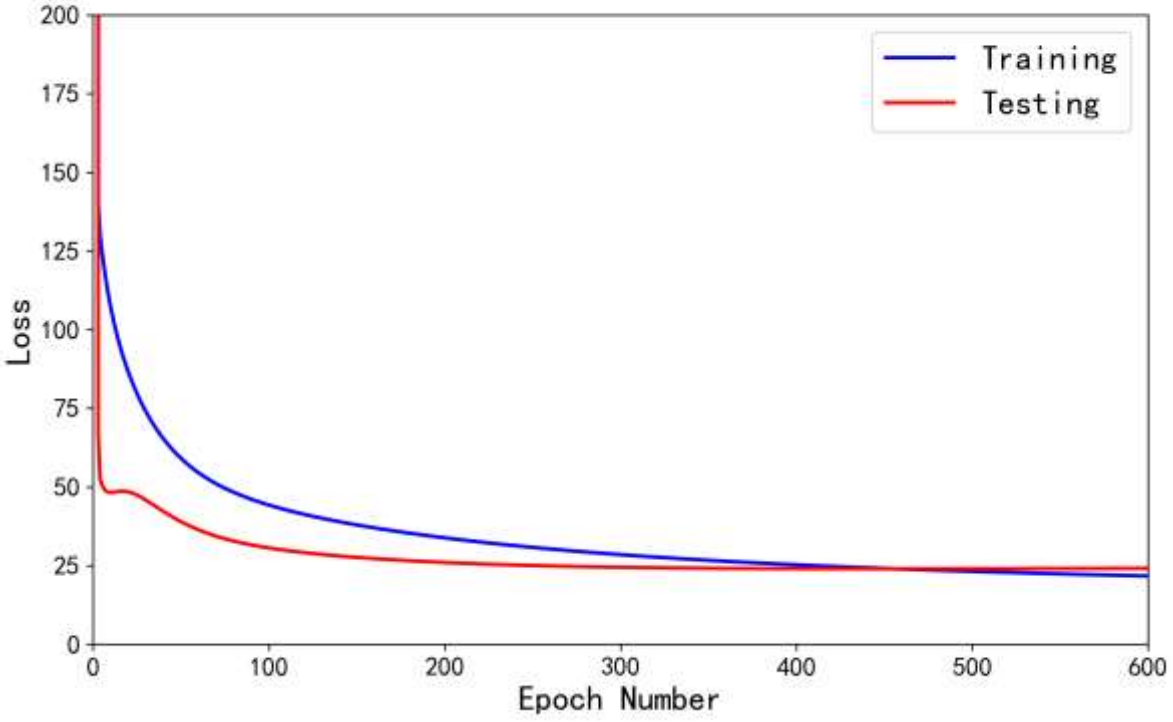
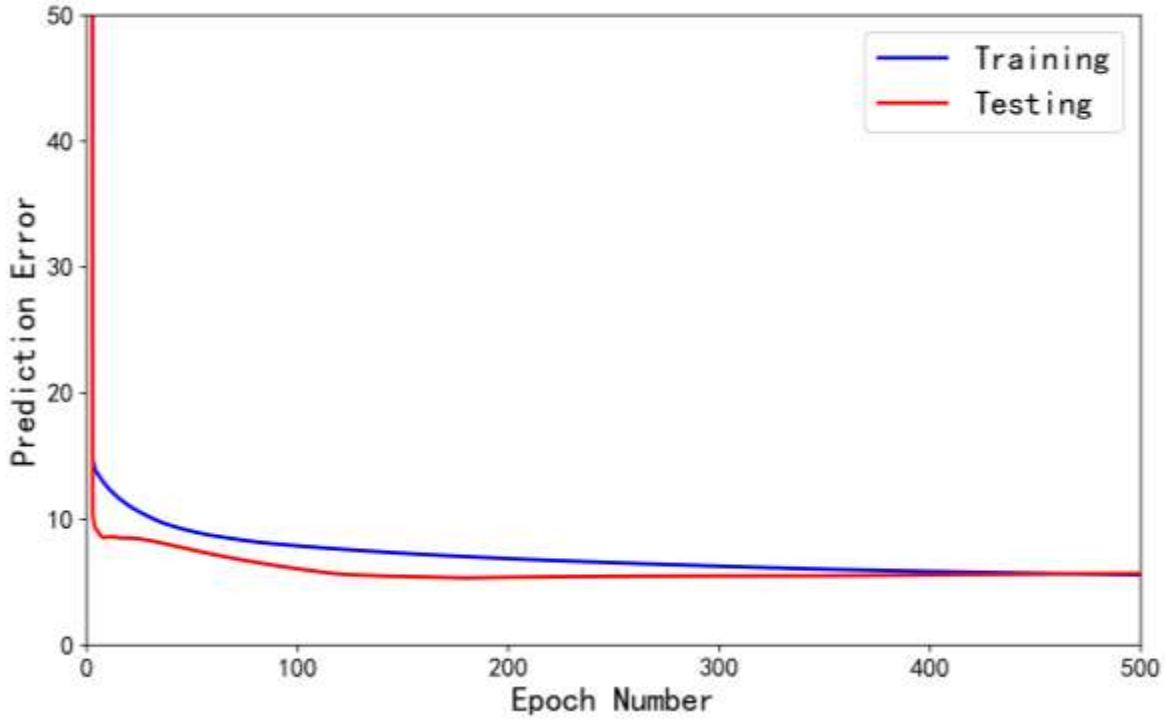


4.6 过拟合及解决思路⑤：在发生过拟合之前结束训练

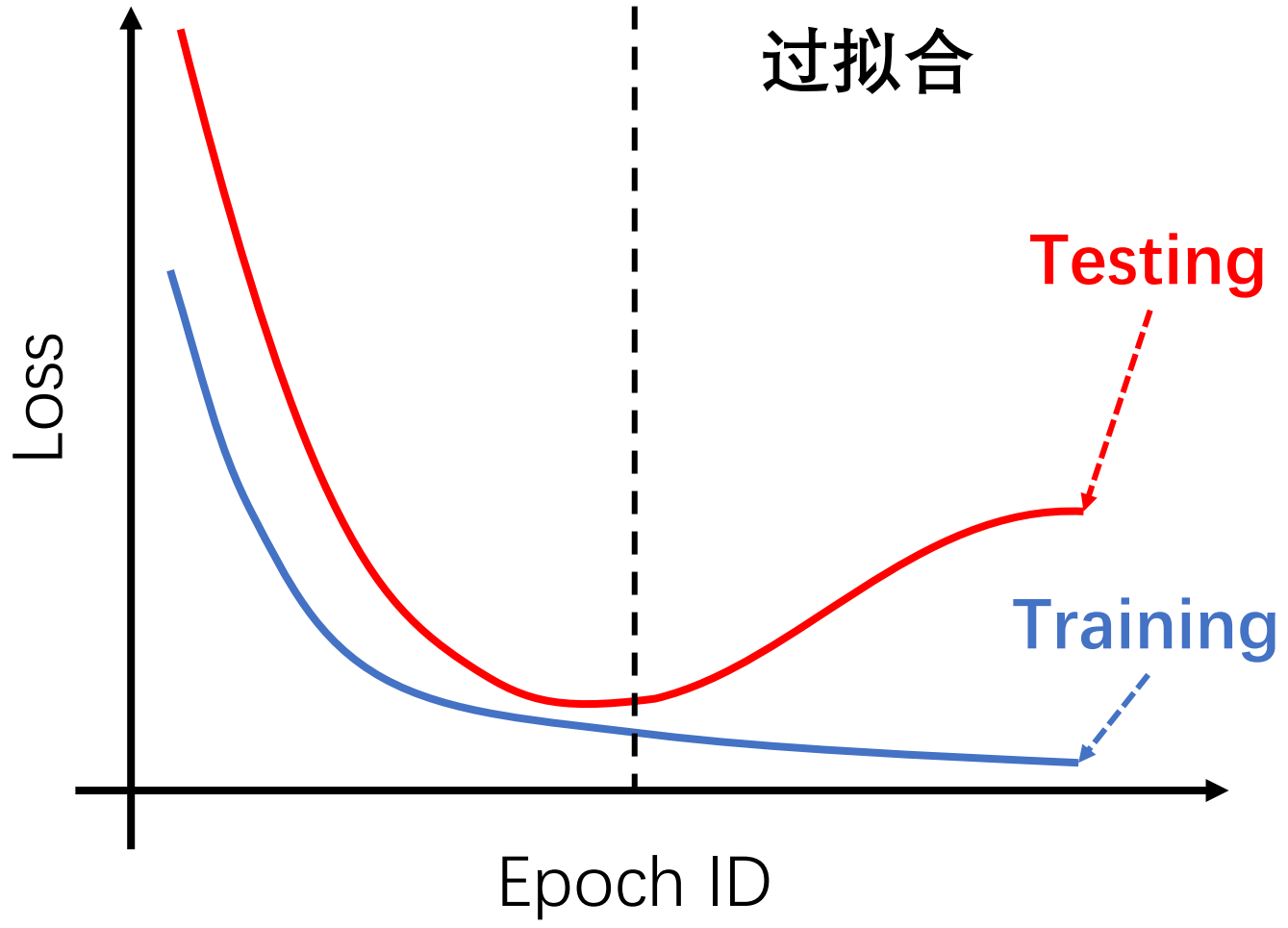
利用**学习曲线**判断泛化能力

理想情况：训练和验证的损失（误差） 以相似的趋势减小并收敛。





4.6 过拟合及解决思路⑤：在发生过拟合之前结束训练

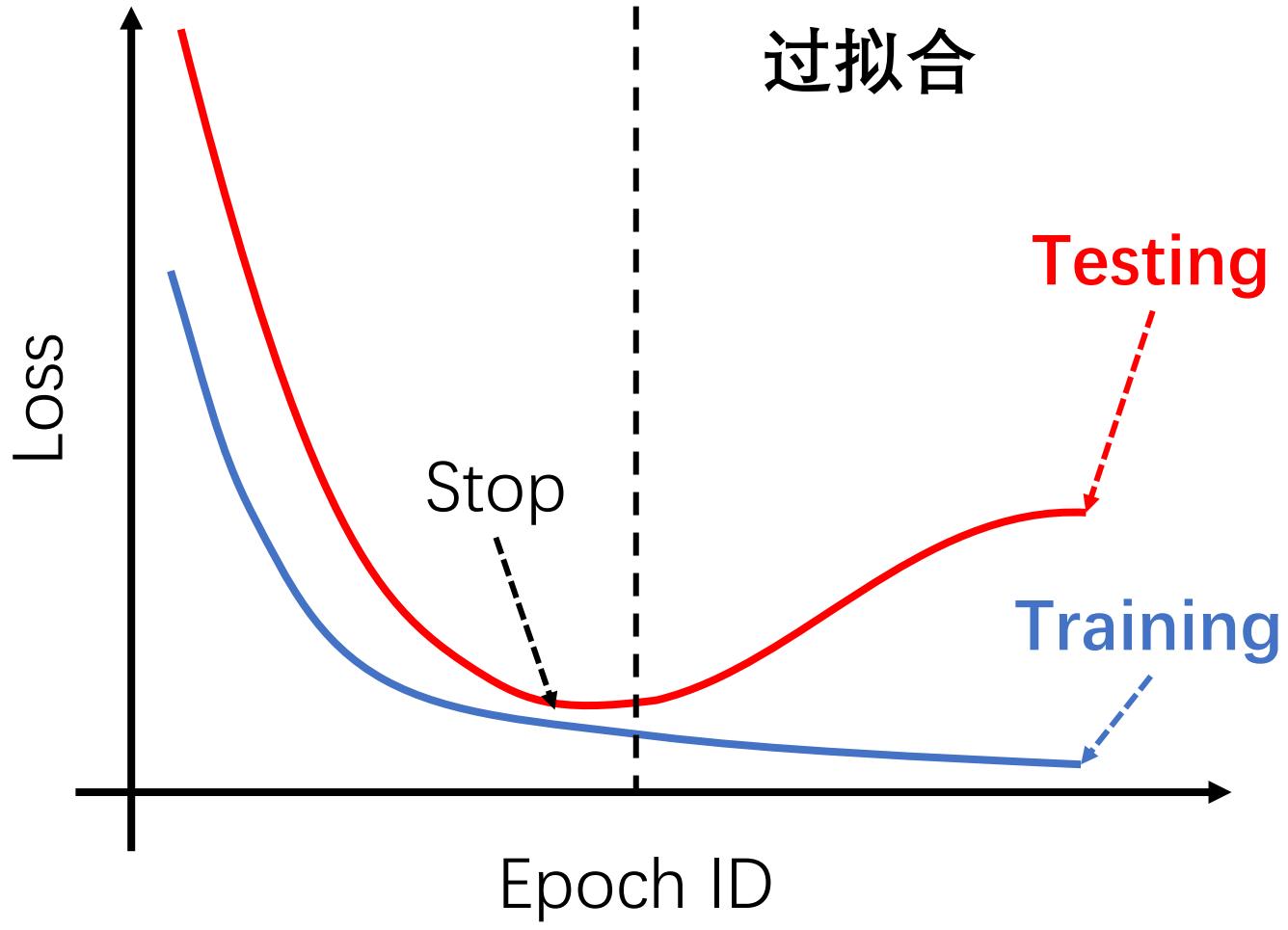


经过一段时间的训练

在训练集上表现很好

在验证集上越来越差

4.6 过拟合及解决思路⑤：在发生过拟合之前结束训练

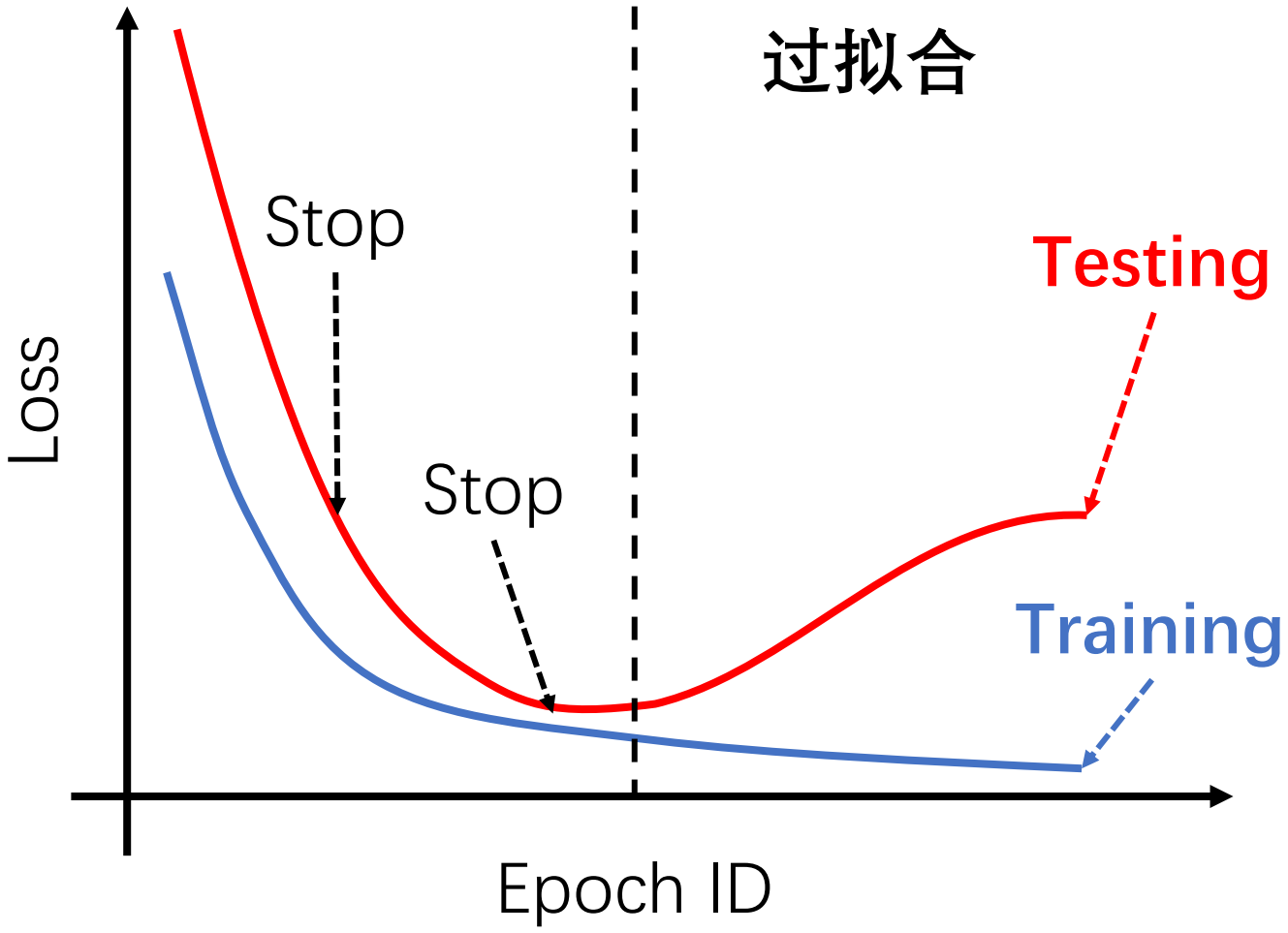


早停法（Early Stopping)

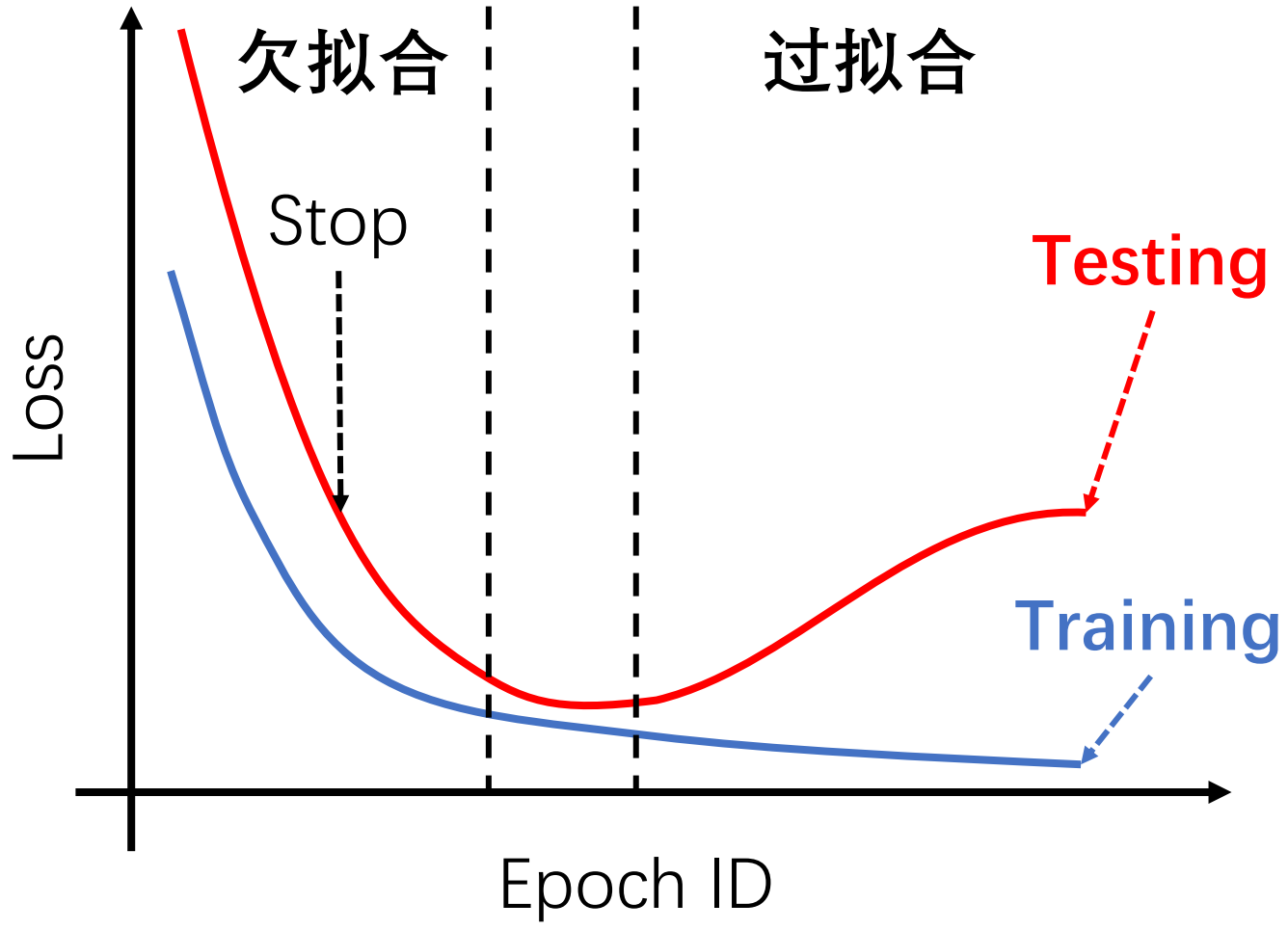
一段时间内，在测试集上的表现不再提升则停止训练。

4.6 过拟合及解决思路⑤：在发生过拟合之前结束训练

如果一不小心停早了会怎么样?



4.6 过拟合及解决思路⑤：在发生过拟合之前结束训练



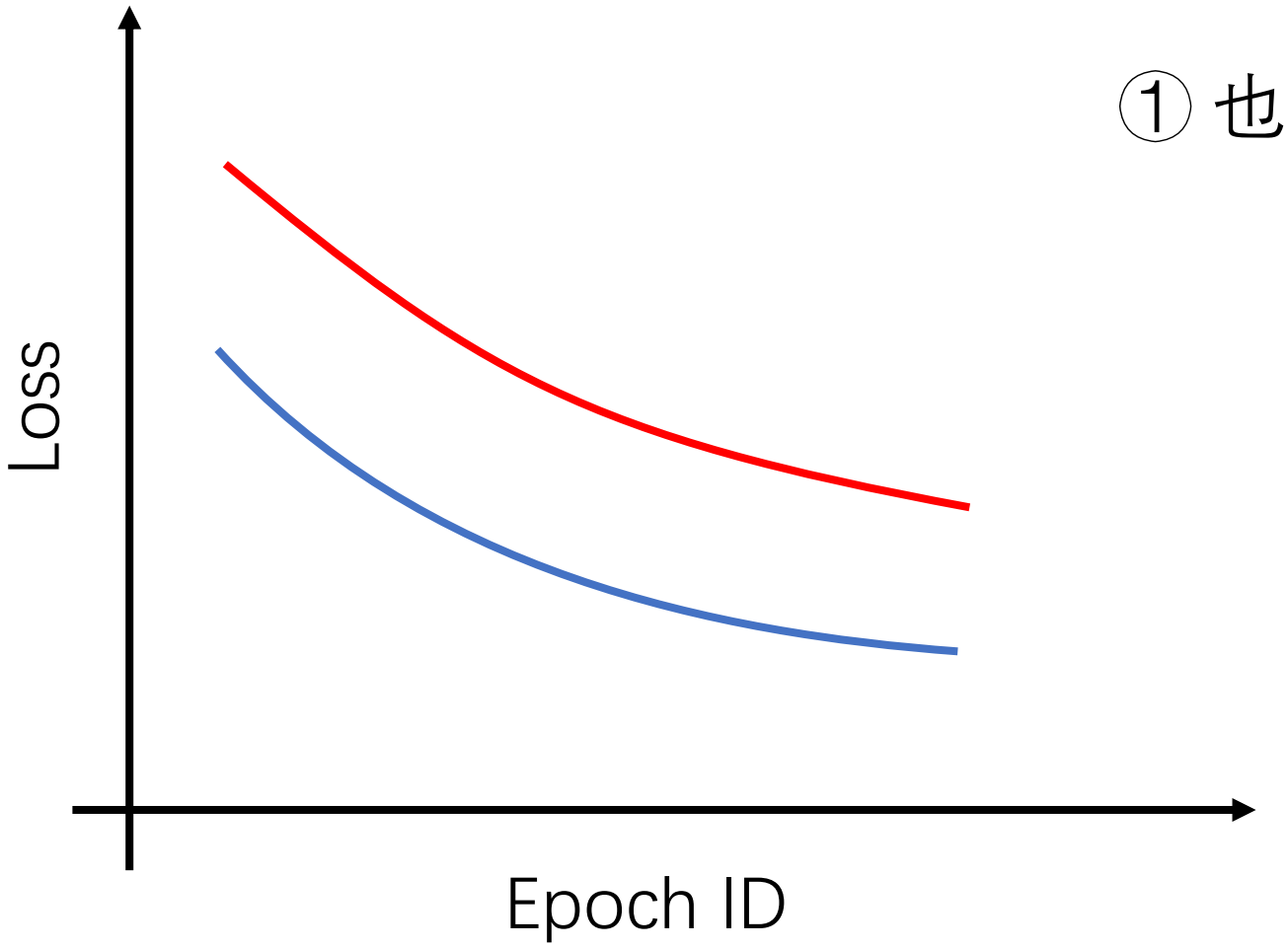
如果一不小心停早了会怎么样?

在训练集上表现差

在测试集上表现差

欠拟合

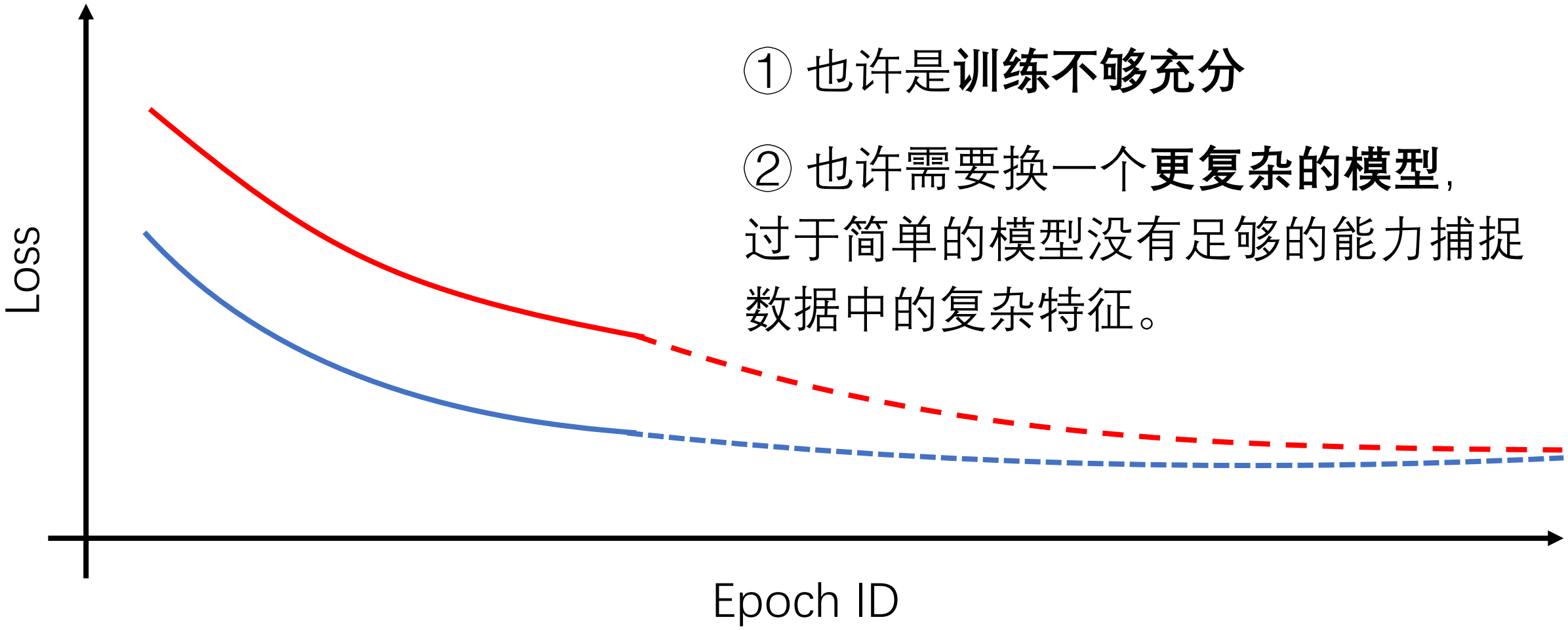
4.6 过拟合及解决思路⑤：在发生过拟合之前结束训练



发生欠拟合的原因

① 也许是**训练不够充分**

4.6 过拟合及解决思路⑤：在发生过拟合之前结束训练

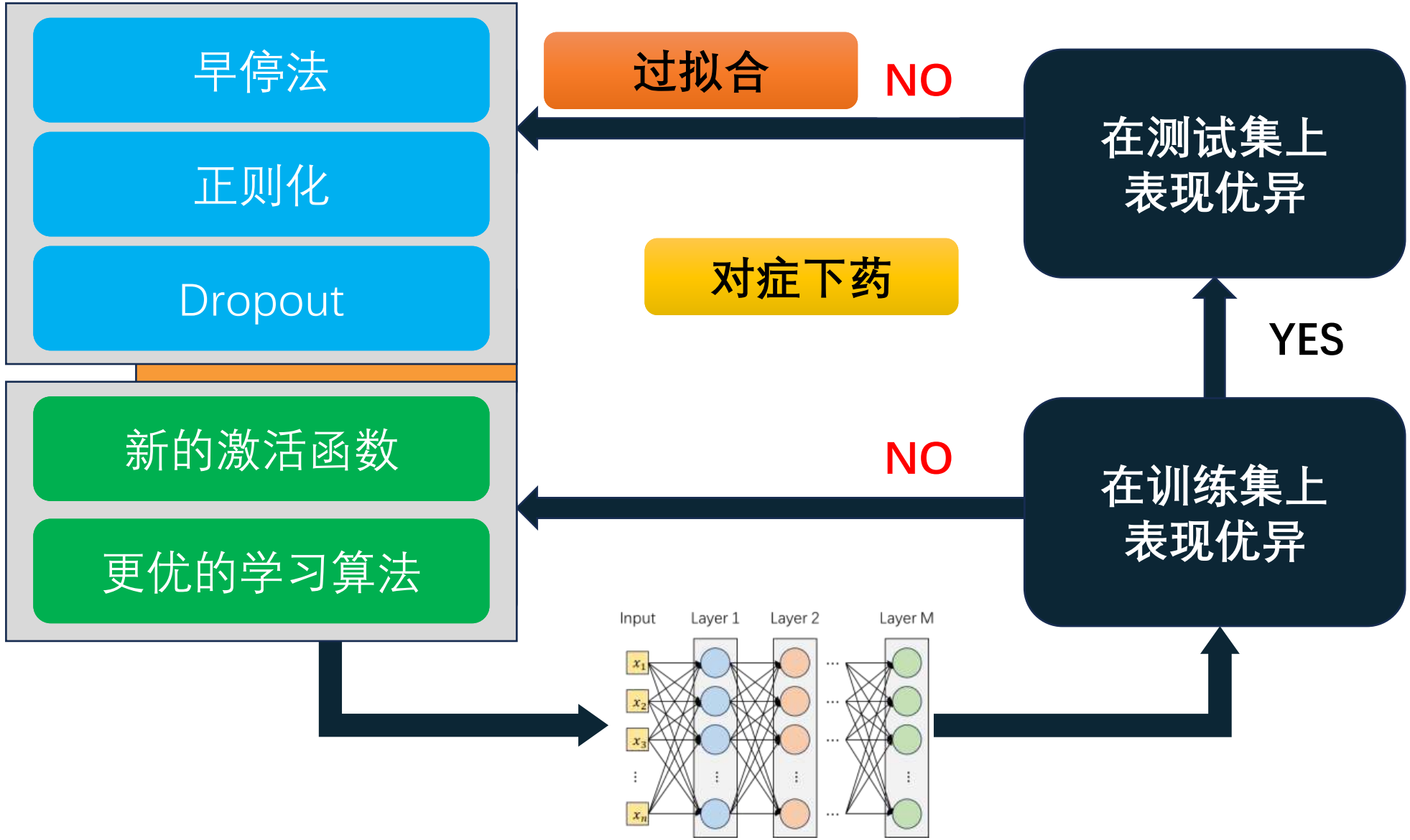


发生欠拟合的原因

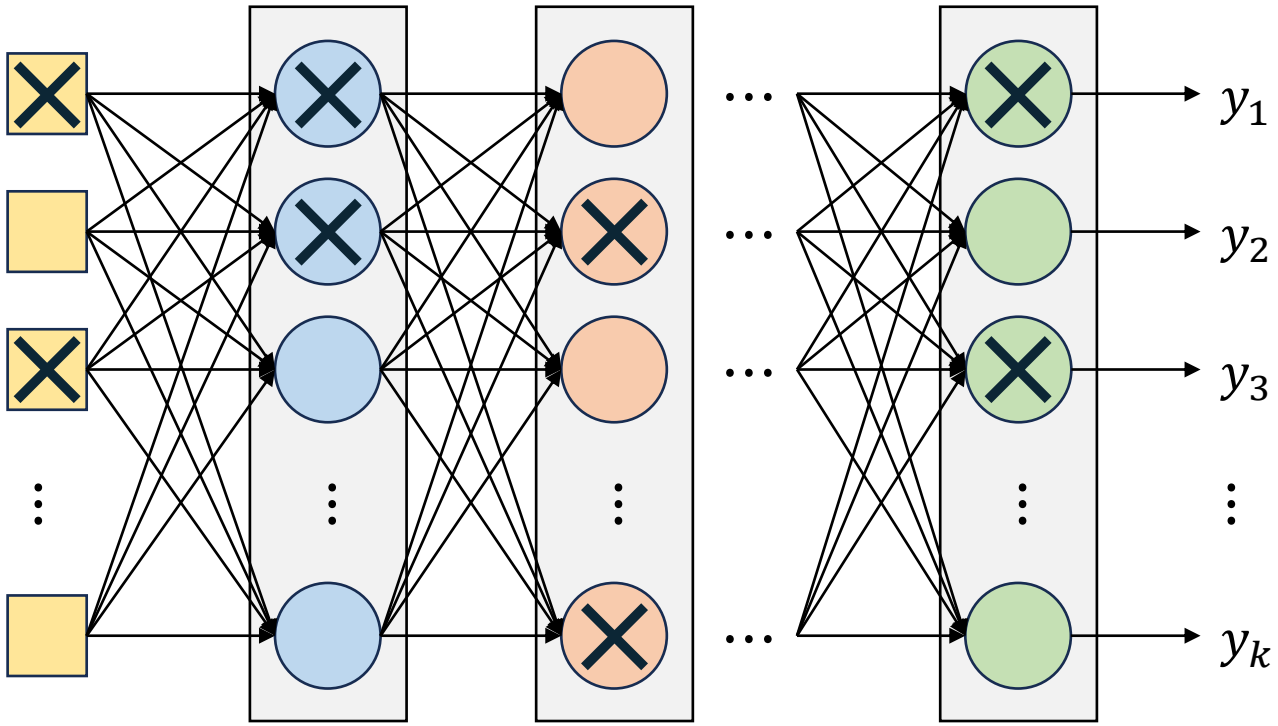
① 也许是**训练不够充分**

② 也许需要换一个**更复杂的模型**，过于简单的模型没有足够的能力捕捉数据中的复杂特征。

训练深度神经网络一点经验



4.7 过拟合及解决思路：Dropout



Dropout以 $p = 0.5$ 的概率丢弃特征或节点

！！！每次更新都需要重新Dropout！！！

4.7 过拟合及解决思路：Dropout为什么有用？

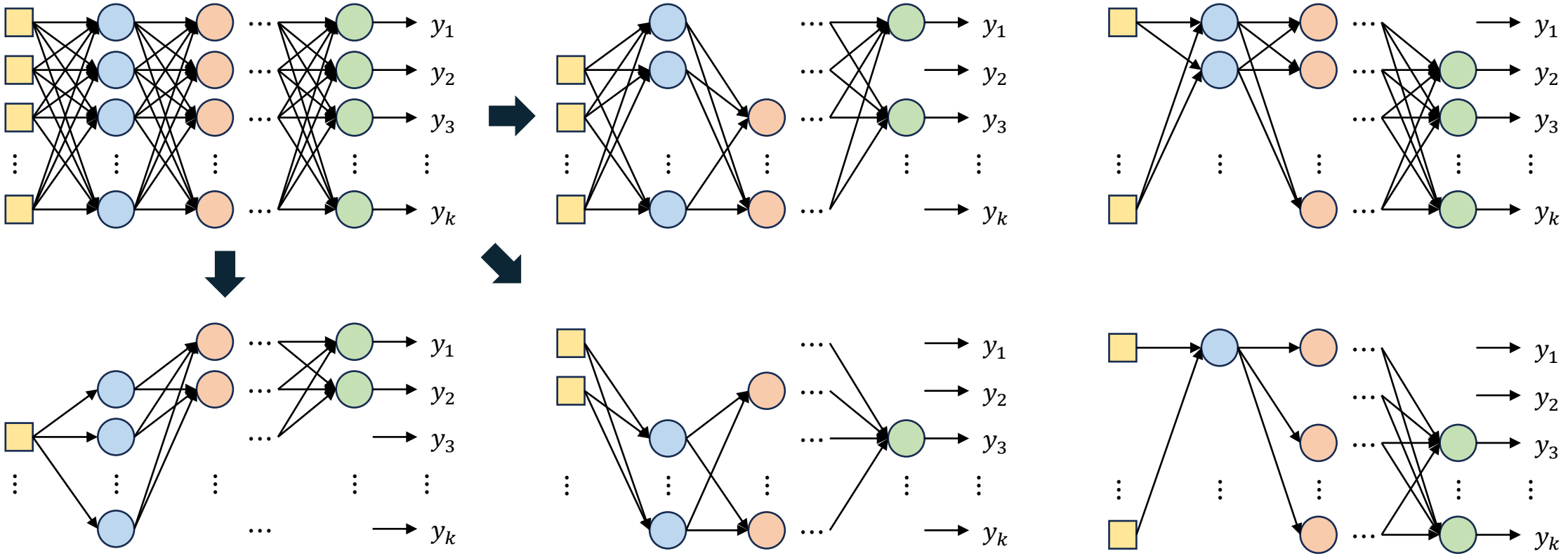
Dropout使网络变得极为简单

每个结点需要“加倍努力”变得更强

！！！每次更新都需要重新Dropout！！！

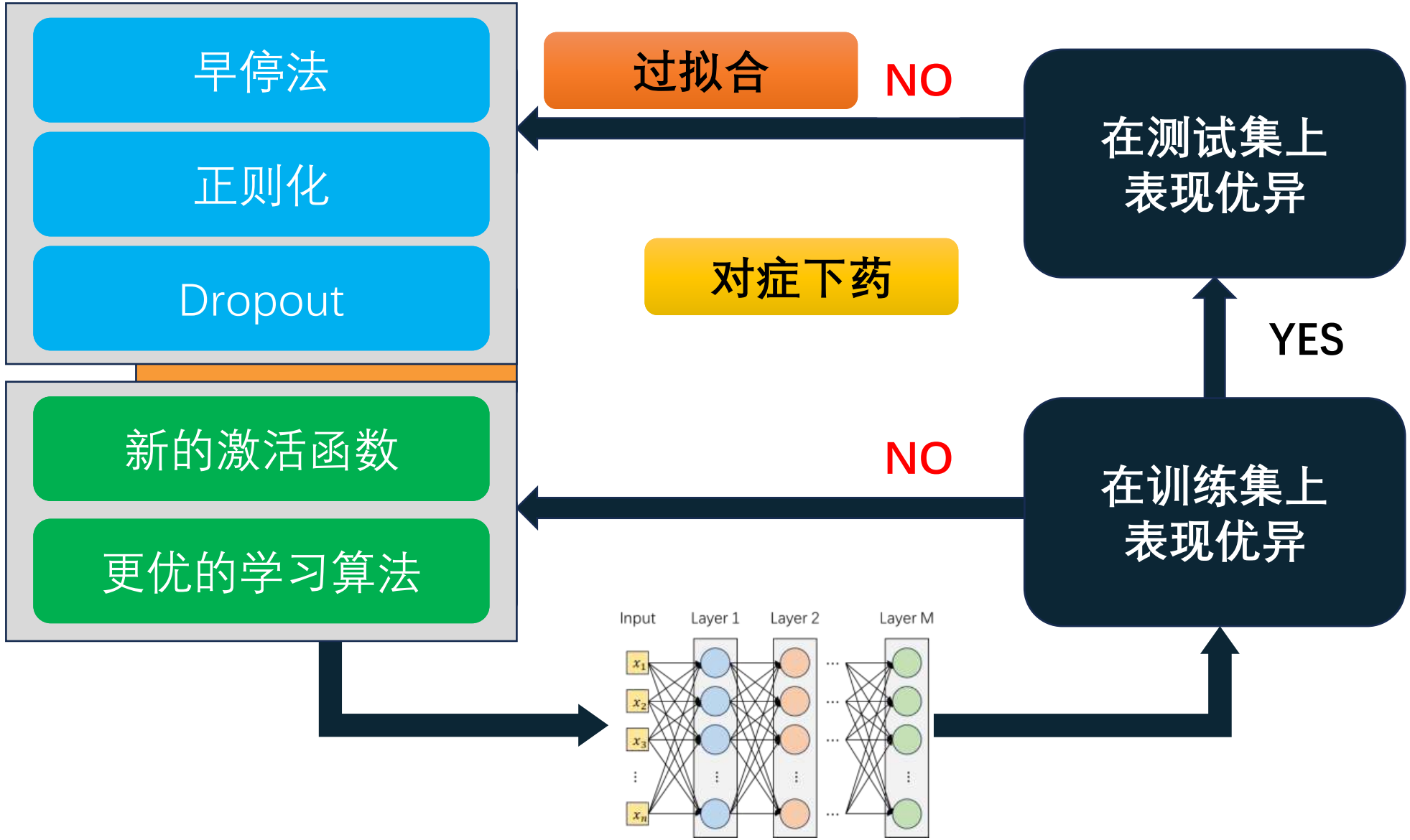
4.7 过拟合及解决思路：Dropout为什么有用？

Dropout相当于训练了极多个不同的简单神经网络



融合多个弱分类器的结果，能够达到极好的效果，并具有更强的泛化能力

训练深度神经网络一点经验



4.8 最后梳理一下

- 1958：感知机 Perceptron（linear model）
- 1969：Perceptron has limitation
- 1980s：多层感知机 Multi-layer Perceptron
  - Do not have significant difference from DNN today
- 1986：反向传播算法 Backpropagation
- 1989：1 hidden layer is “good enough”, why deep?
- 2006：RBM initialization（breakthrough）
- 2011：Start to be popular in speech recognition
  - 发现GPU效果非常好
- 2012：Win ILSVRC image competition
  - 深度学习+GPU=碾压之势

Geoffrey Hinton

图灵\诺贝尔物理学奖得主

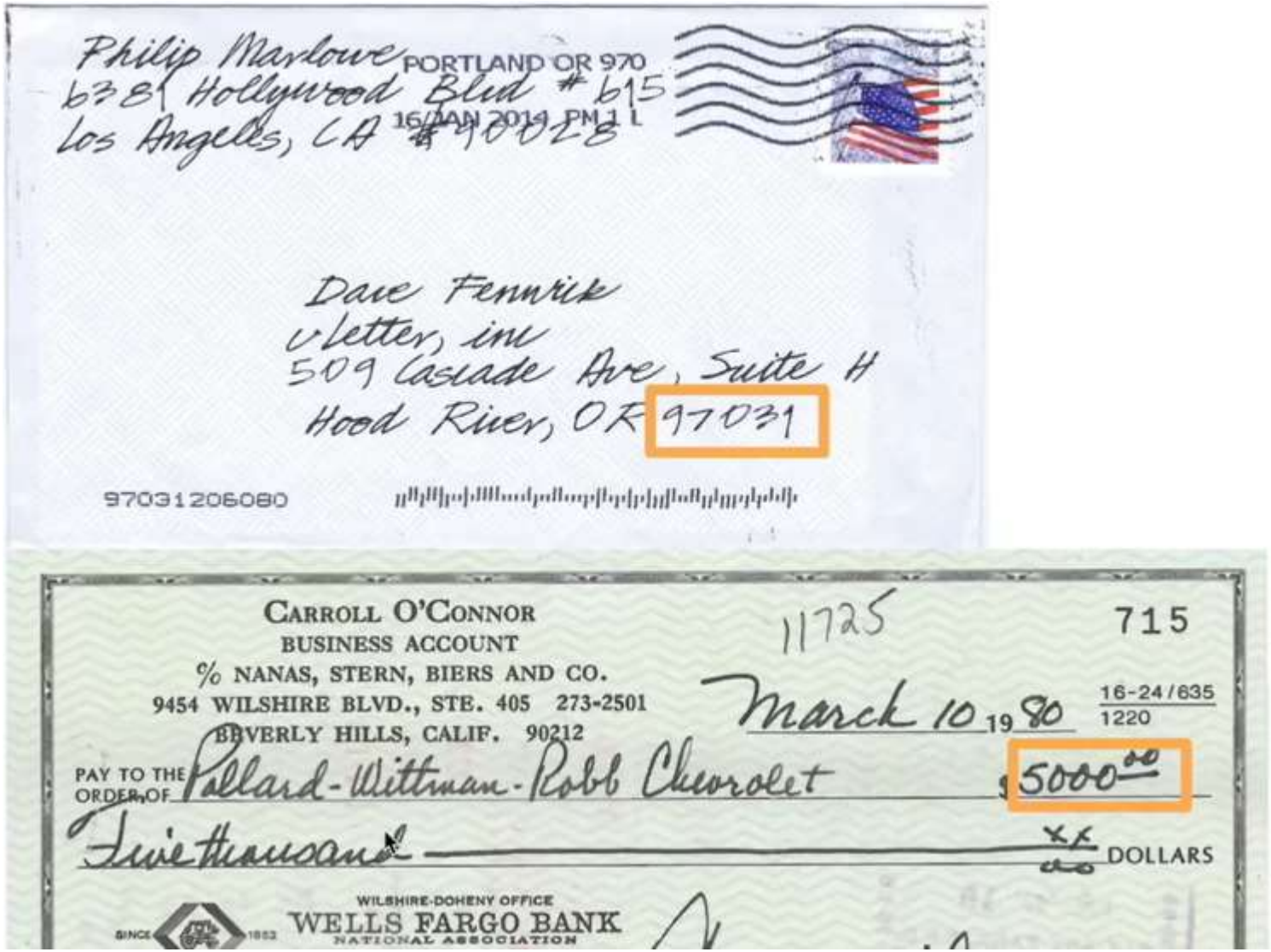
李飞飞

ILSVRC竞赛创始人

黄仁勋

Nvidia CEO

4.8 最后梳理一下

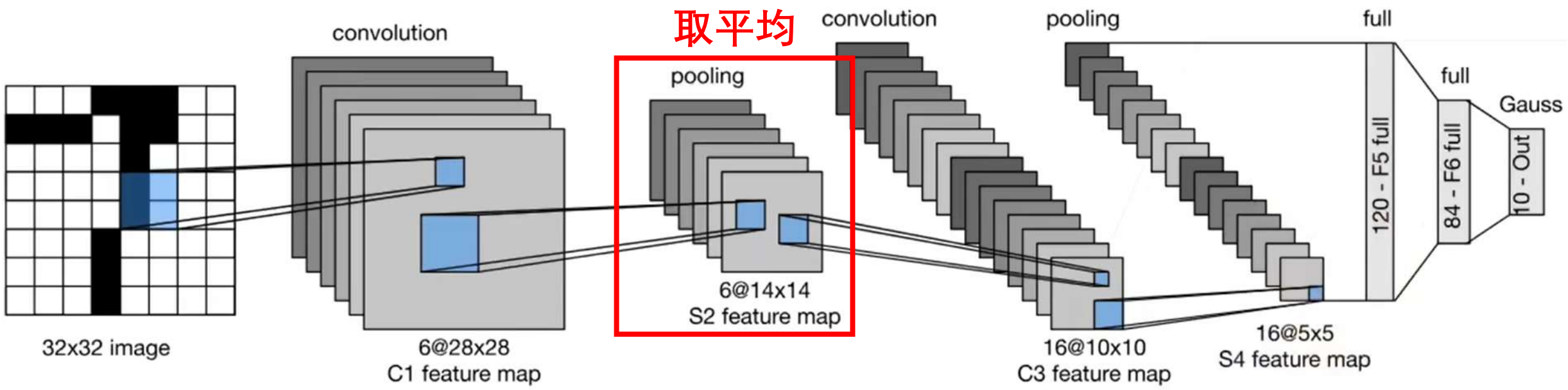


Yann LeCun

Pioneer of CNN

Yann LeCun, Corinna Cortes, and Christopher J.C. Burges. The MNIST database of handwritten digits. https://yann.lecun.com/exdb/mnist/

4.8 最后梳理一下



- LeNet是早期成功的神经网络
- 先使用卷积层来学习图片空间信息
- 然后使用全连接层来转换到类别空间

4.8 最后梳理一下

李飞飞

数据集	IMAGENET	MNIST
图片类型	自然物体的彩色图片	手写数字的黑白图片
图片大小	469 X 387	28 X 28
样本数量	1200000	60000
类数	1000	10

4.8 最后梳理一下

ImageNet Classification with Deep Convolutional Neural Networks

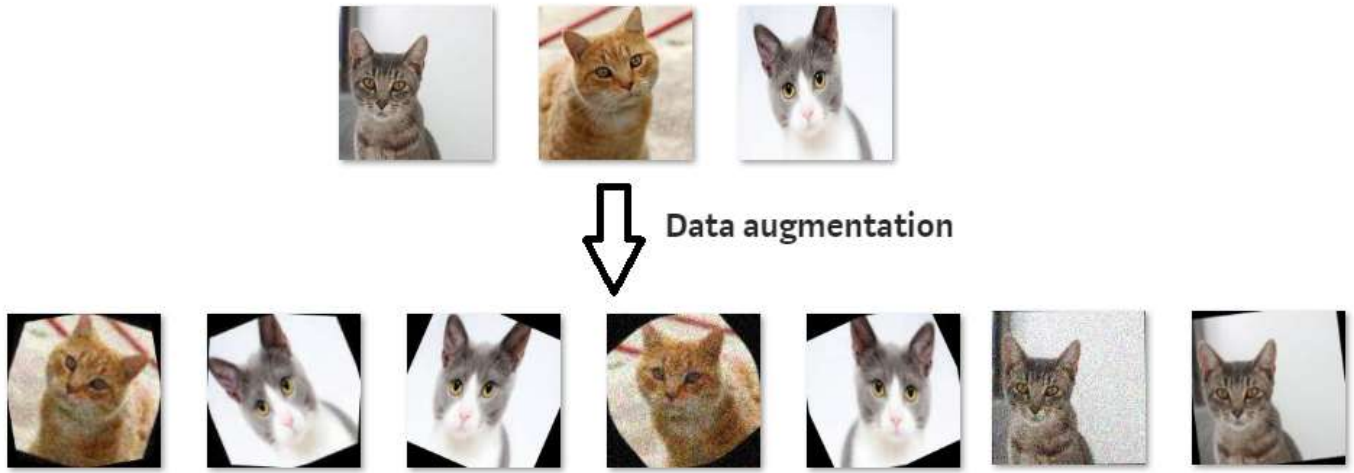
- AlexNet 赢了 2012 年 ImageNet 竞赛
- 更深更大的 LeNet
- 主要改进：
  - 丢弃法 Dropout
  - ReLu
  - MaxPooling
- 计算机视觉方法论的改变

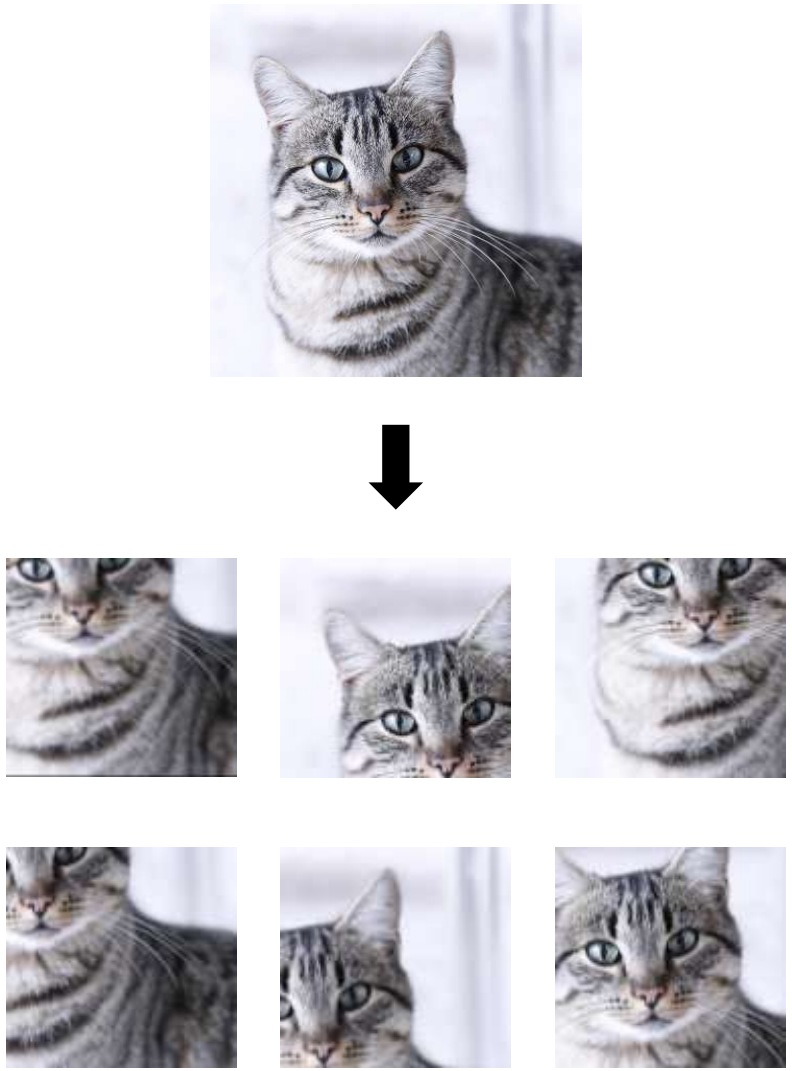
人工提取特征 → 自动提取特征

4.8 最后梳理一下

AlexNET的一些细节

- 数据增强
- 隐藏全连接层后加入了Dropout
- 激活函数从Sigmoid变为了ReLU





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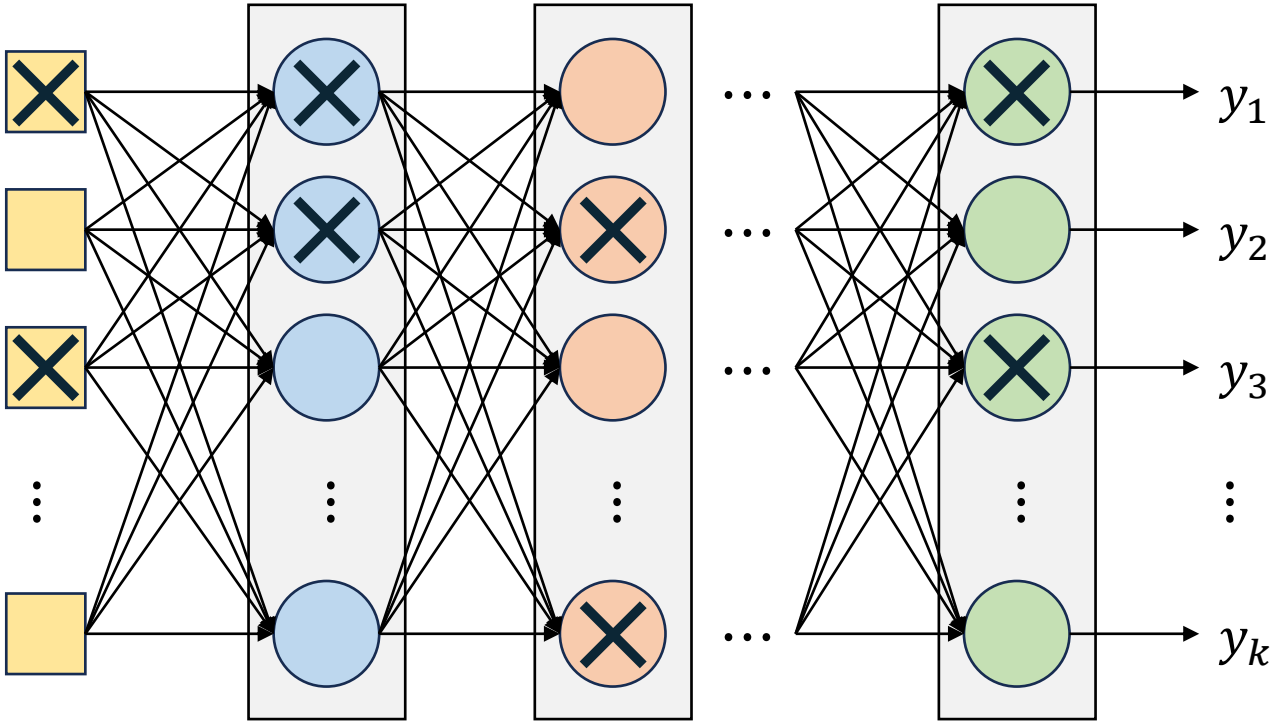
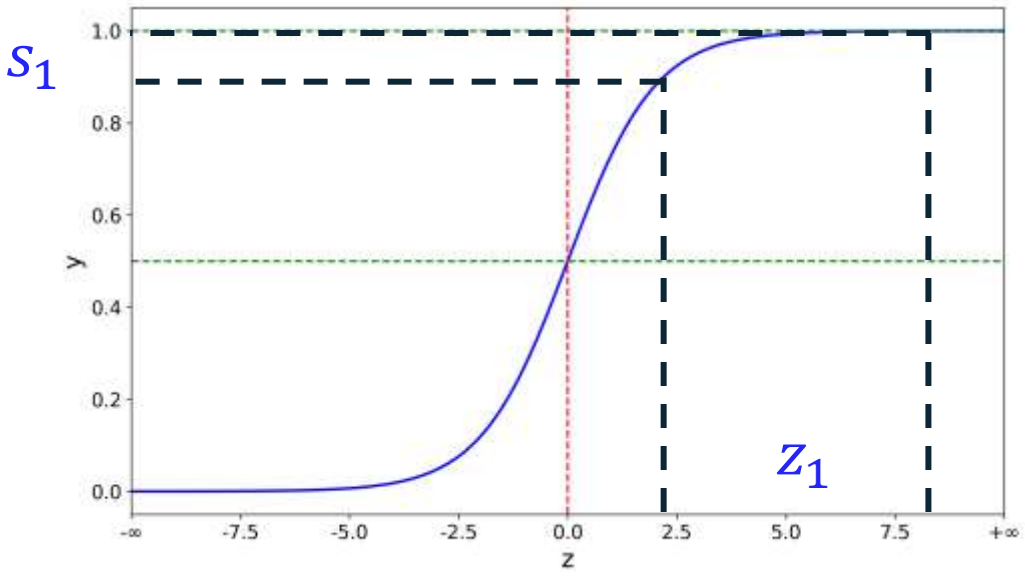


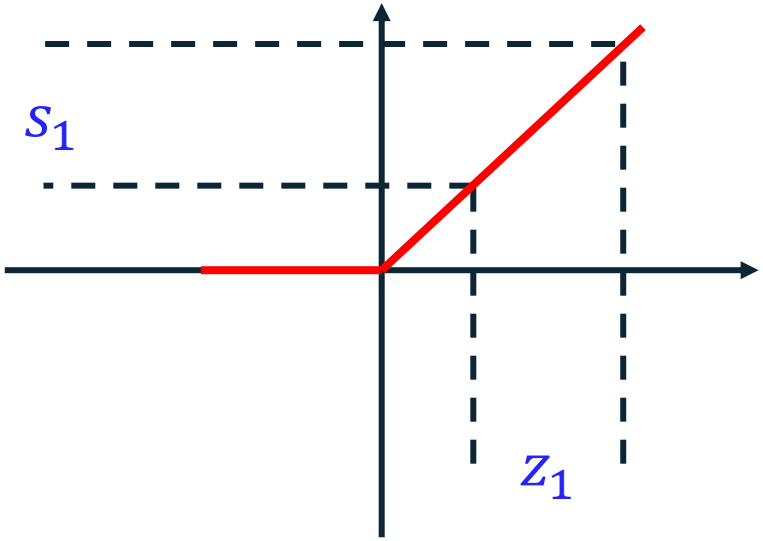
Dropout以 $p = 0.5$ 的概率丢弃特征或节点

！！！每次更新都需要重新Dropout！！！

4.8 最后梳理一下

- 数据增强
- 隐藏全连接层后加入了Dropout
- 激活函数从Sigmoid变为了ReLU（缓解**梯度消失**问题）

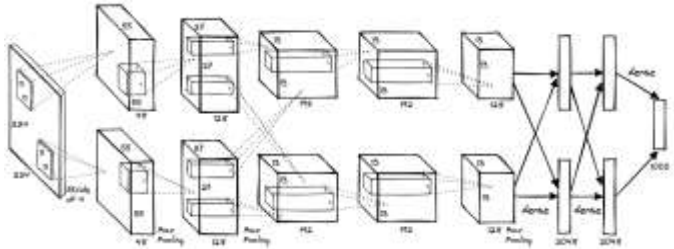




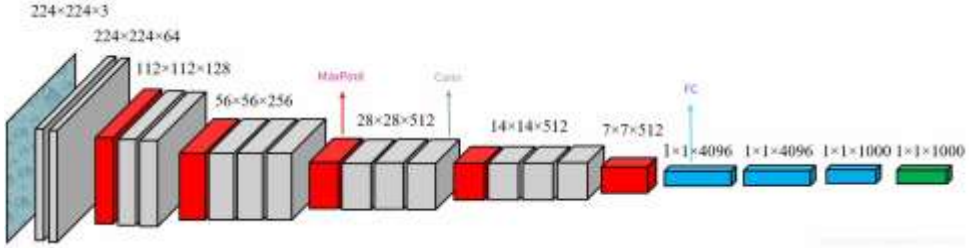
4.9 One More Thing

神经网络的能力和隐藏层数正相关

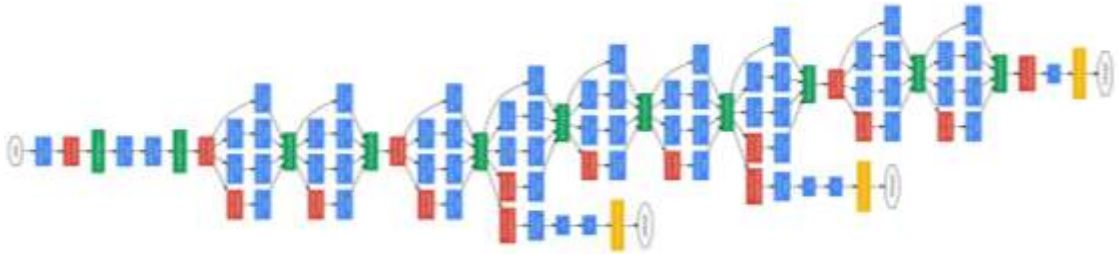
AlexNet'12
8 Layers
16.4%



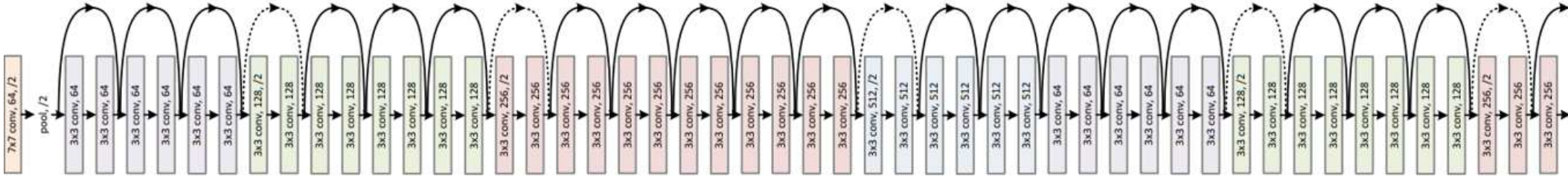
VGG'14
19 Layers
7.3%



GoogleNet'14
22 Layers
6.7%

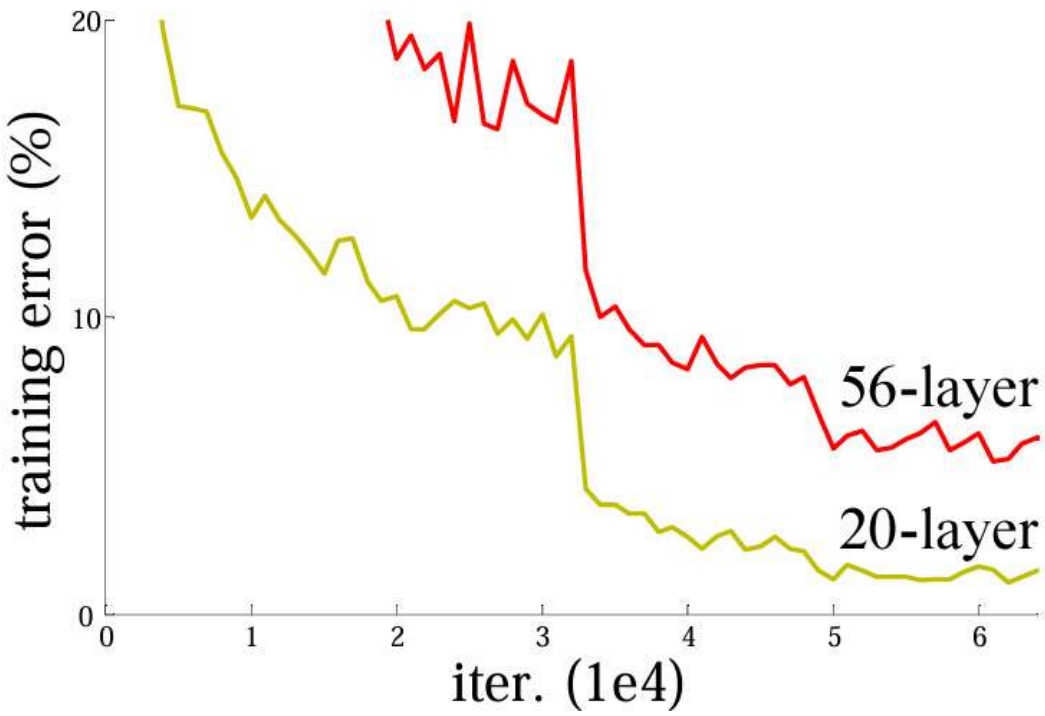


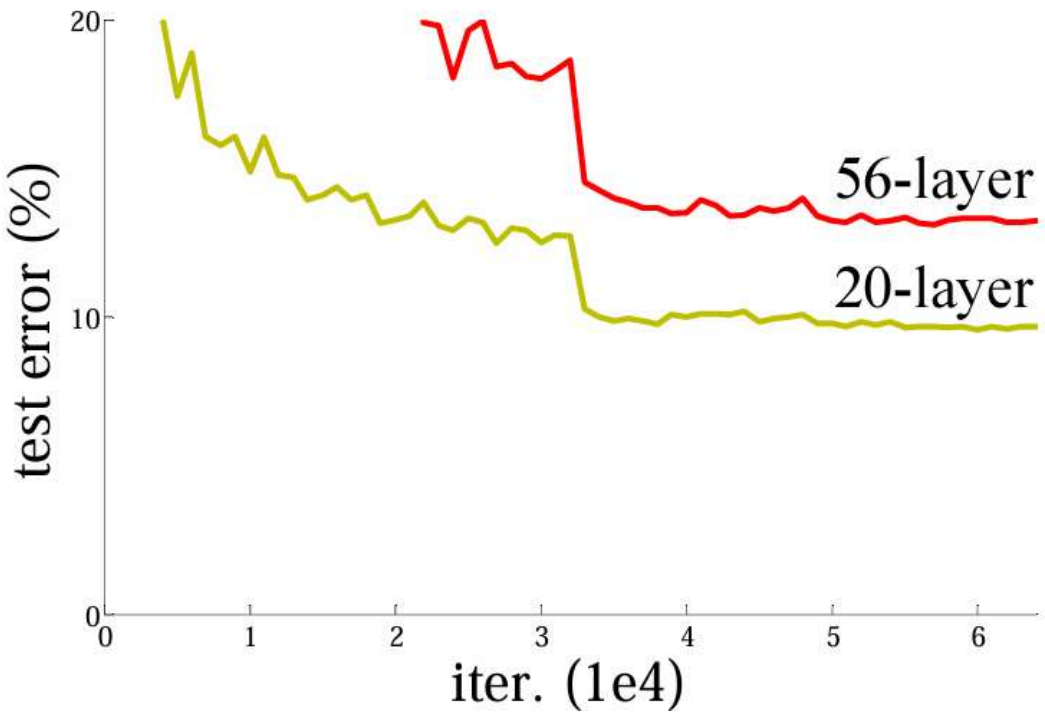
Residual Net'15
152 Layers
3.57%



ResNet让训练
真·深度神经网络
成为可能

4.9 One More Thing

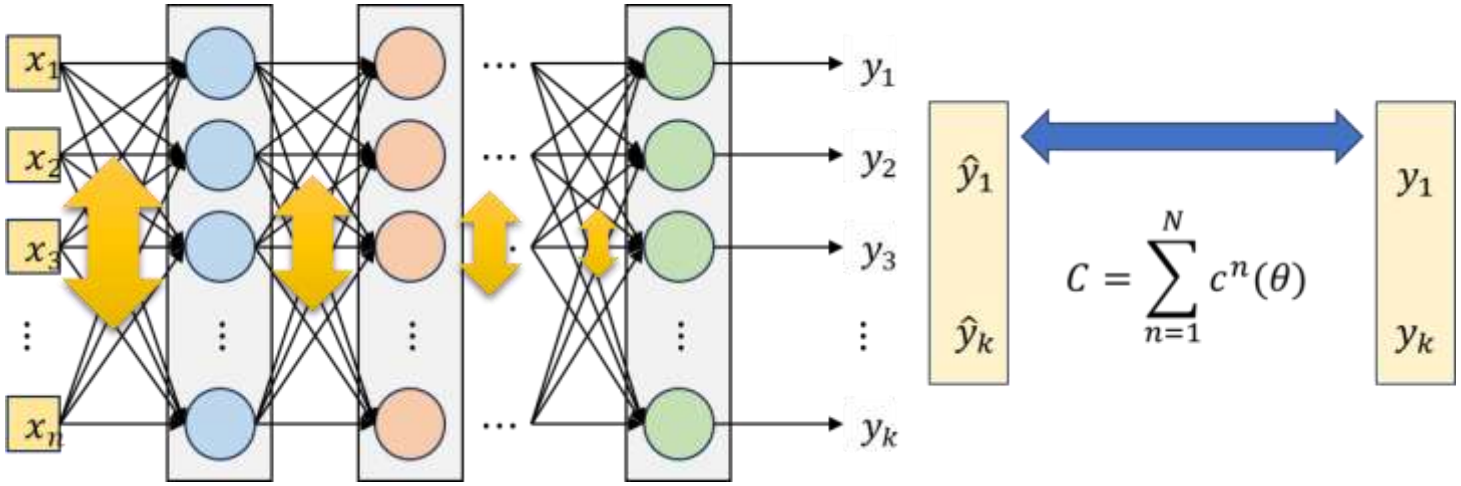




并非发生过拟合，而是训练出了偏差

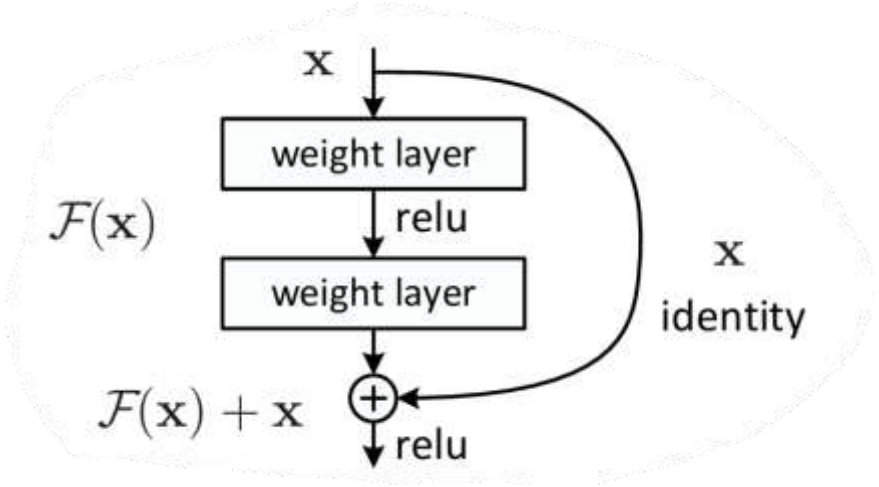
https://arxiv.org/pdf/1512.03385

4.9 One More Thing

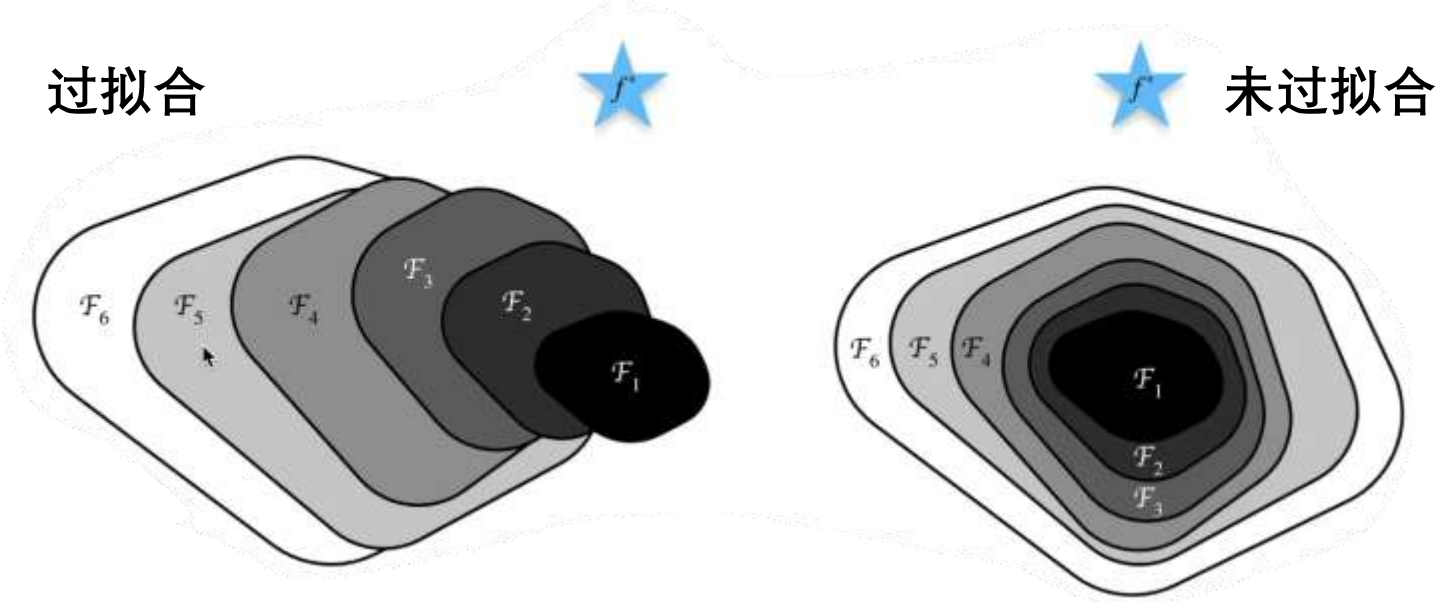


ResNet朴素的愿望是

深模型至少不会比浅模型差



过拟合

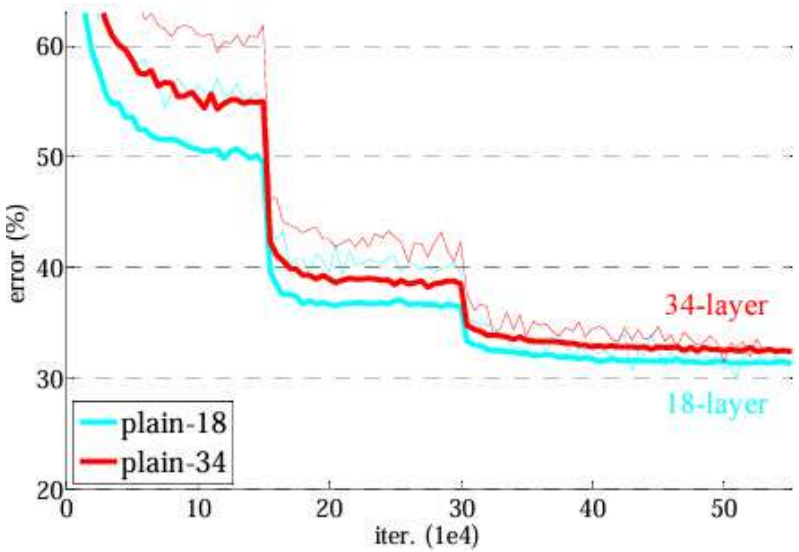


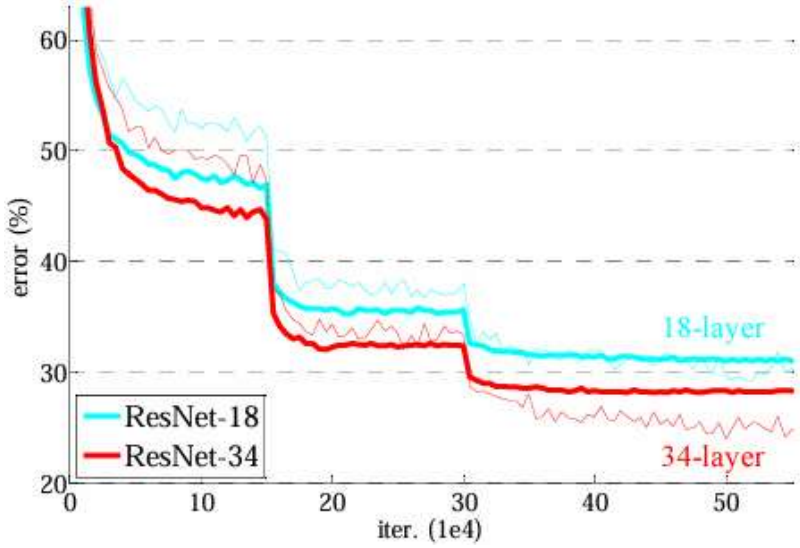
未过拟合

即使 $F(x)$ 学“偏”了

x 也不影响后续层的学习

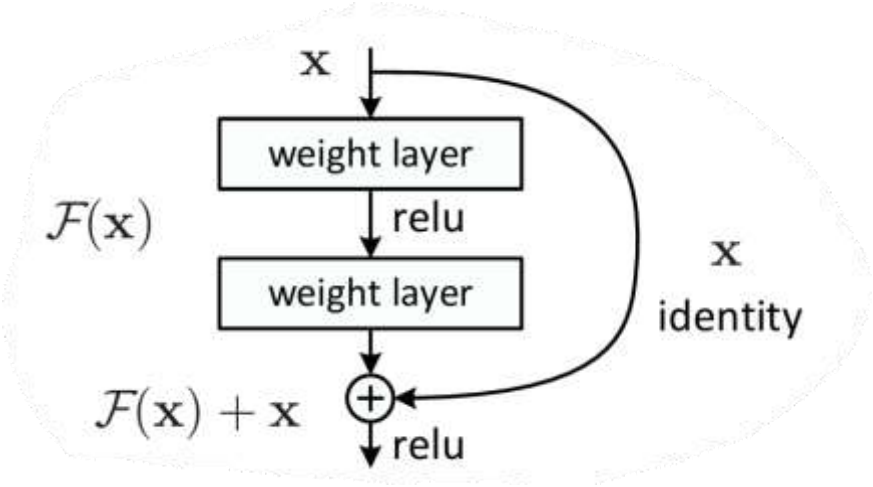
4.9 One More Thing





ResNet朴素的愿望是

深模型至少不会比浅模型差



即使 $F(x)$ 学“偏”了

x 也不影响后续层的学习

信息直达高层，就不会跑歪了

在进入LLM章节之前思考一下……

“**文字接龙**”是回归还是分类?

用机器学习或者深度学习怎么做?